NEXTCYCLE MICHIGAN

MICHIGAN 2025
GAP ANALYSIS

PREPARED BY:

RRS

COMMISSIONED BY:

EGLE MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

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EXECUTIVE SUMMARY

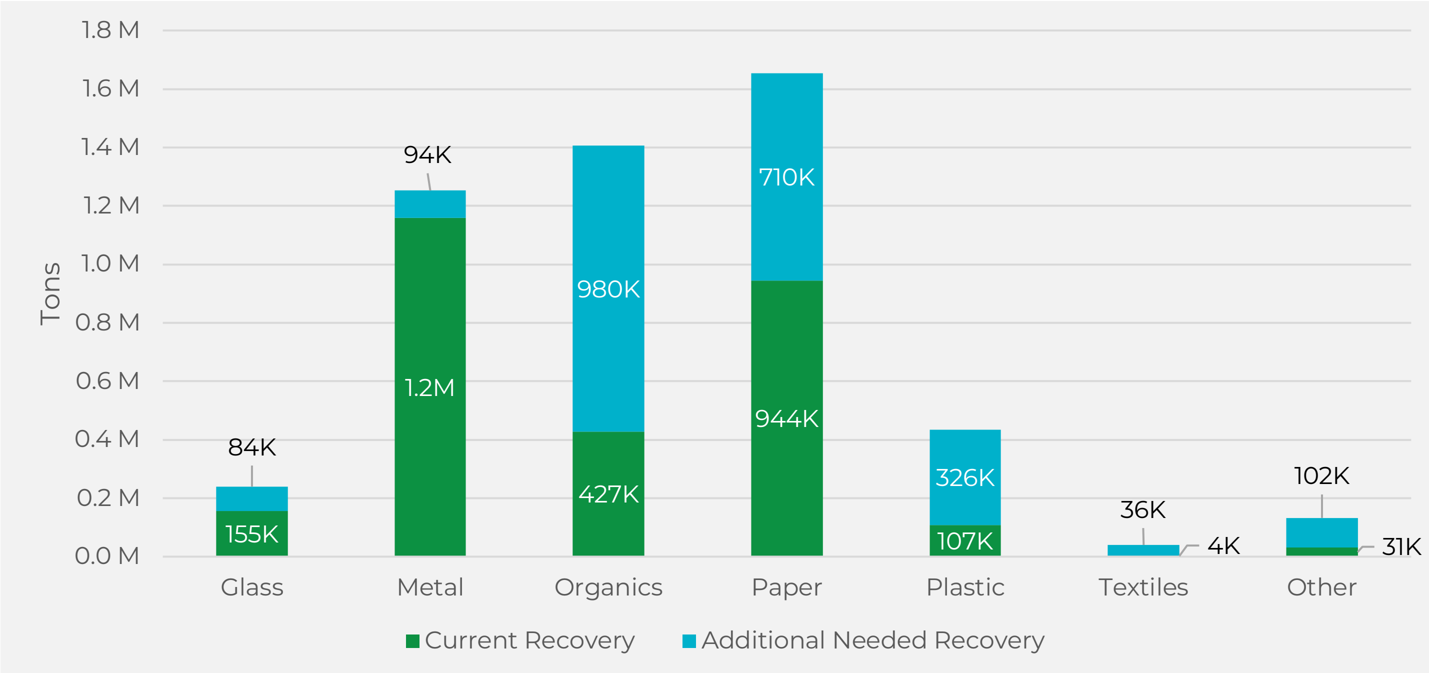
The 2025 Gap Analysis builds on prior reports from 2021 and 2023, identifying critical needs and opportunities in Michigan's materials management system as the state continues its transition from a linear waste model to a circular economy aligned with decarbonization and climate goals. Under the 2022 legislative framework, Michigan is pursuing a statutory 30% recycling rate by 2029, with the ultimate goal of a 45% Municipal Solid Waste (MSW) recycling rate, supported by Renew Michigan Fund investments driving growth in recycling, composting, and end-market development.

Since the 2023 Gap Analysis, Michigan has achieved a 25% recycling rate. This progress has been made possible in large part by sustained EGLE investment. Between 2019 and 2024, more than $60 million in direct EGLE investments were distributed, contributing to over $442 million in total circular economy investments statewide. In 2025, EGLE announced an additional $11.8 million in new grant awards. These investments target priority materials highlighted in this report, including food rescue and diversion, commercial cardboard recycling, scrap metal and wood waste, and expanded access to comprehensive drop-off recycling programs. Beyond grant funding, EGLE's programming and educational support in the areas of innovation, infrastructure, and market development remain essential at the local, regional, and statewide levels.

These investments are producing real results in communities across Michigan. In Holland, a grant-funded transition from recycling bags to carts nearly doubled recyclables collected per household while cutting landfill-bound solid waste nearly in half. In Calhoun County, a mobile drop-off program brought recycling access to rural residents and public events, diverting an additional 86 tons of material in its first full years of operation. In Michigan's Upper Peninsula, Partridge Creek Compost used grant funding to launch a food scraps collection program that now serves an estimated 9,500 residents, local restaurants, and educational institutions. The City of Lansing developed a comprehensive drop-off site accepting materials well beyond standard curbside recyclables, open daily and accessible to all residents. And at two Oakland County material recovery facilities, AI-powered robotic sorting technology is improving operational efficiency while serving a combined population of 1.2 million people. These examples reflect the range of programs EGLE investments are supporting and the tangible impact that targeted funding can have when paired with strong local implementation.

Despite this progress, Michigan disposed of 8.5 million tons of MSW in 2024, and the state's per capita disposal rate of 4.6 pounds per person per day remains higher than neighboring states with stronger recycling programs, such as Wisconsin (4.1 pounds per person per day) and Minnesota (3.3 pounds per person per day). Closing this gap will require sustained and targeted efforts across every material category. In total an additional 2.3 million tons of material, 27% of 2024 disposal, will need to be diverted to reach the 45% recycling rate goal. That includes 980,000 tons of organics, 710,000 tons of paper (includes all fiber materials such as newsprint, magazines, office paper, paperboard, corrugated cardboard), 326,000 tons of plastics, 94,000 tons of metals, 84,000 tons of glass, 36,000 tons of textiles, and 102,000 tons of other materials such as electronics, construction and demolition (C&D) debris, tires, and household hazardous waste (Figure 1). For more detail on consumer battery generation and recycling — a subset of household hazardous waste — see the companion Battery Gap Analysis report.

Figure 1: Current Diversion from Disposal and Additional Diversion Needed to Reach the 45% Recycling Rate Target[1]



FOCUS MATERIALS

As mentioned above, this report highlights several focus materials for a deep dive analysis.

Food Scraps and Surplus Food

Food scraps and surplus food alone represent 1.5 million tons (18% of all disposal) and remain the single largest opportunity for recovery in Michigan's waste stream. The state is moving in a positive direction, with additional curbside and drop-off programs and increased hauler access expanding organics diversion across Michigan. Despite this progress, 67% of Michigan residents still lack convenient access to food scrap diversion programs, and much of the access that does exist relies on private subscription services that require extra steps to enroll and often come at an added cost. EGLE's 2025 grant awards include nearly $1.6 million directed toward organics programs and infrastructure, supporting collection programs, food rescue operations, and processing capacity. To reach the state's 45% recycling rate goal, an additional 766,500 tons of food scraps and surplus food must be diverted from Michigan's disposal stream.

Commercial Corrugated Cardboard

Another key target for recycling is commercial corrugated cardboard, the third largest category of disposal in the commercial sector, representing 10% of total commercial waste. Cardboard is a priority because it appears in such high volumes in the disposal stream, is highly recoverable, and typically retains strong market value. The primary challenge lies in collection. A survey of small businesses found that 25% do not separate cardboard from trash, citing lack of access to haulers, limited space for on-site separation, and the high cost of recycling. Encouraging local haulers and businesses to expand service availability, providing dumpster right-sizing and staff training, and supporting cost-effective recycling options can help ensure businesses across Michigan can recycle cardboard effectively.

Scrap Metal

In 2023, scrap metal was identified as a highly recyclable material and a potentially undercounted component of the state's recycling system. Despite progress in recovering metal with an estimated 1.2 million tons recycled in 2024, gaps remain. Residential scrap metal recycling totaled about 181,500 tons in 2024, yet a survey found residents are still holding onto an additional 94,000 tons of unused items in their homes. One-third of respondents said these items remain only because they lacked knowledge of proper recycling methods or convenient collection options.

[1] Paper refers to all fiber materials such as newspaper, magazines, office paper, paperboard, and corrugated cardboard

Wood Waste and Storm Debris

Wood waste, consisting of tree stumps and branches, painted or treated materials, and untreated structural debris such as fence posts, siding, and roofing, continues to be a challenging material across Michigan. In March 2025, a major ice storm struck the northwest corner of the Lower Peninsula, damaging trees and utility poles across 12 counties and generating an estimated 192,000 tons of wood waste. For many of the affected communities, this single event produced double or more of the typical annual volume of wood waste. This storm-related debris came on top of the 554,700 tons of wood waste already discarded in Michigan landfills each year.

DIVERSION PROGRAM ACCESS

Capturing the additional tons needed to reach the 45% recycling rate requires residents and businesses to have access to a full range of diversion programs: single-family curbside recycling, onsite multi-family recycling that is as convenient as trash service, and drop-off recycling programs. Drop-off programs serve two essential functions. First, they supplement curbside service by providing additional access to traditional recyclables such as glass bottles and jars, aluminum and steel cans, plastic bottles and tubs, paper, and corrugated cardboard. Second, they capture materials that cannot be collected curbside, including plastic film, bulky plastics, scrap metal, expanded polystyrene, electronics, wood waste, and over recoverable materials.

Tracking recycling program access across the state is inherently challenging. Programs change regularly, and a significant portion of access is provided by the private sector, making it difficult to obtain definitive information without a specific address. Despite these challenges, meaningful insights can be drawn about recycling access in Michigan, including the progress made in recent years and the gaps that remain.

A total of 80% of single-family households in Michigan have some level of access to curbside recycling. However, access to curbside service is only one piece of what is needed to ensure adequate recycling opportunities for Michigan residents. Larger population centers are most likely to have universal curbside service bundled with other services, organized by local governments through municipal collection or contracted haulers. Rural residents, by contrast, face sparse access and are often limited to subscription-based haulers. Subscription-based programs are less reliable because they require residents to take proactive steps to enroll and typically impose an added fee. They can also be discontinued at the discretion of the service provider. Contracted or municipal programs, on the other hand, are more likely (though not always) to enroll residents automatically and are less likely to be shut off abruptly.

Multi-family residents (in buildings with five or more units) face even steeper challenges. Best practice calls for onsite recycling containers that are as convenient to access as trash containers. Yet only 20% of multi-family households live in a municipality that even addresses the need for onsite recycling at multi-family properties. Even within those municipalities, actual uptake of onsite recycling service is unknown and likely well below 20%. As a result, relatively few apartment residents have access to convenient onsite recycling. Service is either less convenient than trash, if relying on drop-off access existing nearby, or unavailable altogether.

Residential access to drop-off programs appears high at first glance. Sixty-three percent of Michigan residents have access to drop-off recycling through their county or municipal government, and 72% live within a 25-minute drive of a drop-off site open to the public. However, having access to a drop-off site is only one component of a high-functioning drop-off recycling system. Drop-off programs must provide both convenient and comprehensive access and not just for traditional recyclables, but for materials that cannot be accepted curbside. When county and municipal drop-off programs are examined more closely, only 26% of residents have access to drop-offs that accept materials beyond what a typical curbside program accepts, such as plastic film, bulky plastics, scrap metal, and expanded polystyrene. While some of these materials may be available at private drop-off locations such as retail plastic bag takeback programs or scrap metal yards, this requires residents to visit multiple sites and may not be reliably or equitably accessible. Reaching the state's 45% recycling goal requires capturing more than 306,000 tons of recyclables that curbside programs cannot accept, making the expansion of comprehensive drop-off access a high priority.

Some of these shortfalls will be addressed through progress towards the state's benchmark recycling standards. The standards set statewide minimums to ensure that most residents have convenient access to recycling, requiring that 90% of single-family homes in urban areas (by 2026), and in communities over 5,000 people (by 2028) have curbside recycling. The standards also push counties to provide adequate drop-off options for households in counties with populations under 100,000 by 2032, requiring one drop off for every 10,000 residents without curbside recycling service.

For counties with populations exceeding 100,000 without curbside recycling, one drop off for every 50,000 residents is required by 2032.

Finally, this report focuses primarily on residential recycling access, but commercial recycling access warrants acknowledgment given the significant role of the commercial sector in Michigan's waste stream. As noted earlier, commercial corrugated cardboard is a major component of the waste stream, accounting for more than 431,000 tons of disposal annually. In most municipalities and townships across Michigan, commercial recycling is provided entirely by the private sector at an added cost above standard trash collection. As a result, many of the same challenges that affect residential subscription recycling programs apply equally to commercial recycling, such as the need for businesses to proactively seek out service, bear the added cost, and face the risk of service discontinuation.

POLICY OPTIONS TO INCREASE DIVERSION

While there are many ways to increase recycling, policy can be one of the most impactful and universal tools a state, region, county or city can implement. Local policies can be very effective at requiring service provision for material types and sectors. This report reviews key policy areas and offers options for state-level adoption related to organics, extended producer responsibility (EPR) for compostable packaging and electronics, and Michigan's bottle deposit system. Organics, including food scraps and surplus food, account for 38% of the state's disposal stream, making supportive policy essential to achieving the 45% recycling rate. EPR policies are among the most effective tools for generating sustained funding and supporting robust recycling infrastructure. Seven states have enacted EPR for paper and packaging, but some do not include compostable packaging. As a result, producers of compostable products must pay into systems that do not support their material type. As Michigan considers its own legislation, both recyclability and compostability could be considered, ensuring producers' contributions are aligned with the infrastructure needed to manage their packaging.

Electronics represent another growing challenge in the waste stream. Rapidly changing product types and the increasing prevalence of rechargeable embedded batteries pose both management difficulties and fire risks for landfills, material recovery facilities, and throughout the system. Diverting electronics away from disposal—or "wish-cycling," where residents place them in recycling bins without a viable pathway for recycling—requires a more structured approach. A convenience-based EPR model for electronics, building on Michigan's current e-waste law, Part 173 which requires free and convenient takeback programs for TV's, computers, and printers, can ensure residents have easy and consistent access to collection opportunities year-round, offering an effective pathway for increasing recycling of additional electronic devices and improving safety.

Michigan's bottle deposit system has long delivered some of the nation's highest recycling rates for beverage containers, but redemption has fallen in recent years, reaching a record low of 70% redemption rate in 2024. Strengthening the bottle bill will require a combination of reforms that could include expanding the range of covered containers, adopting universal redemption so all retailers accept all brands, and redirecting a portion of unredeemed deposits to support infrastructure, retailers, and material recovery facilities. Enhancing convenience through more redemption options, updating the deposit value to restore its motivational impact, and investing in transparency, education, and community engagement are all potential policy directions Michigan could consider to enable the long-term success of the deposit law system.

LOOKING BEYOND THE 45% GOAL

Reaching Michigan's 45% recycling rate is an essential milestone, but the state's long-term vision extends further. The final section of this report examines opportunities to align Michigan's new Materials Management Planning (MMP) process with the broader goals of the MI Healthy Climate Plan (MHCP). In 2024, EGLE launched the MMP process to replace traditional county solid waste plans, requiring all 83 counties to develop plans that prioritize reduction, reuse, recycling, and composting before disposal. While the MMP process and the MHCP share common goals, neither plan currently requires counties to calculate or report the carbon benefits of their diversion strategies. Strengthening the connection between the two planning processes presents a meaningful opportunity, particularly around food scraps and surplus food reduction, for which the MHCP sets a 50% reduction goal, and which represents the single largest category in Michigan's disposal stream. Additional alignment opportunities include disaster debris planning, which is increasingly relevant as Michigan faces more frequent and severe weather events, sustainable purchasing policies, and the development of standardized carbon metrics to measure and report the climate impact of materials management activities across the state.

TERMINOLOGY, DEFINITIONS, AND ACRONYMS

Acronyms

AD	Anaerobic Digestion
ASP	Aerated Static Pile
ASTM	American Society for Testing and Materials
BPI	Biodegradable Products Institute
C&D	Construction and Demolition
CAA	Circular Action Alliance
COGs	Council of Governments
DEQ	Department of Environmental Quality
DNR	Michigan Department of Natural Resources
DPA	Designated Planning Agencies
EGLE	Michigan Department of Environment, Great Lakes, and Energy
ELI	Environmental Law Institute
EPR	Extended Producer Responsibility
EV	Electric Vehicle
FEMA	Federal Emergency Management Agency
GAAMP	Generally Accepted Agricultural and Management Practice
HDPE	High-density polyethylene
HHW	Household Hazardous Waste
HVAC	Heating, Ventilation, and Air Conditioning
MEC	Michigan Environmental Council
MHCP	Michigan Healthy Climate Plan
MMP	Materials Management Plan
MRF	Materials Recovery Facility
MSW	Municipal Solid Waste
NAICS	North American Industry Classification System
NRDC	Natural Resources Defense Council
PE	Polyethylene
PET	Polyethylene terephthalate
PFAS	Per- and Polyfluoroalkyl Substances
PP	Polypropylene
PPAs	Power Purchase Agreements
PRO	Producer Responsibility Organization
RCI	Residential, Commercial, and Institutional
RTFA	Michigan's Right to Farm Act
TPY	Tons Per Year
TRP	The Recycling Partnership
U.S. EPA	United States Environmental Protection Agency
WARM	EPA's Waste Reduction Model
WDS	Waste Data System

Definitions

Aerated Static Pile (ASP): A composting method in which organic materials are piled and aerated using a system of perforated pipes or blowers, without the need for turning.

Anaerobic Digestion (AD): A process in which microorganisms break down organic material in an environment without oxygen. During the digestion process, microorganisms produce biogas (methane) which can be converted to natural gas as a renewable source of energy.

Commercial: The commercial sector refers to MSW and recycling generated and collected from the commercial and institutional sectors.

Compostable Products: Utensils, food service containers, and other packaging and products that are certified by the Biodegradable Products Institute (BPI), or an equivalent recognized third-party verification body, as meeting either of the following requirements: (a) ASTM D6400, "Standard Specification for Labeling of Plastics Designed to Be Aerobically Composted in Municipal or Industrial Facilities", by ASTM International; or (b) ASTM D6868, "Standard Specification for Labeling of End Items that Incorporate Plastics and Polymers as Coatings or Additives with Paper and Other Substrates Designed to Be Aerobically Composted in Municipal or Industrial Facilities", by ASTM International (Act 451 1994).

Contracted service: Provided by a franchised hauler under agreement with the local government. Recycling services may be automatically provided to residents or residents may still be required to sign up or opt-in to receive a recycling cart or bin.

Construction and Demolition (C&D): Waste generated during the construction, renovation, repair, and demolition of buildings, roads, and other structures.

Council of Governments (COGs): Multi-County organizations in Michigan that support activities such as grant writing, economic and community development, coordination of services, and any other services beneficial for counties to organize together.

Food Scraps and surplus food: The inedible portions of animal or vegetable matter generated during food preparation, cooking, or consumption, including pits, cores, bones, peels, rinds, shells, eggshells, coffee grounds, and tea bags.

Extended Producer Responsibility (EPR): A policy approach that requires producers to take financial and/or physical responsibility for management of the products and/or packaging they produce at the end of their useful life.

Household Hazardous Waste (HHW): Wastes generated which are potentially dangerous or harmful to human health or the environment.

Hub and Spoke: A waste collection and processing model in which smaller collection sites gather materials and transport them to a central location for sorting, processing, or further distribution.

In-Vessel Composting: A controlled composting process where organic materials are enclosed in a vessel, drum, or container to optimize decomposition and control temperature, moisture, and odor.

Local Government Program: A program run by either a municipality, solid waste district, county, or specified communities.

Materials Recovery Facility (MRF): A solid waste plant that sorts and processes materials to prepare them for recycling.

Materials Recovery Facility (MRF) Compatible Recyclables: Refers to materials that can be sorted and marketed successfully at MRFs such as mixed paper, newsprint, corrugated cardboard, plastic bottles, jugs, tubs, glass containers, aluminum and steel bottles and cans.

Multi-Family Units: Residential buildings containing multiple separate housing units, such as apartments, condos, or townhomes. This is flexible dependent on community and typically includes 4 units and above properties unless noted.

Municipal service: Recycling services managed directly by a city or county public works department.

Municipal Solid Waste (MSW): MSW refers to household waste, commercial waste, waste generated by other nonindustrial locations, waste with characteristics like that generated at a household or commercial business, or any combination thereof. MSW does not include municipal wastewater treatment sludges, industrial process wastes, automobile bodies, combustion ash, or construction and demolition debris (Act 451 1994).

NAICS Codes: Standardized codes from the North American Industry Classification System (NAICS) used to classify business establishments by industry.

Organics and Compostables: Organic material such as food scraps and surplus food, yard clippings, wood waste, and compostable paper that can be converted to finished compost. In the State of Michigan, compostable material comprises class one compostable material and class two compostable material.

Other Recyclables: Refers to materials that are recyclable but generally are not compatible with MRF processing such as plastic film, bulky plastics, scrap metal, rubber, textiles, electronics, and other materials that may be recycled but cannot be sorted at MRFs.

PFAS (Per- and Polyfluoroalkyl Substances): A group of human-made chemicals used in a wide range of products for their water-, grease-, and stain-resistant properties.

Power Purchase Agreements (PPAs): Long-term contracts between an energy producer and a buyer for the purchase of electricity at a predetermined price.

Producer Responsibility Organization (PRO): An entity created by manufacturers and producers to fulfill their obligations under Extended Producer Responsibility (EPR) laws.

Recycling Rate: The amount of recyclable material that is diverted from landfills through processes like recycling or composting.

Recovery: The act of diverting material from the disposal stream either through prevention, reuse, recycling, or organics recovery processes.

Recycled Content: The proportion of a product made from recovered materials rather than virgin materials.

Recycling Rate: The amount of total waste that is specifically collected, processed, and reused as raw material to make new products.

Re-TRAC: An online data management platform used to track, analyze, and report waste and recycling data.

Single Family Units: Residential single-unit buildings designed to house one family, typically a standalone home with its own entrance and utilities.

Subscription Service: When residents/property owners in the community must select a hauler operating in the area. The residents/property owners choose to enroll in a subscription curbside recycling program.

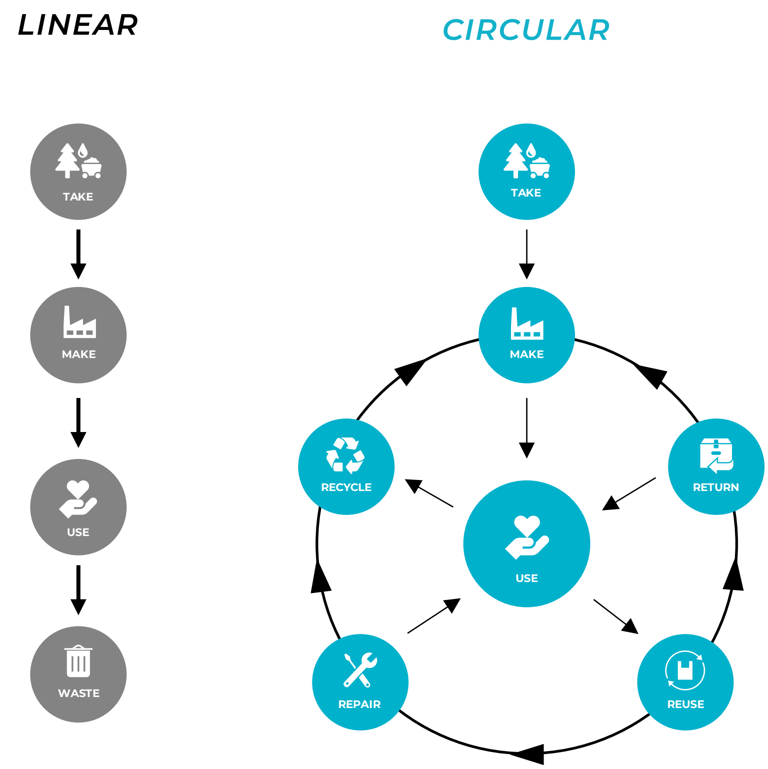
Super Drop-off: A designated recycling location that accepts a wide range of materials not typically collected through regular curbside recycling programs.

Surplus Food: Animal or vegetable matter that was used or intended for human or animal food or that results from the preparation, use, cooking, dealing in, or storing of animal or vegetable matter for human or animal food if the accumulation is or is intended to be discarded.

Windrow Composting: A long, narrow pile of organic materials arranged for composting, typically turned periodically to maintain aerobic conditions.

INTRODUCTION

The following report provides the 2025 update to Michigan's Gap Analysis, building on the 2021 baseline and the 2023 update. Since the 2023 report, Michigan has continued to implement the solid waste and materials management reforms enacted under the state's landmark legislative overhaul, which codified the 45% recycling rate into law and shifted Michigan's framework from disposal management to comprehensive materials management. These reforms, coupled with continued investment from EGLE and partners across the state, are advancing the transition to a circular economy where resources are conserved, reused, and recycled.



Michigan has made notable progress in diversion, reaching a 25% recycling rate, with over $60 million in EGLE investment tracked towards supporting programs. In addition to grant programs, the state launched the Materials Management Planning process which will further scale efforts. However, challenges remain. Michigan's 2024 per capita disposal rate is greater than other high performing diversion states in the Great Lakes, such as Minnesota and Wisconsin. Business and residential outreach about recycling routinely finds that Michiganders want more information on how to recycle properly and greater convenience in access.

This 2025 Gap Analysis evaluates Michigan's current position relative to its statutory recycling goal, quantifies the additional recycling required, and identifies priority actions in recycling of specific materials, diversion program access, and policy.

KEY REPORT TAKEAWAYS AND ACTION ITEMS

DISPOSAL AND ADDITIONAL NEEDED RECYCLING TO REACH 45%

- **Michigan's 2024 MSW Disposal:** Michigan has made progress increasing the state's recycling rate to 25%, but work remains. 8.5 million tons of MSW were disposed of in 2024. More than half of disposed material is recoverable food scraps and surplus food, paper, plastics, and metals; targeting these high-volume streams through expanded prevention, rescue, recycling, and composting is essential to cut disposal and reach recycling goals.
- **Reaching the 45% Recycling Rate Goal:** Michigan must divert an additional 2.3 million tons of material by expanding residential and commercial recycling access, scaling up organics collection and processing, capturing more paper, plastics, and metals, and supporting markets for harder-to-recycle materials to reach the state's 45% recycling target.

SPECIFIC MATERIAL GAPS

- **Focus materials:** This section highlights major diversion gaps in corrugated cardboard, scrap metal, food scraps and surplus food, and wood waste, materials that together represent some of the largest opportunities to increase recycling and reduce disposal in Michigan's MSW stream.
- **Key takeaway:** Significant gains in cardboard recycling for small businesses, expanded comprehensive drop-off options, scaled food scraps and surplus food diversion, and stronger wood reuse and recovery markets will be critical for Michigan to reach its 45% recycling rate goal.

DIVERSION PROGRAM ACCESS

- **Rural curbside gap**: Rural areas often lack curbside recycling or rely on opt-in and sometimes costly subscription services which are known for lower participation rates
- **Multi-family onsite access gap:** Fewer than 20% of multi-family residents live in municipalities that mention onsite recycling access at multi-family properties—that is, recycling service provided to residents within multi-family buildings at a level of convenience comparable to trash service. Actual uptake of these services is likely far lower, meaning that for the vast majority of multi-family residents in Michigan, recycling is either less convenient than trash service or not available at all.
- **Drop-off shortfalls**: Currently, only 63% of residents have access to a dedicated local government or municipal drop-off site, yet only 26% of those residents can drop-off materials beyond traditional curbside recyclables. This creates a significant barrier to responsibly managing other recyclable materials such as bulky rigids, expanded polystyrene (EPS/"Styrofoam"), plastic bags and film, and scrap metal and hard-to-recycle materials such as electronics, HHW, and C&D debris. Because these specialized items are typically prohibited from curbside bins due to contamination risks or processing complexities, drop-off sites represent the only viable pathway for their recovery. When these sites are unavailable or overly restrictive, it results in a significant loss of potentially recyclable resources and increased environmental risk.

INVESTMENT IMPACTS AND CONTINUED SUPPORT

- **Investments:** From 2019–2024, EGLE provided over $60 million in funding leveraged by other public and private investment for a total of $443 million in total circular economy investments tracked statewide. Building on this, in 2025, EGLE announced more than $11.8 million in new grants focused on priority materials such as food recovery, cardboard recycling, scrap metal and wood waste recycling, and expanded drop-off access.
- **Case Studies:** EGLE's investments are producing measurable results across Michigan, from doubled recycling collection rates in Holland to a food scraps program serving 9,500 residents in the Upper Peninsula.

POLICY REVIEW AND RECOMMENDATIONS

- **Policy focus:** This section examines Michigan's organics policies, EPR frameworks for compostable packaging and electronics, and the bottle deposit system.
- **Key takeaway:** Advancing diversion will require strengthening local composting access and zoning, integrating compostable packaging into EPR through statewide needs assessments, shifting electronics takeback to a convenience standard model, and modernizing the bottle bill with expanded coverage, universal redemption, updated deposit values, and improved infrastructure and transparency.

MICHIGAN'S MATERIAL MANAGEMENT PLANNING AND ALIGNMENT WITH MICHIGAN'S CLIMATE ACTION INITATIVES

- **Align Materials Management Plans (MMPs) with climate goals:** EGLE's new MMP framework requires all counties to pursue the 45% recycling goal. Aligning these plans with the MI Healthy Climate Plan, which includes 50% food scraps and surplus food reduction, disaster-debris planning, and standardized carbon metrics, will accelerate diversion, cut the state's largest disposal stream, and improve climate resilience.

MUNICIPAL SOLID WASTE GENERATION AND RECYCLING

SECTION SUMMARY

This section of the report provides an overview of MSW management in Michigan, highlighting current recycling and disposal trends. It summarizes total tons managed by method, examines the composition of disposed materials, and compares Michigan's per capita disposal to that of other Great Lakes states.

Key findings include:

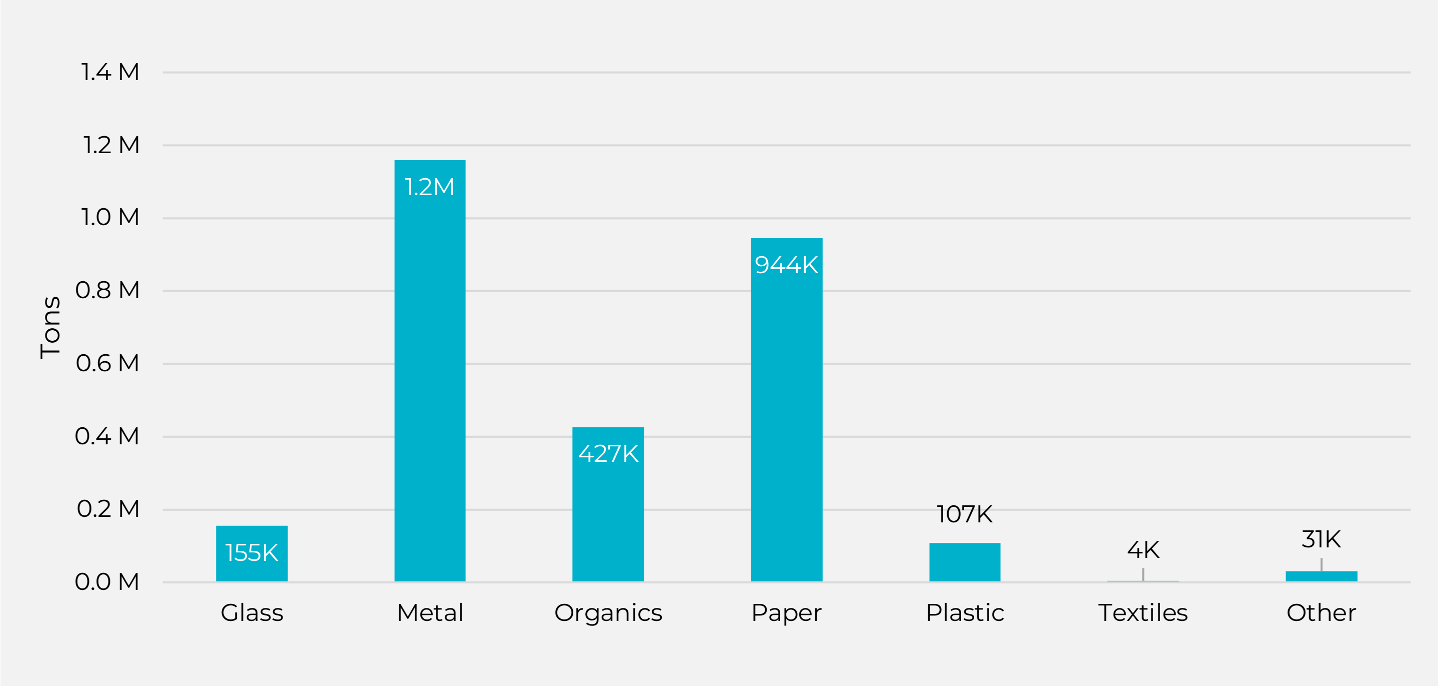
- Disposal and recycling: In 2024, Michigan disposed of 8.5 million tons of MSW and recovered 2.8 million tons through food rescue and recycling, resulting in a 25% recycling rate.
- Regional comparison: Michigan's MSW disposal rate is higher than other Great Lakes states, such as Minnesota and Wisconsin, which have more robust recycling programs and policies.
- Top residential disposal categories: Food scraps and surplus food, compostable paper, plastic film, textiles, and mixed paper account for more than half of all residentially disposed material.
- Top commercial disposal categories: Food scraps and surplus food, plastic film, corrugated cardboard, compostable paper, and mixed paper are the leading commercial disposal categories, accounting for more than half of all commercially disposed material.
- Small truly unrecoverable fraction: Only about 10% of Michigan's MSW disposal stream is considered truly unrecoverable.
- Priority action: Achieving higher diversion will require focusing on recovering both high-volume and high-value materials, particularly food scraps and surplus food, paper, plastics, and metals, through expanded prevention, rescue, and recycling infrastructure supported by diversion focused policies and funding.

RECOVERY AND RECYCLING FROM MSW

EGLE's efforts to expand recycling access, increase public awareness, and improve data tracking resulted in Michigan reporting a record-high recycling rate of 25% in 2024 (Michigan Department of Environment, Great Lakes, and Energy, 2025). This recycling reflects a wide range of sources, including residential and commercial curbside and drop-off programs that collect mixed paper, corrugated cardboard, aluminum and steel cans, glass bottles and jars, and plastic bottles, jugs, and tubs. The recycling rate also incorporates the recycling of scrap metal (including metals from incinerator ash), white goods, plastic film, electronics, textiles, scrap tires, C&D debris, and organics such as yard trimmings and food scraps and surplus food.

Recycling data is reported to the state through a combination of material recovery facilities (MRFs) and direct reporting from private businesses participating in Michigan's recycling economy. In total, 2.8 million tons of material were diverted for recycling in 2024. Beyond this diversion, an additional 49,000 tons of food were rescued and redirected to feed people in need. Figure 2 presents the total tons of material recycled in Michigan in 2024.

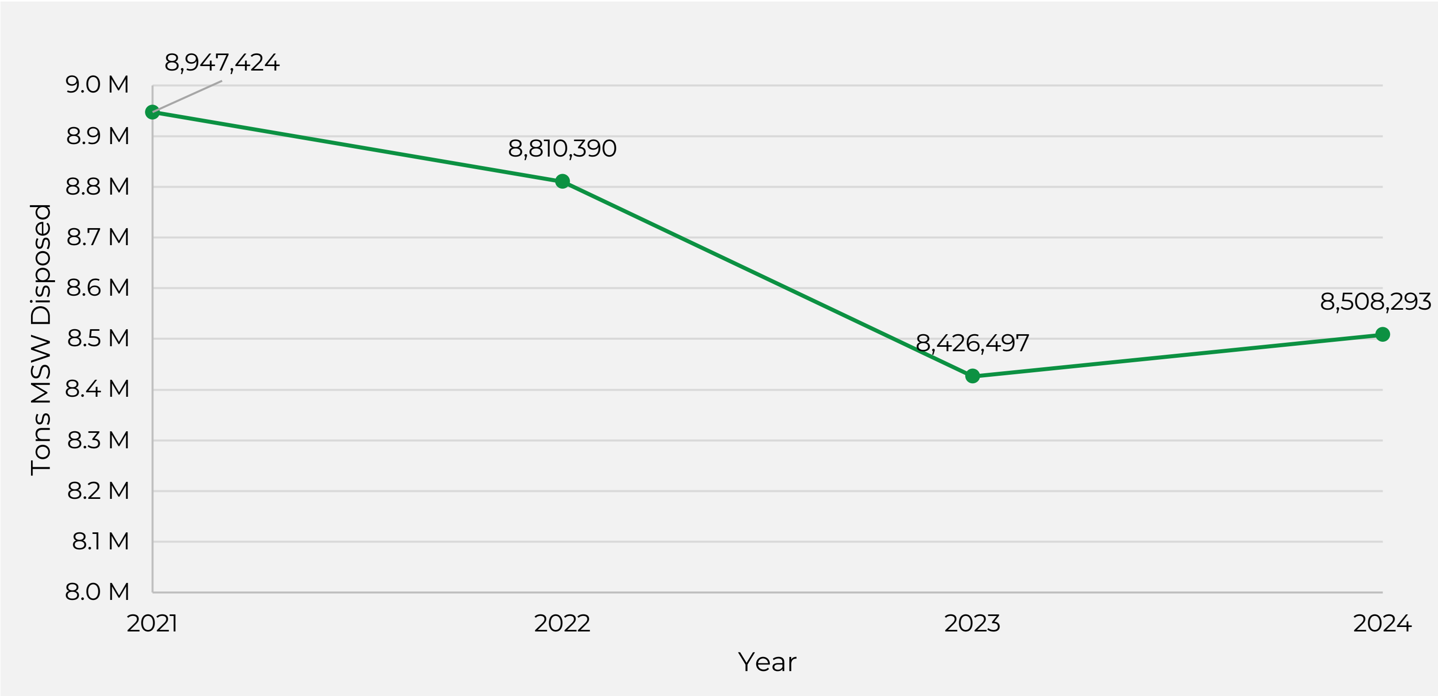
Figure 2: Total Tons of Recycled Material in Michigan in 2024[2]



TOTAL MSW DISPOSAL

In 2024, Michigan disposed of 8.5M tons of MSW throughout the state, a slight increase from 2023 where 8.4M tons of MSW was disposed of, but a decrease from 2021 and 2022 levels (Figure 3).

Figure 3: Total MSW Disposed from Michigan from 2021 to 2024 (Tons)



[2] Paper refers to all fiber materials such as newspaper, magazines, office paper, paperboard, and corrugated cardboard

PER CAPITA DISPOSAL COMPARISON

Table 1 and Figure 4 compare per capita MSW disposal rates across Michigan and four neighboring states: Indiana, Minnesota, Ohio, and Wisconsin, and in FY2024,

Table 2 provides a snapshot comparison of diversion program requirements between the states.

Minnesota distinguishes itself as the state with the most robust recycling requirements, particularly around the Twin Cities, and is the only comparison state to implement commercial diversion requirements. The Twin Cities metropolitan area, which accounts for more than half of the state's population, drives strong diversion with requirements that all businesses have recycling services in place. As a result, Minnesota has the lowest per capita disposal rate among the comparison states, at 3.3 pounds per person per day.

Wisconsin enforces a "universal recycling law" that mandates access to recycling services for all residents and bans the disposal of specific recyclables in landfills. However, Wisconsin does not have strong policies around commercial recycling, and the state has no policies that require food scraps and surplus food diversion. Wisconsin has the second lowest per capita disposal rate of the region at 4.1 pounds per person per day.

Ohio requires residential access to recycling for at least 80% of the population and mandates recycling opportunities for commercial entities. Organics diversion is limited to a yard waste landfill ban. Despite moderate policies, Ohio has a relatively high disposal rate of 5.2 pounds per person per day, suggesting room for enhanced diversion programming.

Michigan and Indiana both have limited diversion requirements. Neither state mandates residential or commercial recycling access, though both ban yard waste from landfills. Notably, Michigan has a per capita disposal rate of 4.6 pounds per person per day, while Indiana's is the highest among the five states at 5.9 pounds per person per day, reinforcing the link between limited diversion policies and higher waste generation.

Additional details on differences in state reporting requirements and granting data are provided in the Appendix section Regional State Reporting and Funding Snapshots.

Table 1: Disposal MSW per Capita

STATE	MSW DISPOSED (TONS)[3]	POPULATION	PER CAPITA (LBS/PERSON/DAY)	DATA YEAR	SOURCE
MI	8,508,293	10,077,331	4.6	FY2024	(Department of Environment Great Lakes and Energy 2024)
IN	7,433,127	6,862,199	5.9	2023	(Indiana Department of Environmental Management 2023)
MN	3,425,845	5,737,915	3.3	2023	(Minnesota Pollution Control Agency (MPCA) 2023)
OH	11,127,767	11,736,358	5.2	2023	(Ohio Environmental Protection Agency 2023)
WI	4,473,716	5,910,955	4.1	2023	(Wisconsin Department of Natural Resources 2024)

[3] Disposed waste includes MSW sent to landfill or waste-to-energy for disposal.

Figure 4: Disposal MSW per Capita.

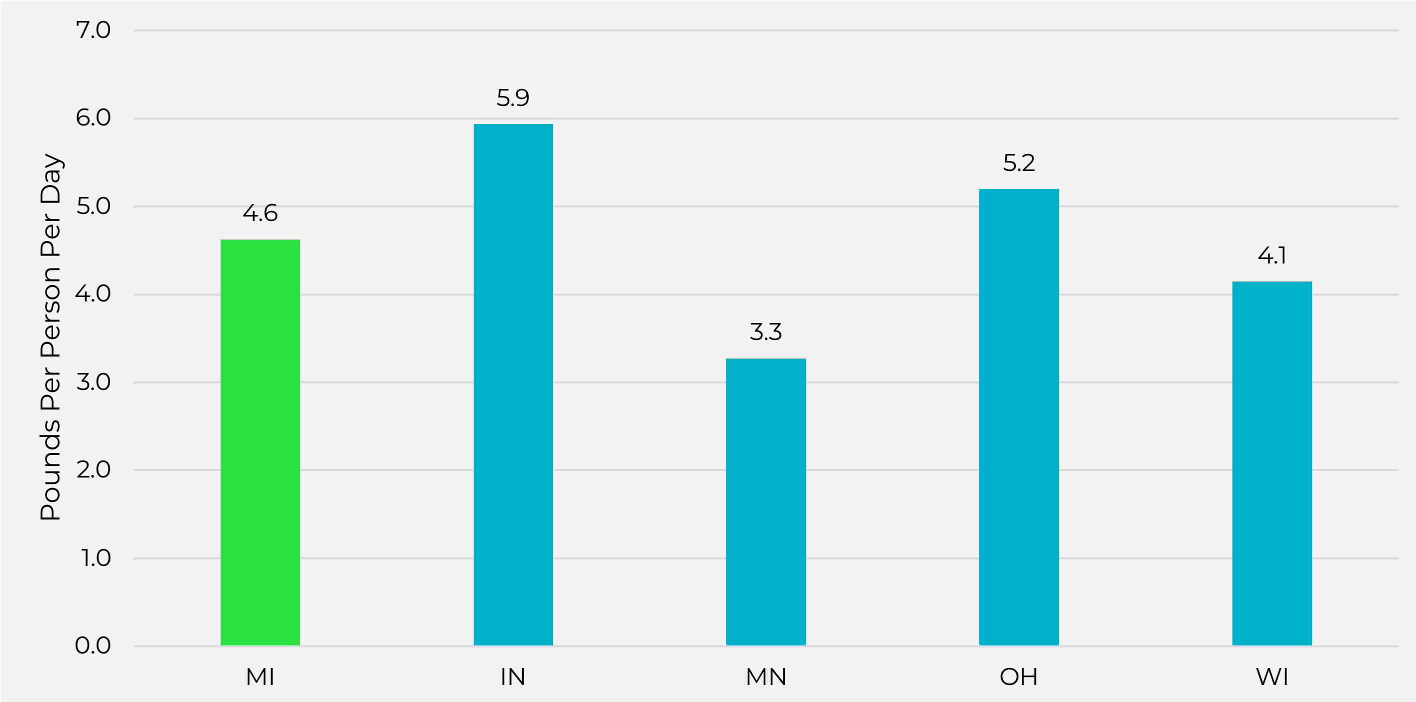


Table 2: Cross-State Program Comparisons

	RESIDENTIAL DIVERSION PROGRAM REQUIREMENTS		**COMMERCIAL DIVERSION PROGRAM REQUIREMENTS**	
MI	None	Yard Waste Landfill Ban	None	Yard Waste Landfill Ban
IN	None	Yard Waste Landfill Ban	None	Yard Waste Landfill Ban
MN	Required Curbside or Drop-Off Opportunities to Recycle Additional requirements in Hennepin County for onsite multi-family property recycling services	Yard Waste Landfill Ban	Twin City metro counties require all commercial business to have recycling services and recycle at least three materials, including food scraps	Hennepin County requires commercial food scraps and surplus food recovery for businesses generating above a certain volume of waste
OH	Required Curbside or Drop-Off Opportunities to Recycle for at least 80% of residential population	Yard Waste Landfill Ban	Required Opportunities to Recycle	Yard Waste Landfill Ban
WI	Wisconsin's "universal recycling law" requires that all residents have access to recycling services (curbside or drop-off), and certain recyclables (e.g., cardboard, plastics #1 and #2, glass, etc.) are banned from landfill disposal	Yard Waste Landfill Ban	Businesses must comply with the statewide disposal bans on recyclables.	Yard Waste Landfill Ban

MSW COMPOSITION

Table 3 provides an estimated breakdown of tons of MSW landfilled by general category for the commercial, multi-family, and single-family generating sectors in Michigan. The composition estimate is based on the 2024 Economic Impact Potential and Characterization of Municipal Solid Waste in Michigan report along with several other comprehensive statewide reports. The full methodology of the waste characterization analysis is presented in the Appendix.

Table 3: MSW Sent to Landfill by Material Type (Tons)[4]

GENERAL CATEGORY	SINGLE-FAMILY	MULTI-FAMILY	COMMERCIAL	TOTAL
Glass	93,440	21,004	79,174	193,618
Metal	142,100	31,942	186,437	360,479
Organics	1,275,321	286,677	1,635,962	3,197,961
Paper (Including Cardboard)	496,890	111,695	945,046	1,553,631
Plastic	519,618	116,804	907,842	1,544,264
Textiles	182,168	40,949	137,447	360,564
Other	600,588	135,005	562,184	1,297,777
Total	**3,310,125**	**744,077**	**4,454,091**	**8,508,293**

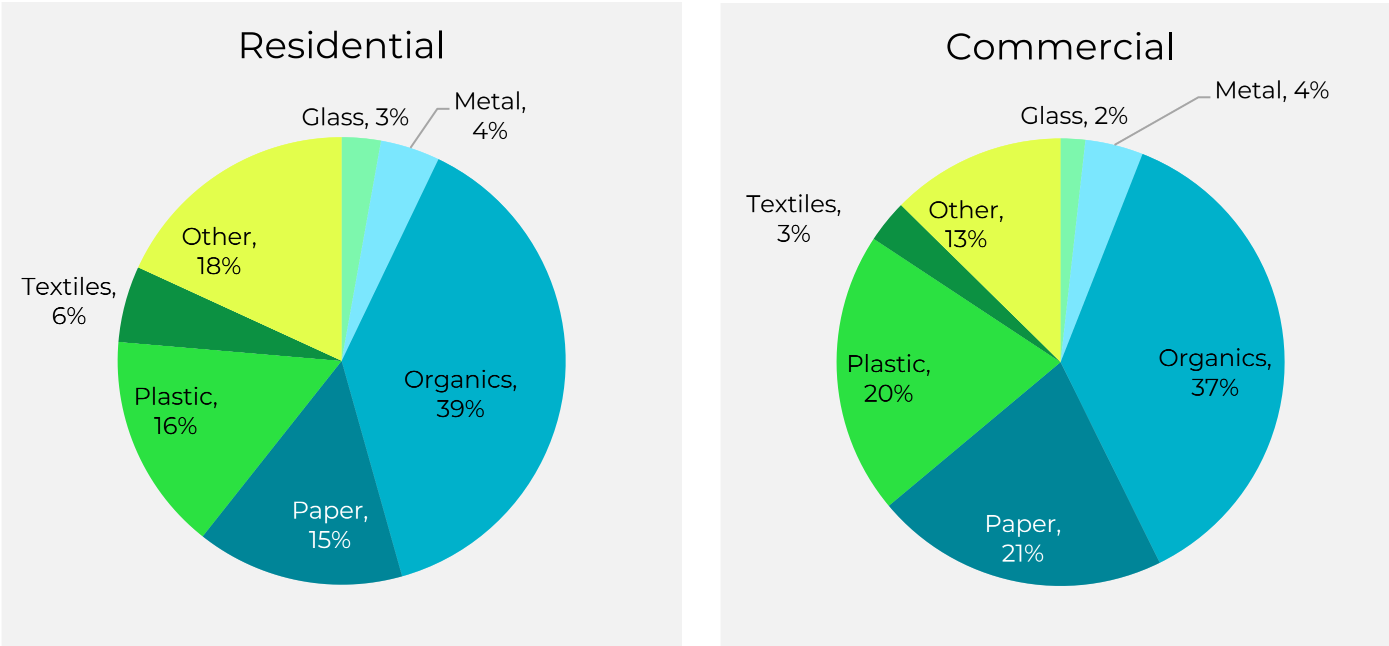
Figure 5 provides the proportion of general categories by the generating sectors - residential and commercial. The largest portion of MSW is composed of organics, accounting for 39% of the residential stream and 37% of the commercial stream. The second and third largest categories for both the residential and commercial streams are paper followed by plastic at 15% and 21%, and 16% and 20% of composition, respectively.

Other materials constitute 18% of the residential stream and 13% of the commercial stream and may include miscellaneous materials such as electronics, C&D debris, household hazardous waste (HHW), bulky waste, tires, rubber, inert rock, soil, and fines, and residue.

Textiles accounts for 6% of the residential stream and 3% of the commercial stream. Finally, metal, including aluminum and steel cans, as well as ferrous and non-ferrous scrap metal, account for 4% of both streams.

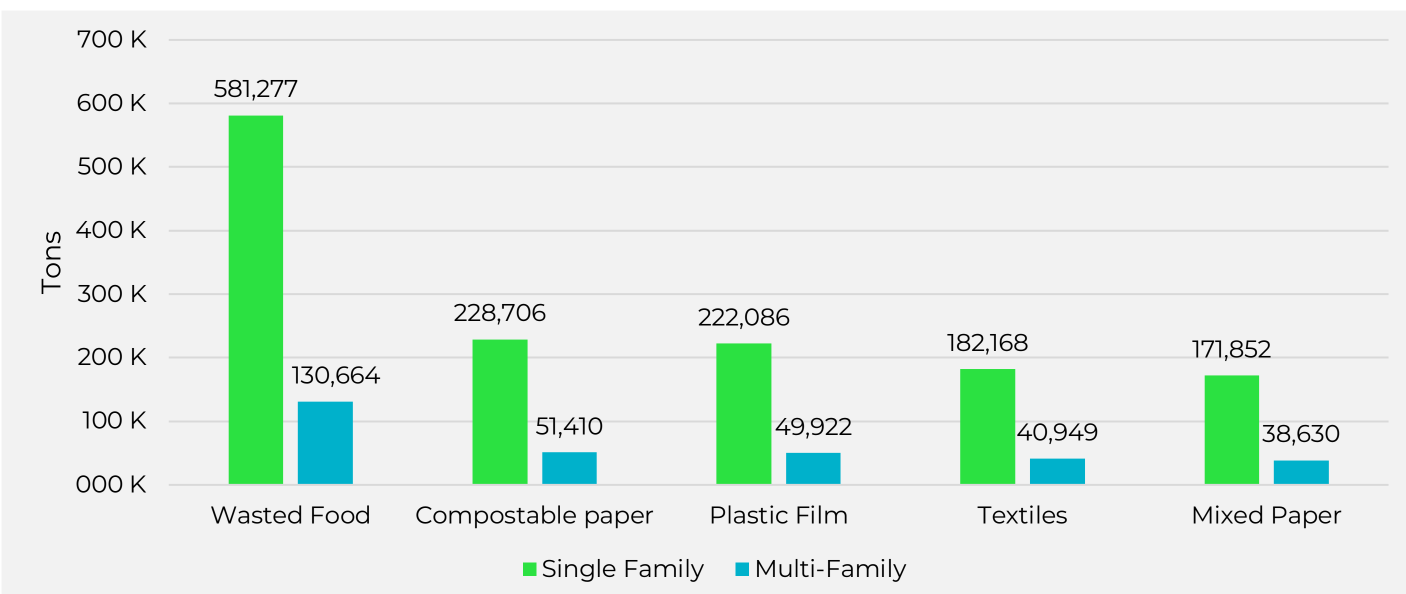
[4] Due to rounding, the totals presented in this table may differ from the sum of individual values by several tons.

Figure 5: Composition of Total MSW Stream by Material, Residential and Commercial[5]



The top five disposal categories for the residential sector broken down by single and multi-family sectors are shown in Figure 6. Notably, the four largest categories in the residential disposal stream are not recoverable through traditional curbside or single-stream recycling programs. The largest two categories, accounting for more than 992,000 tons collectively, are food scraps and surplus food and compostable paper and are divertable through organics collection programs such as curbside and drop-off programs for organics. The third largest category is plastic film (272,000 tons total). Recycling of plastic films from the residential sector is generally limited to store drop-off programs. Textiles is another high volume disposal item, accounting for more than 223,000 tons, and like films, the recovery of textiles is generally limited to private sector drop-off options.

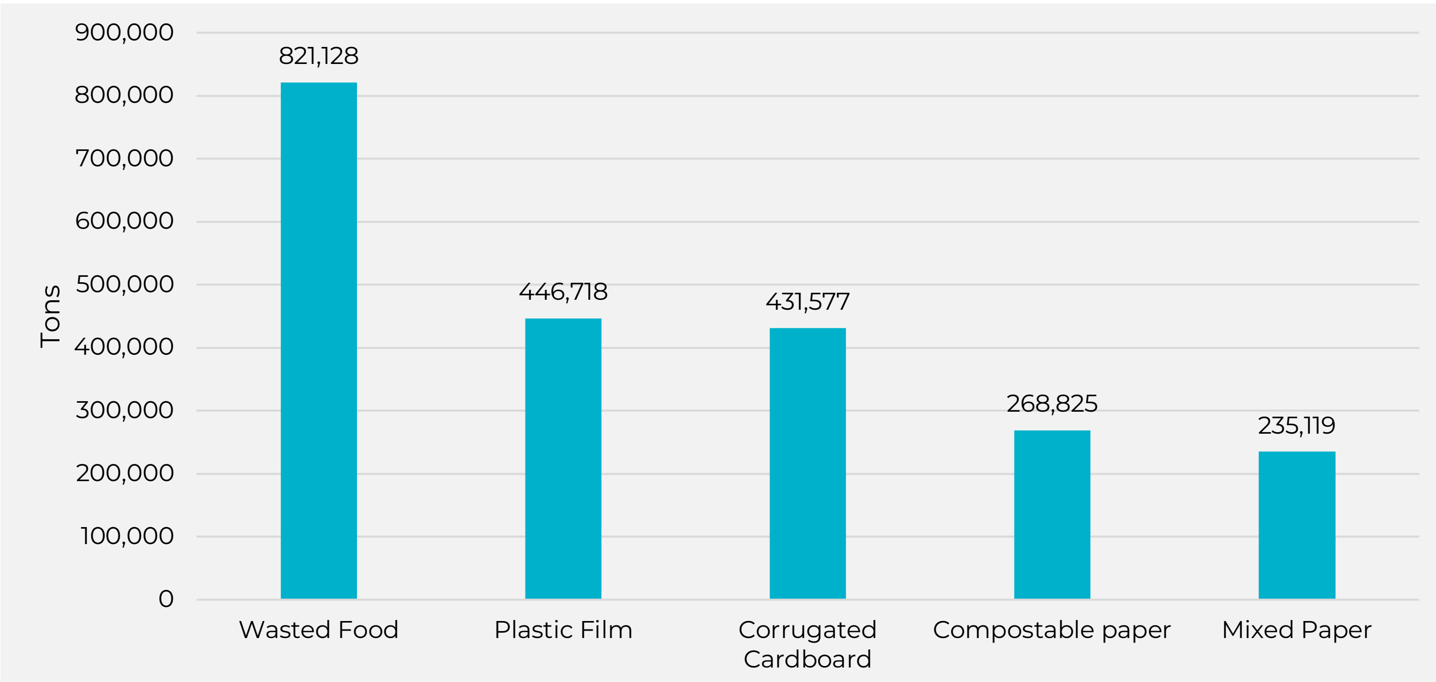
Figure 6: Top Five Disposal Categories in the Residential Sector



[5] Paper refers to all fiber materials such as newspaper, magazines, office paper, paperboard, and corrugated cardboard

Some of the same material categories shown in Figure 6 also appear in Figure 7, which presents the top five disposal categories for the commercial stream, including wasted food, compostable paper, plastic film, and mixed paper. Although plastic film appears on both the top five disposal categories for residential and commercial streams, the volume of plastic film from the commercial sector is double the residential. Plastic film is an often-recovered item in the commercial sector—particularly clean PE films such as shrink wrap—however, there is still significant opportunity in Michigan to capture film from the commercial sector. Corrugated cardboard is another top disposal category from the commercial sector that is also highly recoverable.

Figure 7: Top Five Disposal Categories in the Commercial Stream



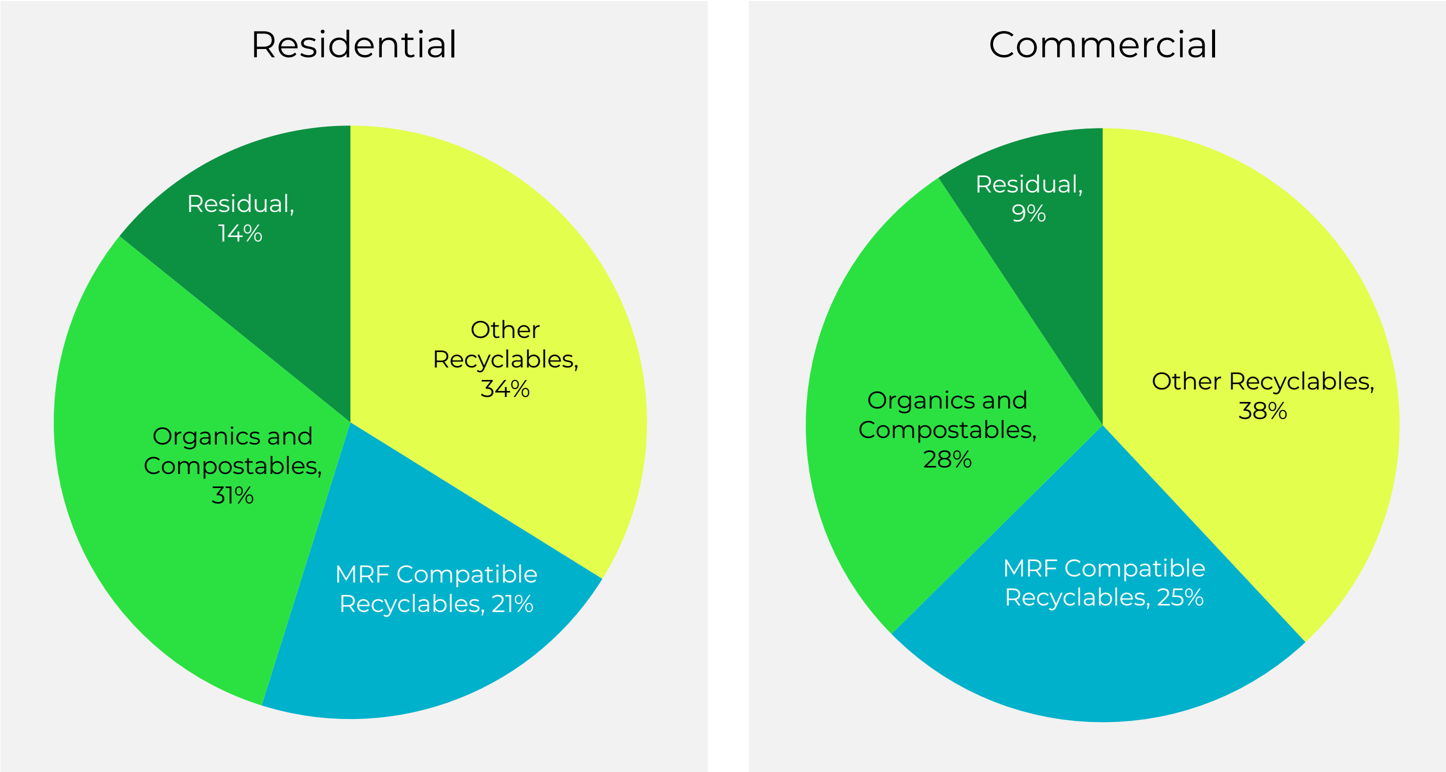
The proportion of organics, plastics, and paper in the MSW stream highlights the need for increased diversion opportunities such as curbside, drop-off, and organics recycling programs.

Figure 8 identifies the potential recycling pathway by material type for the aggregated MSW stream. Approximately 85% of the MSW stream may have the opportunity for potential diversion and can be grouped as follows:

- **MRF Compatible Recyclables:** Refers to materials that can be sorted and marketed successfully at MRFs such as mixed paper, newsprint, corrugated cardboard, plastic bottles, jugs, tubs, glass containers, aluminum and steel bottles and cans.
- **Other Recyclables:** Refers to materials that are recyclable but generally are not compatible with MRF processing such as plastic film, bulky plastics, scrap metal, rubber, textiles, electronics, and other materials that may be recycled but cannot be sorted at MRFs.
- **Organics and Compostables:** Organic material such as food scraps and surplus food, yard clippings, wood waste, and compostable paper that can be converted to finished compost. In the State of Michigan, compostable material comprises class one compostable material and class two compostable material.
- **Non-Recyclable:** Material that cannot be recycled at this time.

Food scraps collection programs through curbside and drop-off services and drop-off recycling programs are necessary for handling materials that are not easily recoverable via single- or dual-stream curbside programs, such as food scraps, expanded polystyrene, plastic film, bulky plastics, and textiles like clothing, sheets, towels, rope, household linen, and leather products.

Figure 8: Proportion of the Disposal Stream That is Divertible, Residential and Commercial

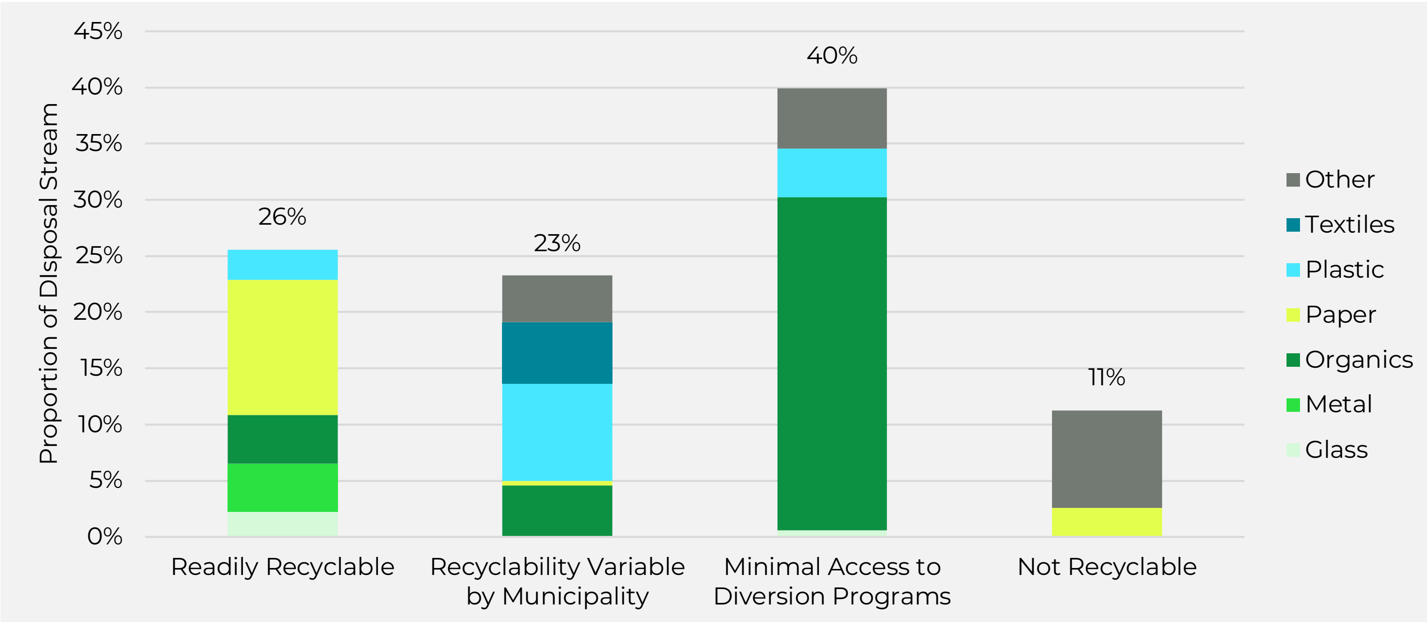


While approximately 85% of the disposal stream could potentially be recovered, the ease of recoverability varies based on the material. To break down the estimated level of difficulty recovering, the project team categorized materials based on the following criteria:

- **Readily recyclable through commonly accessible programs:**
  - » Items including glass bottles and jars, aluminum cans and foil, steel cans, corrugated cardboard, paperboard, kraft paper, newsprint, magazines, white office paper and mixed paper, plastic PET bottles and non-bottles, and plastic HDPE colored and natural, yard waste, and ferrous and non-ferrous scrap metal.
- **Recyclability variable by municipality:**
  - » Plastic film, mixed plastics, polypropylene, cartons, wood and C&D waste.
- **Minimum Access to Recycling:**
  - » Non-container glass, food scraps, other organics, and compostable paper and plastics, expanded polystyrene, durable and bulky plastic items, mattresses, carpet, other bulk items, and tires.
- **Not Recyclable:**
  - » Broken glass, inert materials like rocks, soils, fines, diapers, and other products & packaging contaminated with adhesives or other non-recyclable materials, and animal waste.

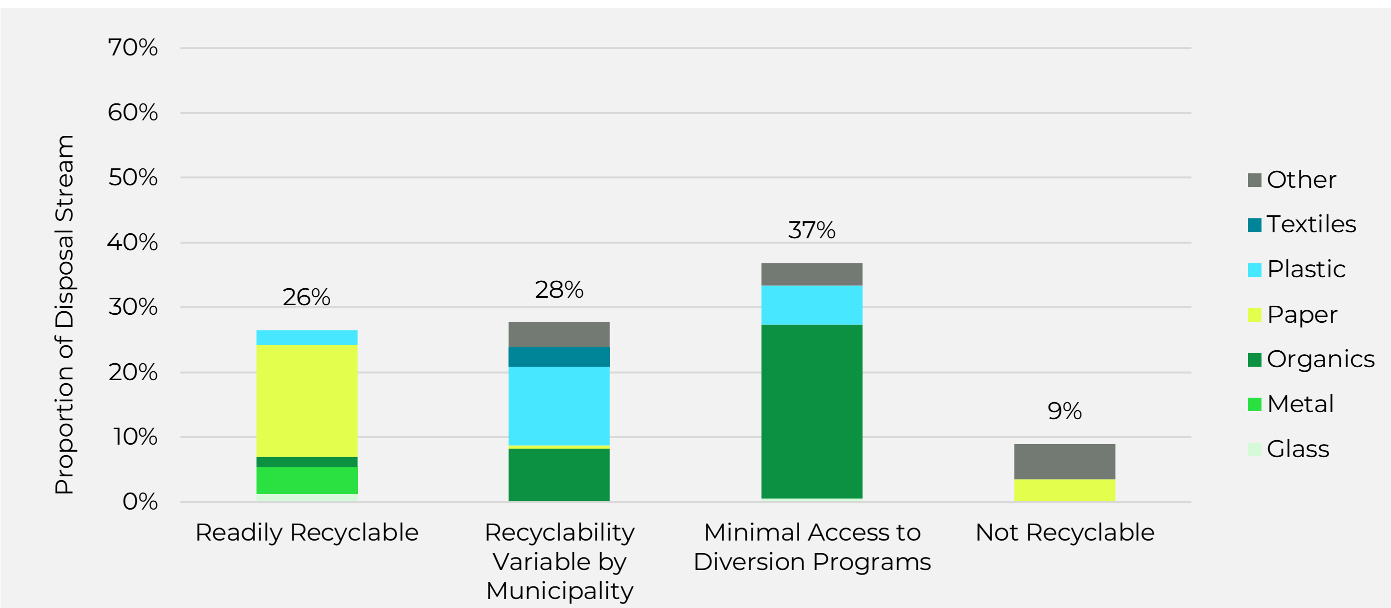
Figure 9 breaks down the residential composition of materials by recoverability. Approximately a quarter of the residential disposal stream is readily recoverable through commonly accessible programs in Michigan including curbside and drop-off recycling programs, yard waste collection programs, and scrap metal recycling businesses. Nearly another quarter of materials, including plastic films and textiles, are currently recyclable, but access to recycling is variable by municipality and often relies on private sector businesses, such as retail grocery stores and resale shops, to accept certain items. Another 40% of the disposal stream falls into recoverable items with minimal access to diversion programs, with most of that material being organic including food scraps and surplus food and compostable paper.

Figure 9: Ease of Recyclability for General Material Categories Based on Aggregated Waste Composition - Residential Waste Stream



A similar breakdown of readily recyclable, variably recyclable, and minimal access appears in the commercial waste stream (Figure 10).

Figure 10: Ease of Recyclability for Material Categories Based on Aggregated Michigan Waste Composition- Commercial Waste Stream



The results from this analysis align with the results from the 2024 Economic Impact Potential and Characterization of Municipal Solid Waste in Michigan, which found that approximately 80% of Michigan's waste stream was recoverable. The additional analysis conducted allowed the project team to conduct a deeper dive assessment into wood waste generation and estimated recovery presented in the next section.

NEEDED DIVERSION TO REACH 45 PERCENT DIVERSION

SECTION SUMMARY

This section summarizes the additional tons the state needs to recover to reach Michigan's 45% recycling goal. Additional diversion is grouped by material type and management strategy.

Key findings include:

- **Reaching the 45% recycling rate goal:** Michigan must divert an additional 2.3 million tons of material from disposal to achieve the 45% recycling target—equivalent to 27% of the state's MSW disposal stream.
- **By generator:** Meeting the goal will require diverting 867,700 tons from single-family households, 195,000 tons from multi-family households, and 1.3 million tons from commercial generators.
- **By collection and processing type:** Progress will depend on capturing 1.1 million tons of MRF-compatible recyclables (e.g., glass, metals, paper, cardboard, and #1–2 plastics), 306,600 tons of other recyclables not typically processed at MRFs (e.g., plastic film, bulky plastics, scrap metal, rubber (primarily tires), textiles, and electronics), and 956,000 tons of organics (food scraps and surplus food, yard waste, compostable paper, and compostable plastics).
- **By material type:** The largest diversion gains must come from paper (710,000 additional tons), organics (1.0 million tons), plastics (326,000 tons), and metals (94,000 tons).
- **Priority action:** Achieving the 45% recycling rate goal will require a diverse portfolio of strategies, including expanding residential and commercial recycling access, scaling up organics rescue, collection and processing infrastructure, strengthening cardboard and paper recycling, and supporting markets and end uses for plastics, textiles, and other difficult-to-recycle materials – all supported by local and state-wide recycling focused policies and funding.

The 2023 Michigan Gap analysis estimated the total tons of material needed to be diverted from disposal for Michigan to reach the state's 45% recycling rate goal. The needed diversion was broken into three categories:

- **MRF compatible Recyclables**
  - » Mixed paper, newsprint, corrugated cardboard, plastic bottles, jugs, tubs, glass containers, aluminum and steel bottles and cans.
- **Other Recyclables**
  - » Plastic film, bulky plastics, scrap metal, tires, textiles, electronics, and other materials that may be recycled but cannot be sorted at MRFs.
- **Organics and Compostables**
  - » Organic material such as food scraps and surplus food, yard clippings, wood waste, and compostable paper that can be converted to finished compost. In the State of Michigan, compostable material comprises class 1 compostable material and class 2 compostable material.

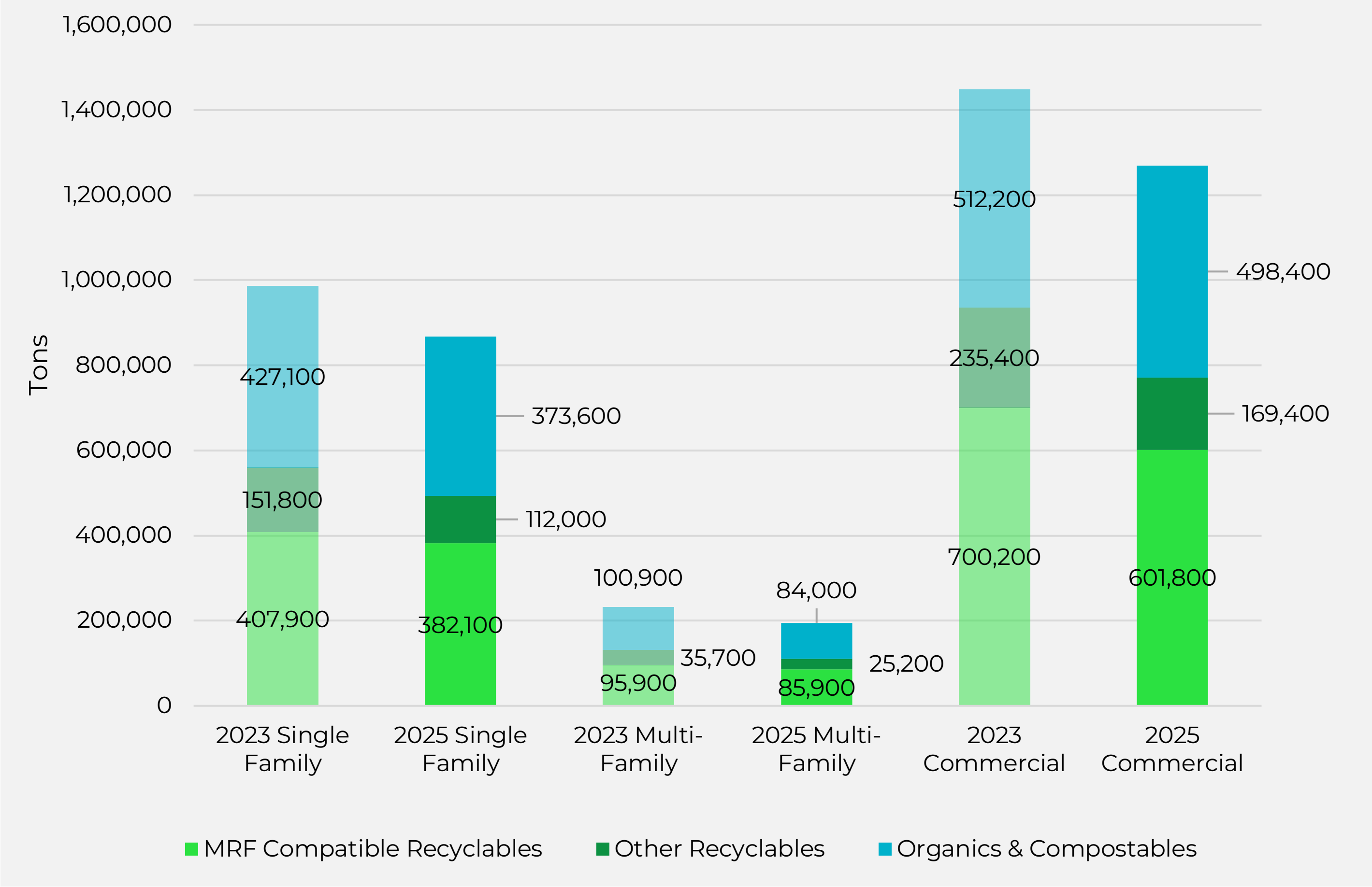
Diversion from disposal could include prevention, reuse, and recovery of materials for recycling or composting. The 2023 model estimated that an additional 2.7 million tons of material needed to be diverted from Michigan's disposal stream to reach the state's 45% recycling rate goal. Since that study was completed, the state has gathered additional waste composition data as well as reported recycling data. Given additional data, the project team updated the needed diversion model for Michigan to reach the 45% recycling goal. The updated additional needed tons for diversion are lower in the 2025 model, reflecting the state's progress towards the 45% goal and the updated waste characterization data. Table 4 and Figure 11 provide a comparison of the 2023 and 2025 additional diversion tons to reach the state's recycling goals, broken down by generating sector – single-family, multi-family, and commercial generators – and stream type – MRF compatible recyclables, other recyclables, and organics and compostables.

Overall, reaching the state's 45% recycling rate goals will require Michigan to divert a total of 867,700 tons from single-family households, 195,100 tons from multi-family households, and 1.3 million tons of commercial sector generators.

Table 4: Comparison of total additional tons for diversion to reach the state's 45% recycling rate goal from 2023 to 2025.

GENERATING SECTOR	STUDY YEAR	MRF COMPATIBLE RECYCLABLES	OTHER RECYCLABLES	ORGANICS & COMPOSTABLES	TOTAL
Single Family	2023	407,900	151,800	427,100	**986,800**
Single Family	2025	382,100	112,000	373,600	**867,700**
Multi-Family	2023	95,900	35,700	100,900	**232,500**
Multi-Family	2025	85,900	25,200	84,000	**195,100**
Commercial	2023	700,200	235,400	512,200	**1,447,800**
Commercial	2025	601,800	169,400	498,400	**1,269,600**

Figure 11: Comparison of total additional tons for diversion to reach the state's 45% recycling rate goal from 2023 to 2025.



ADDITIONAL NEEDED DIVERSION OF MRF COMPATIBLE RECYCLABLES

The 2023 gap analysis estimated that Michigan would need to divert approximately 1.2 million additional tons of MRF-compatible recyclables to meet the state's 45% recycling rate goal. Updated modeling reflecting recent diversion progress and revised waste characterization data indicates that this estimate has decreased to 1.1 million tons.

Of the remaining tons that must be diverted, the commercial sector accounts for the majority (56%), while single-family households contribute 36%, and multi-family households 8%.

The most prevalent MRF-compatible materials requiring diversion include:

- Mixed paper
- Corrugated cardboard
- Glass bottles and jars
- Plastic bottles and jugs
- Rigid mixed plastics

To meet the revised goal, Michigan must capture approximately 55% of all MRF-compatible recyclables currently being disposed. The materials listed above are frequently found in the MSW stream and represent high-volume or high-tonnage categories that are critical to achieving the state's diversion targets.

Table 5 below summarizes the estimated MRF compatible recyclables tons that need to be diverted across Michigan's Councils of Government (COGs).

Table 5: Additional tons of MRF compatible recyclables needed to divert from disposal in Michigan to reach the 45% recycling rate goal.

COG	SINGLE FAMILY	MULTI-FAMILY	COMMERCIAL	TOTAL
1	182,500	41,800	288,300	512,600
2	11,700	2,100	17,700	31,500
3	20,600	5,600	33,600	59,800
4	11,100	2,100	17,000	30,200
5	22,200	4,400	34,100	60,700
6	17,000	5,200	28,500	50,700
7	30,300	6,000	46,700	83,000
8	44,600	10,700	71,000	126,300
9	6,100	800	8,800	15,700
10	12,700	2,500	19,500	34,700
11	2,100	400	3,200	5,700
12	7,200	1,300	10,900	19,400
13	3,400	500	5,100	9,000
14	10,800	2,600	17,200	30,600
Total	**382,300**	**86,000**	**601,600**	**1,069,900**
Percent	**36%**	**8%**	**56%**	**100%**

ADDITIONAL NEEDED DIVERSION OF OTHER RECYCLABLES

The 2023 gap analysis estimated that Michigan would need to divert an additional 422,900 tons of other recyclables to reach the 45% recycling goal. Like the MRF compatible recyclable, the additional other recyclables needed to reach the state's goal was revised down to 306,800 tons.

In the residential sector, the largest categories of other recyclables for recycling are plastic film and textiles. In the commercial sector, wood waste such as painted and treated wood and untreated wood from fencing, roofing, and siding are the most prevalent items for diversion along with plastic film and textiles.

Reaching the 45% goal will require capturing 10% of other recyclables out of the disposal stream for recycling. The recycling targets for these materials was set significantly lower than MRF compatible recyclables due to the additional challenges to collection for these items and for some items such as textiles, an expansion in processing infrastructure.

Table 6 below summarizes the estimated other recyclables tons that need to be diverted across Michigan's Councils of Government (COGs).

Table 6: Additional tons of other recyclables needed to divert from disposal in Michigan to reach the 45% recycling rate goal.

COG	SINGLE FAMILY	MULTI-FAMILY	COMMERCIAL	TOTAL
1	53,500	12,300	81,200	147,000
2	3,400	600	5,000	9,000
3	6,000	1,600	9,500	17,100
4	3,300	600	4,800	8,700
5	6,500	1,300	9,600	17,400
6	5,000	1,500	8,000	14,500
7	8,900	1,800	13,100	23,800
8	13,100	3,100	20,000	36,200
9	1,800	200	2,500	4,500
10	3,700	700	5,500	9,900
11	600	100	900	1,600
12	2,100	400	3,100	5,600
13	1,000	200	1,400	2,600
14	3,200	800	4,900	8,900
Total	**112,100**	**25,200**	**169,500**	**306,800**
Percent	**37%**	**8%**	**55%**	**100%**

ADDITIONAL NEEDED DIVERSION ORGANICS AND COMPOSTABLES

Additional organics and compostables for diversion includes wasted food, yard waste, and compostable paper and plastic. Like the other material categories, the additional needed diversion of organics and compostables has also been revised down to 956,100 tons from 1.0 million tons in the 2023 gap analysis.

Wasted food is the largest single category in the organics and compostable stream, making up 78% of the additional organics needed for diversion. In addition to Michigan's 45% statewide recycling rate goal, the 2022 MI Healthy Climate Plan initiative has a stated goal of halving wasted food in Michigan by 2030, amounting to 766,500 tons of just wasted food. Table 7 below summarizes the estimated organics and compostables tons that need to be diverted across Michigan's Councils of Government (COGs).

Table 7: Additional tons of organics and compostables needed to divert from disposal in Michigan to reach the 45% recycling rate goal.

COG	SINGLE FAMILY	MULTI-FAMILY	COMMERCIAL	TOTAL
1	178,400	40,900	238,800	458,100
2	11,400	2,100	14,700	28,200
3	20,100	5,400	27,800	53,300
4	10,900	2,100	14,100	27,100
5	21,700	4,300	28,300	54,300
6	16,700	5,000	23,600	45,300
7	29,600	5,900	38,600	74,100
8	43,600	10,400	58,800	112,800
9	5,900	800	7,300	14,000
10	12,400	2,400	16,200	31,000
11	2,000	400	2,600	5,000
12	7,100	1,300	9,100	17,500
13	3,300	500	4,200	8,000
14	10,600	2,500	14,300	27,400
Total	**373,700**	**84,000**	**498,400**	**956,100**
Percent	**39%**	**9%**	**52%**	

Prevention

Surplus food represents a significant portion of the MSW stream and results in not only squandered food but also wasted energy, water, labor, and emissions associated with its production, transport, and disposal. Prevention strategies offer the highest environmental benefit because they stop food from becoming waste in the first place, rather than managing it after disposal.

Wasted food occurs at various stages of the MSW stream, including retail, foodservice, and residential sectors. According to ReFED, a national nonprofit dedicated to reducing surplus food, the causes of food loss vary across these sectors:

- **Retailers**
    - » Confusion over “best by” date labels leads to premature disposal of safe and edible food.
    - » The expectation that shelves must always appear well stocked and visually appealing often results in overordering and surplus inventory that ultimately goes unsold.
- **Foodservice**
    - » Inadequate storage and difficulty forecasting demand contribute to spoilage.
    - » Food preparation practices may lead to edible food being discarded (e.g., trimming practices or overproduction).
    - » Customer plate waste adds significantly to food loss.
- **Residential**
    - » Poor meal planning and food storage practices contribute to spoilage.
    - » Impulse purchases and bulk buying lead to more food than households can consume before expiration.
    - » Many consumers lack the knowledge or skills to repurpose ingredients or leftovers.
    - » Misunderstandings about “best by” and “use by” labels result in food being thrown away prematurely.

To address these challenges, ReFED’s Solution Database highlights a variety of strategies to prevent surplus food across sectors. For example, markdown alert apps can help retailers reduce surplus inventory by notifying consumers of discounted soon-to-expire items. In the foodservice sector, technologies like waste tracking systems and portion optimization can reduce pre- and post-consumer waste. At the residential level, consumer education campaigns and tools like meal planning apps or clearer date labeling can promote better food management and reduce unnecessary disposal.

Preventing surplus food in Michigan’s MSW stream will require coordinated action across all three sectors. By implementing targeted interventions at the point of loss, Michigan can reduce the environmental impact wasted food while also supporting food security and resource conservation goals.

Rescue

Food rescue is a higher and more beneficial use for surplus food than recycling, as it redirects edible food to people in need rather than to composting or landfilling. Prioritizing rescue not only reduces waste but also addresses food insecurity, creating a dual environmental and social benefit. According to Feeding America, one in six Michiganders and one in five children in the state faces hunger, underscoring the urgent need to expand food rescue infrastructure and programs (Feeding America, 2025).

In 2023, EGLE estimated that approximately 46,500 tons surplus food were rescued statewide, increasing slightly to 49,000 tons in 2024. While these numbers reflect positive progress, there remains substantial untapped potential for rescuing edible food from retailers, institutions, and households.

Policy can play a key role in enabling and expanding food rescue efforts. According to ReFED’s Policy Finder, Michigan offers moderate protections under food donation liability laws. Donations made in the state are covered by the federal Bill Emerson Good Samaritan Food Donation Act, which shields good-faith donors from civil and criminal liability. However, Michigan does not currently offer any state-level tax incentives to encourage donations, relying solely on existing federal tax benefits. Additionally, the state lacks detailed food safety guidance specific to donations—such as clarifying which foods may be donated and under what handling conditions—making it more difficult for potential donors and food recovery organizations to navigate regulatory requirements confidently.

Expanding food rescue in Michigan will require both policy improvements and programmatic investment. Clearer guidance, state-level incentives, and broader awareness of existing protections can help reduce barriers and encourage more stakeholders to participate in food rescue, ensuring that more edible food reaches those who need it.

Recycling

Food scraps and surplus food recycling ideally occurs only after prevention and rescue options have been fully pursued, to ensure that any remaining food in the waste stream is diverted to the highest and best remaining uses. These uses follow the U.S. EPA's food recycling hierarchy and include feeding animals and other value add pathways such as black solider fly processing, small-scale composting, industrial composting, and anaerobic digestion. Each of these strategies plays a critical role in reducing landfill-bound organic waste, cutting methane emissions, and returning valuable nutrients to the soil.

Feeding Animals

Food that is no longer suitable for human consumption, but remains unspoiled, can be a valuable input for livestock feed. According to ReFED, Michigan has a moderate policy environment for food-to-animal diversion. The state allows vegetable scraps to be fed to animals without restriction and permits animal-derived food scraps to be used if they are heat-treated at a licensed facility. A permit is also required to feed such treated material to animals, creating an opportunity for regulated expansion of this diversion pathway.

Black Soldier Fly (BSF) Larvae Processing

Black Soldier Fly larvae processing is an emerging and increasingly commercial pathway for converting food scraps into high-value outputs including protein-rich animal feed, insect oil, and frass fertilizer. BSF larvae can reduce organic waste volume by 50–60% within a matter of days, offering a rapid and efficient alternative to traditional composting or animal feed diversion for materials that may not meet quality thresholds for direct livestock use. The larvae themselves become a sustainable ingredient for aquaculture, poultry, and livestock feed, creating a circular loop in which food scraps re-enter the agricultural system as a valuable protein source.

Backyard Composting

Backyard composting provides a low-cost, decentralized solution for managing residential fruit and vegetable scraps and yard debris. When done correctly, it enables households to reduce their waste while producing nutrient-rich compost for gardening and landscaping. Municipalities can support backyard composting by offering educational workshops, starter kits, and free or subsidized compost bins, especially in communities where curbside organics collection is not yet available.

Community Composting

Community composting offers a scalable, neighborhood-based solution that engages residents and local businesses in organic waste diversion. This model creates closed-loop systems in which local food scraps are transformed into compost that can be used in community gardens, urban farms, or local landscaping. Community composting not only reduces hauling emissions and landfill volumes but also fosters civic engagement and builds local resilience in food systems.

Industrial Scale Composting

Larger composting facilities—such as windrow, aerated static pile (ASP), or in-vessel systems—are essential for managing food scraps and surplus food from large generators including institutions, hotels, restaurants, and residential curbside collection programs. These systems can handle higher volumes and a broader range of materials, including some that may not be suitable for backyard or community composting. However, careful facility siting and operational management are critical to minimizing odors, leachate, and contamination concerns that can affect surrounding communities.

Anaerobic Digestion (AD)

AD is a valuable option for processing large volumes of organic waste, and in particular food, while simultaneously generating biogas and nutrient-rich digestate. AD systems are well-suited for food scraps and surplus food that may be too wet or contaminated for composting and can serve institutional and municipal generators. The captured biogas can be used to generate electricity or upgraded to renewable natural gas, while the digestate can be applied to land or further processed into compost or biochar. Although Michigan has some operational AD capacity, there is significant potential to expand this technology, particularly in partnership with wastewater treatment plants, agricultural operations, or industrial food processors.

ADDITIONAL NEEDED DIVERSION BY MATERIAL

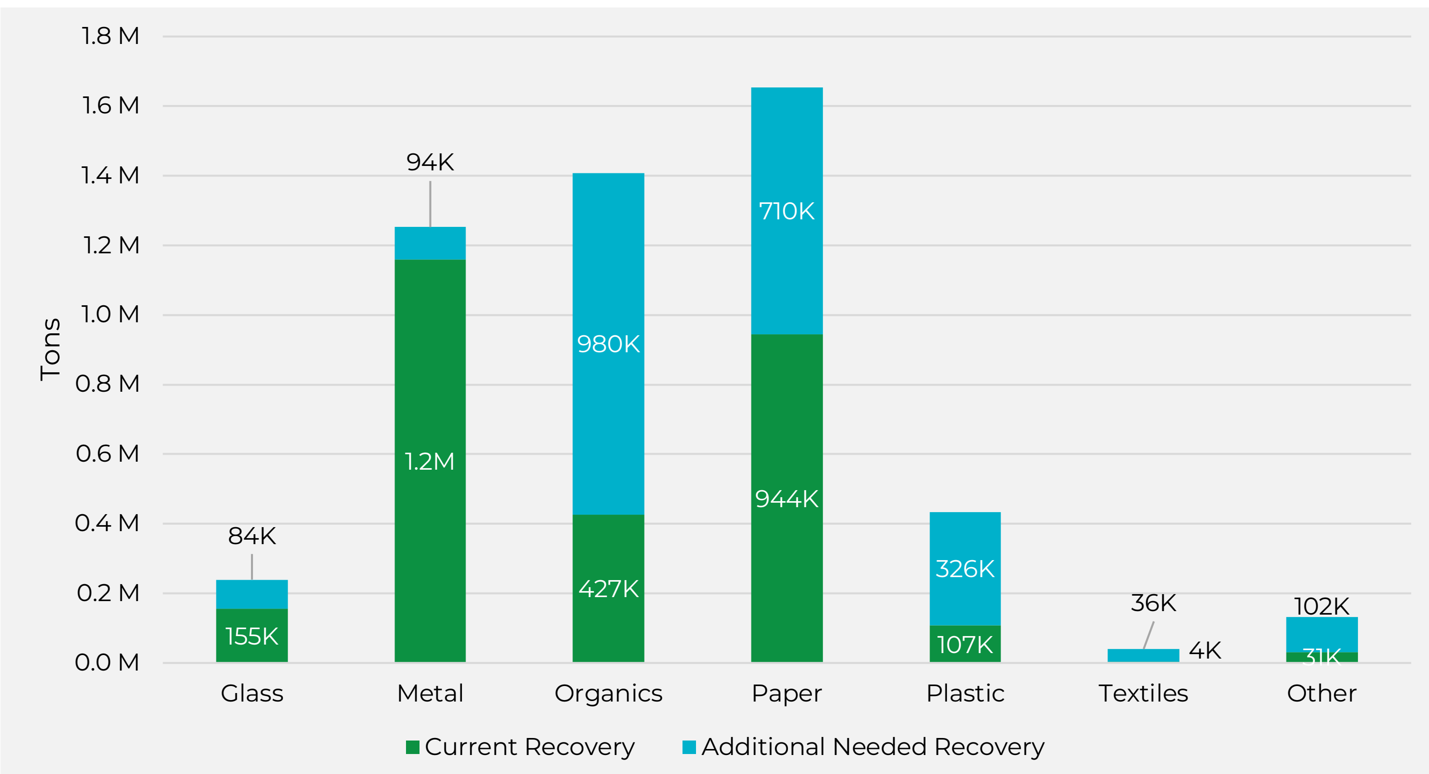
As shown above, reaching the 45% recycling goal will require capturing more of every recoverable material currently ending up in disposal. However, the amount of additional diversion needed varies significantly by material type. Figure 12 illustrates both current diversion levels and the additional tonnage required to meet the target.

Metal and glass stand out as relatively bright spots among the material categories. Michigan already recovers an estimated 1.2 million tons of metal annually, and the additional 94,000 tons still needed represents only an 8% increase over current recovery levels. Glass tells a similar story, with 155,000 tons currently diverted and an additional 84,000 tons needed, a 54% increase that is achievable given the material's strong existing infrastructure.

For all other categories, the increases required are substantially steeper. Organics currently divert 427,000 tons but require an additional 980,000 tons to meet the goal, a 230% increase that will demand significant expansion of composting and organics processing capacity across the state. Plastics face an even greater challenge, with only 107,000 tons currently diverted and 326,000 tons still needed, representing a 305% increase. The "Other" category, which includes electronics, C&D debris, tires, and household hazardous waste, requires an additional 102,000 tons above the 31,000 tons currently recovered, a 329% increase that reflects the limited infrastructure currently available for these harder-to-recycle materials.

Textiles present the most dramatic percentage increase of any category. With only 4,000 tons currently diverted, capturing the additional 36,000 tons needed represents a 900% increase. While the absolute tonnage is modest relative to other categories, it signals a need for significant investment in textile collection and processing infrastructure. Paper presents a different kind of challenge: despite already having the highest current recovery of any category at 944,000 tons, its large share of the disposal stream means an additional 710,000 tons must still be captured, a 75% increase that underscores how much paper remains in Michigan's landfills even with robust existing programs.

Figure 12: Current and Additional Needed Recycling by Material



SPECIFIC MATERIAL GAPS

SECTION SUMMARY

This section of the report examines gaps in the recycling of specific materials within the MSW stream and provides recommendations to increase diversion. The materials analyzed include two with high recycling potential—corrugated cardboard and scrap metal—and two organic materials with greater recovery challenges but substantial disposal volumes—food scraps and surplus food and wood waste. Significant increases in diversion of these materials will be critical to achieving Michigan's 45% recycling rate goal.

Key findings include:

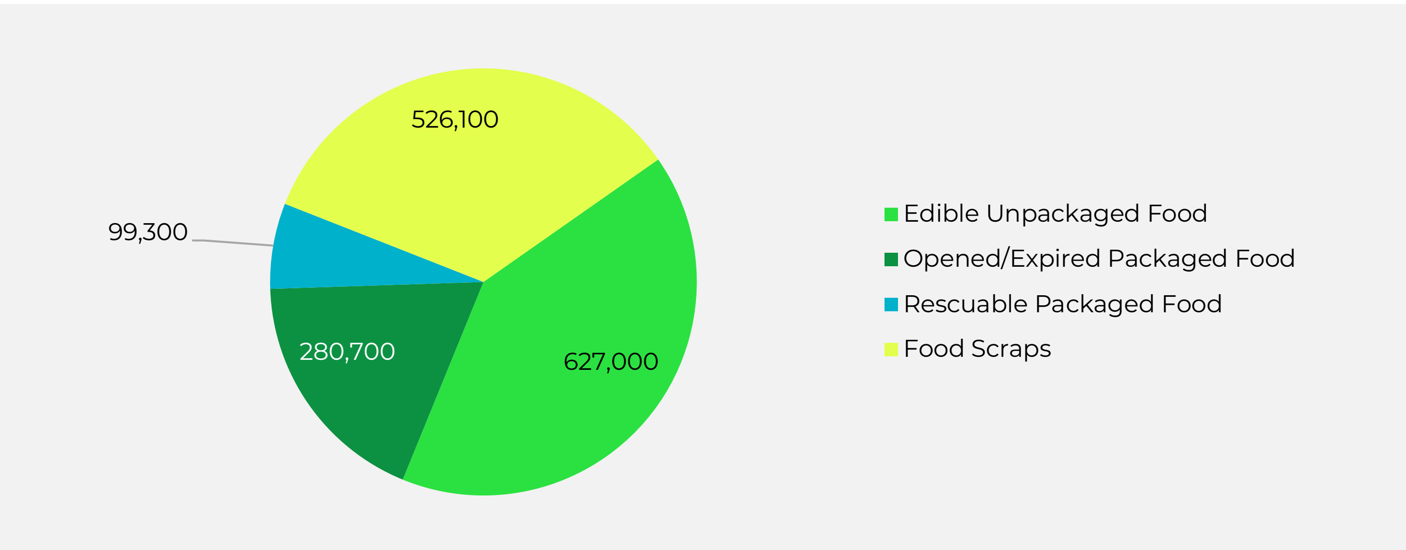
- **Food scraps and surplus food**
  - » 1.5M tons of food scraps and surplus food was disposed in 2024, representing the largest MSW disposal category. Food rescue organizations continue to divert edible food to people in need. Access to organics collection (curbside and drop-off) is expanding, yet roughly two-thirds of residents still lack any food scraps diversion option. Where access exists, it most often takes the form of private, pay-to-participate subscription services. Haulers report outreach and education challenges and seek EGLE's support to deliver consistent, statewide messaging on the benefits and proper use of food-waste programs.
- **Commercial Corrugated Cardboard**
  - » Commercial cardboard disposal exceeds 431,500 tons annually. While large retailers often recycle cardboard through backhaul, consolidating, and direct sale to end markets, many small retailers lack access to similar infrastructure. A survey of small businesses found that one-quarter do not separate cardboard for recycling, citing limited-service availability, lack of storage space, and high costs. EGLE could help expand commercial cardboard recycling by ensuring all businesses have access to recycling services, supporting right-sized collection infrastructure, and promoting staff training to improve separation practices.
- **Scrap Metal**
  - » Scrap metal recycling rates in Michigan are relatively high, estimated at 76%, due to strong market value. Nevertheless, more than 232,700 tons are still landfilled each year. Barriers identified by residents include not generating enough material to justify trips to a scrap yard and lack of nearby facilities. Nearly 80% of surveyed residents indicated they would recycle more scrap metal if it were accepted at comprehensive drop-off sites alongside other recyclables. An additional 94,000 tons of scrap metal remain stored in homes, with one-third of these items retained simply because residents did not know how to properly dispose of them. EGLE could address these barriers by supporting the development of convenient, comprehensive drop-off infrastructure statewide.
- **Wood Waste**
  - » Wood waste from residential, commercial, and institutional sources represents about 8% of total MSW generation, or 554,700 tons annually. Nearly 90% of disposed wood is either painted/treated wood or untreated structural debris such as fencing, siding, and roofing. Diversion strategies will likely require multiple approaches, including comprehensive drop-off programs, expanded salvage and reuse opportunities, and support for resale markets.

FOOD SCRAPS AND SURPLUS FOOD

1.5M+ tons disposed in 2024, most prevalent material in the residential and commercial waste stream; diverting an additional 766,500+ is needed to reach the 45% recycling goal.

In 2024, Michigan disposed of an estimated 1.5 million tons of food scraps and surplus food, accounting for 18% of the state's MSW disposal stream. As the description of food scraps and surplus food implies, this is not a uniform material but can be divided into edible food and inedible food scraps. A 2019–2020 Minnesota food scraps and surplus food characterization study found that approximately 66% of disposed food was once edible, while the remaining 34% consisted of inedible scraps such as peels, bones, and fruit pits. Applying these proportions to Michigan's disposal data indicates that about 1.0 million tons of food scraps and surplus food was edible—comprised of unpackaged food, opened or expired packaged food, and rescuable packaged food—while roughly 526,000 tons were inedible scraps (Figure 13).

Figure 13: Composition of Disposed Food Scraps and Surplus Food in Michigan



As discussed earlier in this report, best-practice management of food scraps and surplus food follows the waste hierarchy: prevention first, then rescue, and finally diversion through animal feed or BSF larvae processing, composting, or anaerobic digestion. The goal is to keep edible food out of landfills by preventing waste at the source or recovering surplus to feed people. In Michigan, an estimated 99,300 tons of disposed food were likely still packaged and unexpired, meaning they could have been rescued, more than double the amount of food actually rescued in 2024 (49,000 tons). With appropriate interventions, additional categories of edible food, such as unpackaged and opened or expired packaged food, could also have been reduced or rescued. Meanwhile, the 526,000 tons of inedible food scraps, while not suitable for prevention or rescue, represent a significant opportunity for diversion to organics processing facilities. Table 8 shows the share of food scraps and surplus food in the MSW disposal stream that could be prevented or rescued, as well as the share suitable for organics processing.

Table 8: Target Focus for Food Scraps and Surplus Food In the MSW Disposal Stream

TARGET FOCUS FOR FOOD SCRAPS AND SURPLUS FOOD	TONS	PROPORTION
Prevention and Rescue Opportunities – Edible Food	1,007,000	66%
Organics Processing – Inedible Food	526,069	34%
Total	1,533,070	100%

Food Rescue Efforts in Michigan

Food rescue organizations throughout the state of Michigan provide valuable services in redistributing food to communities and diverting waste from landfill. Collectively, food rescue organizations keep millions of pounds of food from landfill, providing nutritious meals to thousands of individuals each year. This section includes examples of food rescue organizations operating throughout the state to prevent food scraps and surplus food by feeding people.

Gleaners Community Food Bank

Gleaners Community Food Bank is a food rescue organization in Detroit serving Wayne, Oakland, Macomb, Livingston, and Monroe counties in southeastern Michigan. Gleaners links surplus food from producers to nearly 350 partner organizations, such as soup kitchens, food pantries, shelters, schools, and other agencies. As a supplement to the partner network, Gleaners also hosts mobile distribution sites five days per week that offer fresh food through "no-contact" drive-up pantries. In addition to these food assistance programs, Gleaners also launched a new pilot facility in September 2025, Fresh! By Gleaners, which provides free curbside and in-store "shopping" experiences for community members experiencing food insecurity, offering guests greater choice and flexibility. Gleaners partners with health organizations to distribute food and provides nutrition education programming. In 2024, the organization distributed over 53 million pounds of food, serving over 1 million households.

Forgotten Harvest

Forgotten Harvest is another food rescue organization that serves 587,000 people in Metro Detroit each year. Through programs such as mobile pantries, distribution targeted for children, seniors, and veterans, Forgotten Harvest rescued and distributed nearly 45 million pounds of food—the equivalent of over 37 million meals—in the 2024 fiscal year. Forgotten Harvest rescues surplus food from more than 700 donors, including grocery stores, farms, food processors, manufacturers, warehouses, distributors, dairies, restaurants, caterers, entertainment venues, and sports arenas. Donations collected by Forgotten Harvest are distributed to over 200 partner organizations, such as food pantries, food banks, shelters, and other community nonprofit organizations.

Food Gatherers

Food Gatherers food bank and food rescue program is located in Ann Arbor and serves Washtenaw County. Food Gatherers distributes food to over 140 community partners and operates a community kitchen, Summer Food Service Program, and the Healthy School Pantry Program. The largest source of food donated to Food Gatherers is sourced from the food rescue program, which diverts surplus from food retailers, food wholesalers, and local farmers that would otherwise go to waste. Community partners provide food assistance through schools, clinics, low-income housing complexes, shelters, counseling programs, faith-based organizations, and programs serving seniors, the disabled, people with mental illness, and substance abuse recovery programs. In the 2024 fiscal year, Food Gatherers distributed 10.3 million pounds of food, or 8.5 million meals.

Kalamazoo Loaves & Fishes (KLF)

Kalamazoo Loaves & Fishes (KLF) is the largest independent food bank in Michigan, serving over 38,000 individuals annually in Kalamazoo County and surrounding communities. In 2024, KLF distributed food at 97 different locations through their Grocery Pantry Program, Mobile Food Distributions, School-Based Program, Partner Agency Program, and targeted outreach. KLF rescues food from grocers that is not sold due to "sell by" or "best by" dates that is still safe for consumption, as well as excess inventory from the agricultural industry such as dairy, meat, and produce. In the 2020-2021 fiscal year, KLF distributed nearly 5 million pounds of food.

Food Rescue US

Food Rescue US is a national network of local nonprofit organizations that operate under the Food Rescue US name or use the proprietary Food Rescue US app to facilitate food recovery. The Food Rescue US app allows donors, including grocers like Target, Whole Foods, Trader Joes, and Walmart, to input a pickup time and an estimate of the donation volume. Volunteers are then able to collect donations and distribute food to partner social service agencies, or staff from social service agencies can collect food directly from donors. Food Rescue US has partner organizations in Detroit, the Greater Lansing area, and Traverse City. Food Rescue US – Lansing Communities has provided 347,000 meals since 2023, diverting 416,000 pounds of food from landfill. Donations are sourced from 39 food donor partners and distributed to 37 social service agency partners, including food pantries, shelters, faith-based organizations, and community organizations. In Traverse City, Goodwill Northern Michigan Food Rescue rescues, repacks, and distributes 2.2 million pounds of food to 16,000 people each year. Food is rescued from 210 local organizations, including 82 farms, and distributed to 78 food pantries and community meal sites in the Northwest Food Coalition.

Heartside Gleaning Initiative

Heartside Gleaning Initiative serves the Heartside neighborhood and surrounding communities in downtown Grand Rapids, Michigan, which face high levels of food insecurity. In 2022 and 2023, Heartside Gleaning distributed over 173,000 pounds of food at Fresh Food Fairs, reaching nearly 40,000 individuals. Heartside Gleaning Initiative collects produce donated by local farmers who attend downtown farmers markets. Food is distributed through Heartside Gleaning's food boxes, Fresh Food Fairs, and donations to community partners such as food banks and food pantries.

Residential Access to Food Scrap Diversion Programs

Access to food scrap diversion programs continues to grow in Michigan. Figure 14 illustrates Michigan communities with access to food scrap diversion programs, either through curbside collection or local drop-off options. Access is categorized into four types:

Curbside Verified Program Areas

Represent communities served by municipally or county-contracted curbside food scraps collection programs. These are based on known service boundaries and considered "verified" access.

Curbside Subscription Areas

Include communities located within the self-reported service areas of subscription-based haulers. These are counted as having access for the purpose of this analysis, though actual availability may vary. Service boundaries for these programs are often loosely defined and may extend beyond what is shown on the map.

Municipal/County Drop-Off Areas

Reflect communities with access to government-administered drop-off sites, which are typically open only to local or county residents. These programs are commonly operated in partnership with private haulers.

Private Composter Drop-Off Areas

Represent locations where individual businesses offer drop-off services. Access areas for these sites are approximated using a 25-minute drive-time buffer. These programs may be free, subscription-based, or fee-for-service.

While all mapped areas are considered to have access, further research may be needed to assess program participation barriers, including cost and availability of service to multi-family households. Subscription-based program boundaries are often dynamic and not always precisely delineated.

Figure 14: Areas with Access to Residential Food Scraps and Surplus Food Diversion Programs

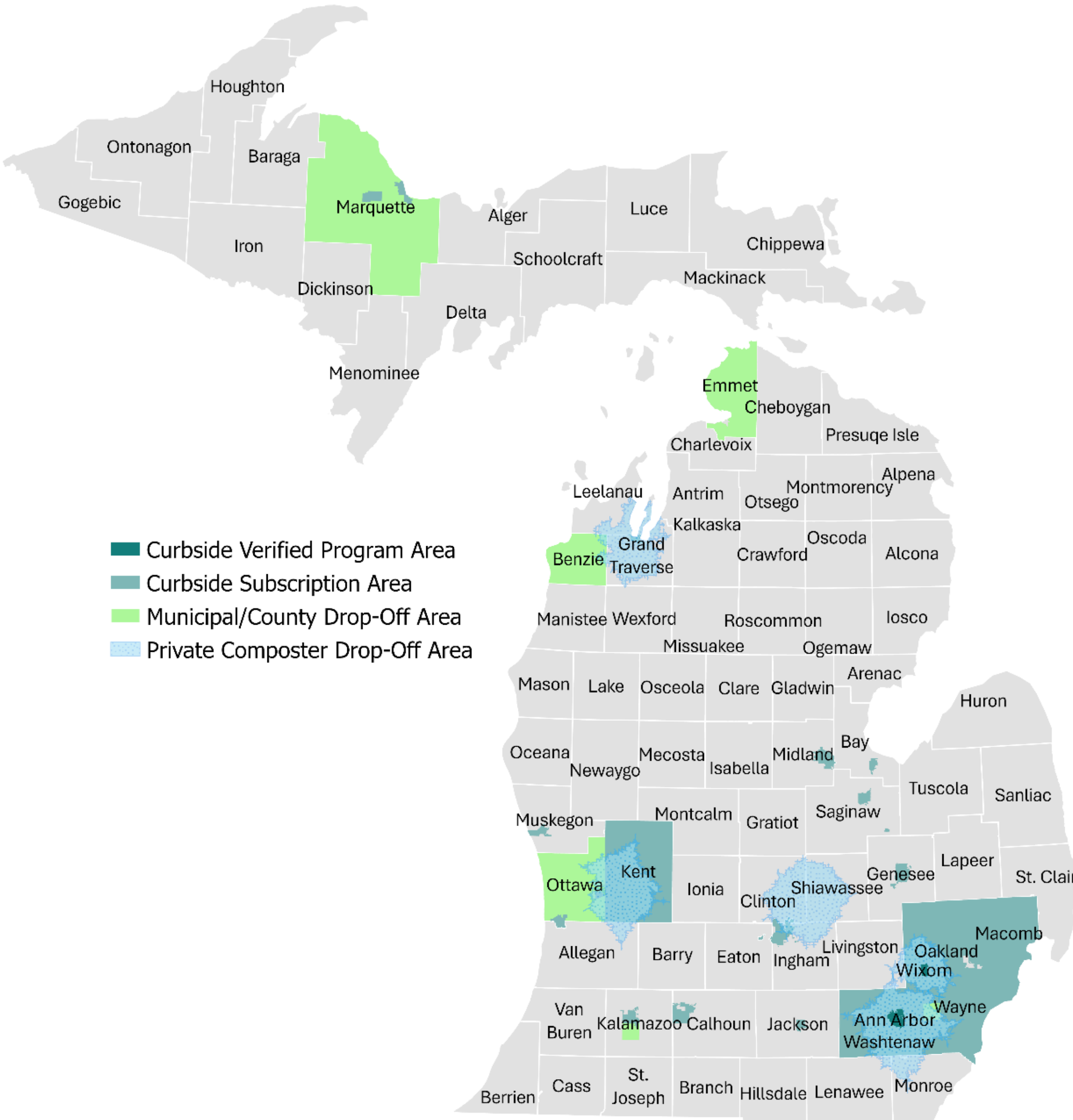


Table 9 summarizes the population with access to residential food scraps diversion programs. Curbside subscription services represent the largest share of access, reaching approximately 32% of Michigan's population, followed by private business drop-off programs at 18%. Despite these efforts, an estimated 67% of the state's population lacks convenient access to either curbside or drop-off food scraps programs. It is worth noting that a small number of communities have overlapping access, such as availability of both a county-administered drop-off site and a subscription-based curbside collection service. Access is likely to grow in the coming years as new programs are rolled out. For example, the City of Wixom recently introduced a pilot curbside pickup program, and the City of Southfield is also considering a pilot curbside pickup program.

Table 9: Percent of State's Population with Food Scraps Diversion Program Access Type

ACCESS TYPE	POPULATION WITH ACCESS	PERCENT OF STATE POPULATION
Drop-Off Municipal/County	898,160	9%
Drop-Off Private Business	1,667,786	17%
Curbside Verified Program	87,967	1%
Curbside Subscription Program	2,645,520	32%
No Convenient Access	5,592,314	67%
Total With Access	**2,654,461**	**32%**

Note: There is access overlap such that some communities have access to multiple types of food scraps collection programs. Curbside population with access includes only single-family residents. Drop-off population with access includes both single-family and multi-family residents.

Municipally Operated Food Scraps Drop-Off Programs

Additional municipal food scraps drop-off locations were added in seven communities since the 2023 gap analysis research was conducted, including programs for the communities of Canton, Portage, Lansing, Traverse City, Kalamazoo, East Grand Rapids, and Madison Heights. There are additional locations found for the existing food scraps programs in the City of Ferndale and Emmet County (Table 10).

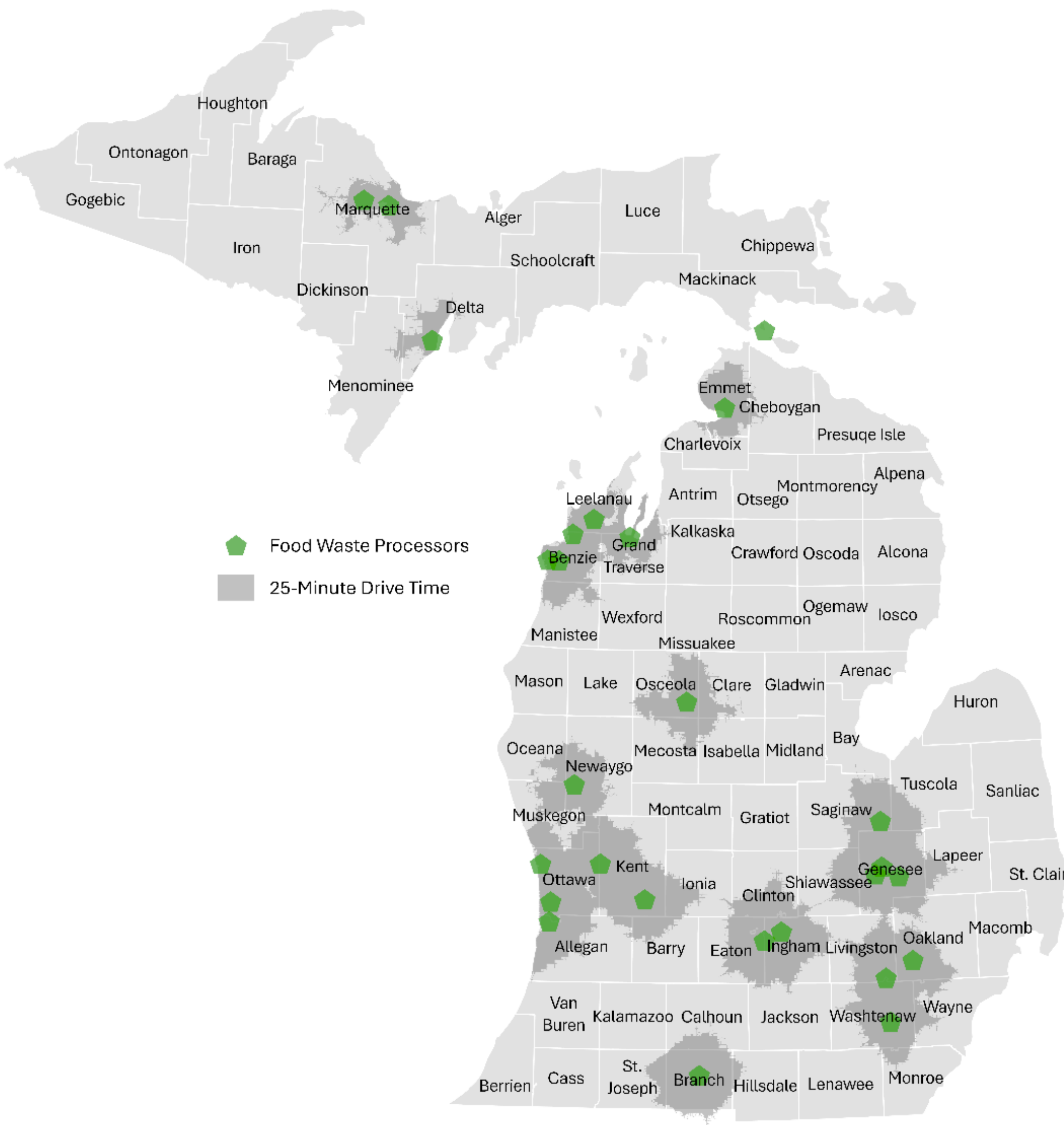
Table 10: Updated Food Scraps Drop-Off Programs by Municipality

MUNICIPALITY	OPERATOR	# OF RESIDENTIAL DROP-OFF SITES	PROGRAM LINK
Canton Township	My Green Michigan	4	https://www.cantonmi.gov/1360/Food-Compost-Pilot-Program
Ferndale	My Green Michigan	5	https://www.ferndalemi.gov/resources/compost-drop-off-program
Royal Oak	My Green Michigan	1	https://www.romi.gov/CivicSend/ViewMessage/message/244524
Bay City	Iris Waste Diversion Specialists	1	https://www.baycitymi.gov/697/Drop-Off-Services
Portage	My Green Michigan	10	https://www.portagemi.gov/933/Food-Scraps-Recycling-Program
Lansing	My Green Michigan	5	https://www.lansingmi.gov/687/News-Events?contentId=37d123ea-c941-4c94-83ca-d6f90b200ed9
Traverse City	Carters Compost	1	https://www.carterscompost.com/farmers-market
Kalamazoo	Organicycle	3	https://www.kalamazoocity.org/Residents/Recycling-Waste/Food-Scrap-Recycling
East Grand Rapids	Organicycle	1	https://www.eastgrmi.gov/243/RefuseRecycleYard-WasteCompost
Madison Heights	My Green Michigan	1	https://madison-heights.org/2010/Food-Scraps

Comparison of Facility Access and Food Scraps Collection Programs

Access to food scraps processing facilities is a foundational requirement for building sustainable diversion programs, as collection efforts cannot succeed without adequate processing capacity nearby. Figure 15 maps food scraps processing facilities across Michigan, with shaded areas indicating the communities within a 25-minute drive of each site. The shaded areas represent 37% of the state's population, indicating that just over a third of Michigan residents are within a reasonable drive time of a food scrap processor.

Figure 15: Food Scraps Processing Facilities With 25-Minute Drive Time Shaded

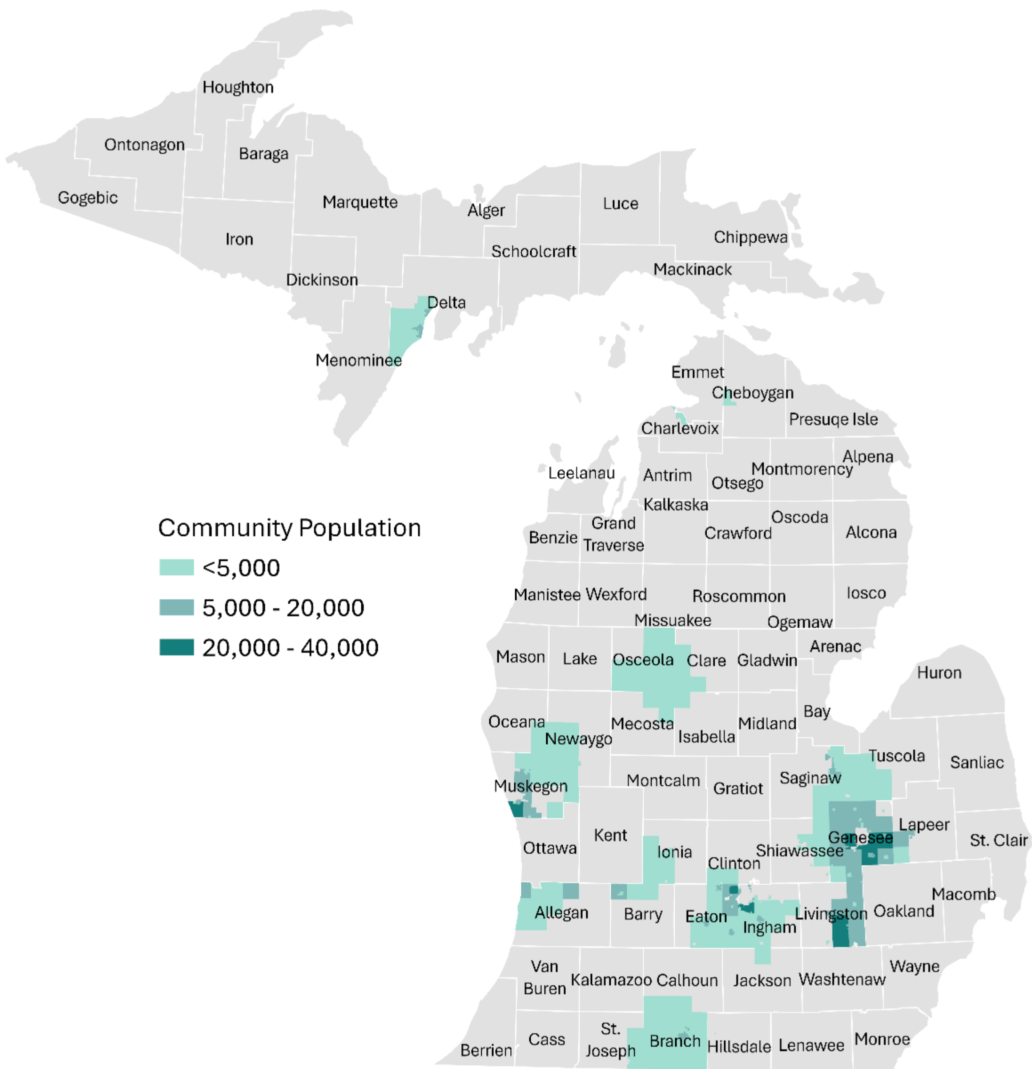


While proximity to a food scrap processor alone cannot guarantee that a community can successfully operate a collection program, it is a strong indicator of potential. Among residents within a 25-minute drive of a food scrap processor, 72% have some form of food scraps collection program available to them, either curbside or drop-off. The remaining 28% of residents within that same drive time have no program available, representing an opportunity to expand program access where processing infrastructure already exists to support it. Figure 16 highlights the high opportunity communities including Livingston, Genesee, and parts of Ingham and Muskegon. In total approximately 10% of Michigan's population live within a 25-minute drive of a food scrap processor but lack access to a food scrap collection program.

Table 11: Number of Communities Within 25-Minute Drive Time of Food Scraps Processor

COMMUNITIES WITHIN 25-MINUTES OF A FOOD SCRAPS PROCESSOR	POPULATION	PERCENT OF TOTAL
Total Communities	3,710,570	37% (of Michigan population)
Communities with Drop-Off or Curbside Access	2,657,928	72%
Communities with No Convenient Access	1,052,651	28%

Figure 16: Selected Communities with No Food Scraps Diversion Programs within 25-minute Drive Time of a Processing Facility



Food Scraps Hauler Interviews

As part of this project, private and public sector food scraps haulers operating across Michigan were interviewed, representing almost half of all the food scraps haulers in the state, to identify gaps, validate existing information, and explore opportunities to improve access to food scraps collection services in Michigan communities. Food scraps haulers provide services via curbside for residents, commercial/institutional onsite collection, and/or drop off site. Some of the questions included:

- What materials do you accept?
- Do you require any certification for accepting materials?
- Who do you provide curbside onsite services to?
- Who can utilize the drop-off sites and how many visitors utilize the sites annually?
- Where do you provide this service?
- How much do you charge for this service?
- How much annual volume do you collect?
- Can you estimate the proportion of your customers that are residential versus commercial?
- How do you provide this collection service
- Where do you take the material you collect for processing and how is the material processed?
- Has your service expanded in the last 3 years and how?
- What is your biggest challenge to expanding your services?
- How do you envision your business changing in the next year?
- How could EGLE provide support for your business?

Table 12 lists current food scraps haulers operating in Michigan, along with updates to their service areas and other program details, highlighting success stories and opportunities for additional growth in this sector. The number of food scraps haulers increased from 16 in 2023 to 18 in 2025. Services areas have also expanded, indicating the increasing demand for access to food scraps collection and recycling services; however, many providers continue to focus on commercial and residential drop-off sites, where the volumes justify the cost of collection.

Food scraps haulers consistently noted that one of their biggest challenges is marketing and educating both the public and businesses on the importance of food scraps composting. Barriers include the high cost of outreach and a general lack of public awareness. Haulers emphasized the need for EGLE to lead a statewide education campaign, like the Recycling Raccoons initiative, for traditional recycling along with dedicated funding to support marketing and outreach for composting programs.

Table 12: Current Food Scraps Haulers

FOOD SCRAPS HAULER NAME	2023 SERVICE AREA	2025 SERVICE AREA UPDATE	OTHER UPDATES
Ann Arbor Municipal	Ann Arbor	Ann Arbor	Stopped accepting BPI-certified compostables, only accepting CMA-W paper products
BARC	The northern Michigan counties of Grand Traverse, Antrim, Kalkaska, Leelanau, Benzie and Manistee	No longer collecting food scraps as Carter's Compost expands	
Carter's Compost	Downtown/West Traverse City: Weekly and Biweekly curbside pickup	BOOM (Base of Old Mission) Neighborhood and Farmers Market in Downtown Traverse City Drop-off Sites	Pay-as-you-throw drop-off option
CoSustainability - Acquired by My Green Michigan	Bay Area	Acquired by My Green Michigan	
Eastside Compost Company	Lansing/East Lansing area	Lansing/East Lansing area	
Emmet County	Emmet County	Emmet County	
Grand Rapids Compost – NEW!	None	Grand Rapids	New in 2025

FOOD SCRAPS HAULER NAME	2023 SERVICE AREA	2025 SERVICE AREA UPDATE	OTHER UPDATES
Growing Hope - NEW!	None	Ypsilanti Farmers Market and Growing Hope Community Garden Drop-offs	New in 2025
Iris Waste Specialists	Birch Run, Bay City, Saginaw	Birch Run, Bay City, Saginaw	
Midtown Composting	Detroit Metropolitan Area	Detroit Metropolitan Area	
My Green Michigan	Major metro areas in Southern Lower Michigan: • West MI - Grand Rapids, Holland, Muskegon • Southwest MI - Kalamazoo, Battle Creek • Mid-MI - Lansing, Jackson • Southeast MI - All of Metro Detroit and Ann Arbor • Bay Area - Flint, Saginaw, Bay City, Midland	Major metro areas in Southern Lower Michigan: • West MI - Grand Rapids, Holland, Muskegon • Southwest MI - Kalamazoo, Battle Creek • Mid-MI - Lansing, Jackson • Southeast MI - All of Metro Detroit and Ann Arbor • Bay Area - Flint, Saginaw, Bay City, Midland	Acquired CoSustainability; offering dumpster service
New Soil	Located in Jenison, Michigan, and serves all around the West Michigan area	West Michigan Area	
Organicycle	Grand Rapids	City of Grand Rapids, Kent, Kalamazoo, Muskegon counties. Planning to expand more in 2026	Phasing out of residential curbside service, servicing more drop off sites
Partridge Creek Compost	Just starting up in 2023	Marquette, Trowbridge Park, Harvey, Negaunee, and Ishpeming	Changed name to Partridge Creek Compost, received a grant to engage with municipalities, servicing two pilot drop off sites
Scrap Soils	Detroit	Detroit	Servicing pilot drop off sites
Turtle Ridge Compost	Southwest Michigan	Southwest Michigan	
Unlimited Recycling	Detroit area	Eastern Metro Detroit area, including Macomb, Wayne, Oakland, and Washtenaw Counties	Purchased a new truck
Wormie's	Grand Rapids	Kent County, greater Grand Rapids area	

Local Government Insights

As part of this project, local governments were interviewed to provide insights on food scraps diversion programs in their regions. The participating counties and authorities represented a range of population types across Michigan and included Delta, Emmet, Kent, Oakland, Van Buren, Washtenaw, and the RRRASOC authority. Several other counties were contacted but did not respond. The key insights from these interviews are summarized below:

- **Rural organics gap:** Rural communities with residential recycling are less likely than urban areas to offer curbside organics collection, though some have third-party yard waste services (e.g., Delta and Emmet Counties).
- **Commercial food scraps focus:** Third-party haulers increasingly serve commercial and institutional food scraps customers while scaling back residential collection (e.g., Kent County/Organicycle).
- **Commingled yard and food scraps:** Municipalities with yard waste collection by cart are beginning to allow food scraps mixing, enabled by partnerships with organics processors.
- **Organics still limited:** Most curbside organics programs remain yard-waste only, with gradual expansion to food scraps dependent on hauler capacity and processing options.
- **Yard waste drop-offs:** Most communities provide yard waste drop-offs, often collected by municipal or third-party haulers; some unmanaged leaf sites risk speculative accumulation (e.g., Van Buren County).

- **Urban food scraps pilots:** Larger cities are adding food scraps drop-offs at downtown areas and farmers markets; grant-funded containers (e.g., Metro Store) are being tested.
- **Washtenaw pilot:** Washtenaw County plans to launch small-scale residential food scraps drop-off to encourage expansion in municipalities without curbside service.
- **Confusing guidelines:** Listing "fruit and vegetables" under yard waste can cause confusion, as it is intended for yard-grown vegetation, not food scraps.
- **Cost barriers:** Pricing at some food scraps sites (e.g., Ann Arbor, charged per yard) discourages residents with small quantities.

Material Gap Takeaways

- **High disposal share:** Food scraps and surplus food represents 18% of Michigan's MSW disposal stream or about 1.5 million tons annually. To meet the state's 45% recycling rate goal, approximately 766,500 tons of food scraps and surplus food must be diverted.
- **Edible food scraps and surplus food:** Two-thirds of disposed food in Michigan was once edible, including unpackaged meals and discarded packaged products. Expanding prevention and rescue programs is essential to reduce this portion of the stream.
- **Food scraps:** One-third of disposed food consists of inedible scraps such as peels, bones, and fruit pits, which are well-suited for diversion to organics processing.
- **Food scraps diversion program access:** Only 32% of Michigan's population has some level of access to food scraps diversion program, and the majority of that access is provided through private sector subscription curbside or drop-off programs. While these programs offer an essential service to residents, there are barriers to participating including extra steps to signing up, additional cost, and inequity in access. Furthermore, 67% of the state's population lacks any access, with some of those residents living in close proximity to food scrap processors.
- **Food rescue efforts:** This report highlights the efforts of six food rescue organizations working across Michigan to feed hungry people with food that would otherwise be wasted. These efforts are making an impact across the state and there is significant opportunity to grow food rescue operations.
- **Access growth opportunities:** Approximately 1.2 million Michigan residents live within a 25-minute drive of a food scraps processing facility but lack access to a food scraps collection program, representing roughly 10% of the state's total population. These residents are a prime opportunity for program expansion, as the processing infrastructure needed to support collection already exists within convenient reach of their communities.
- **Hauler reach:** Food scraps collection haulers are expanding both in number and geographic area in Michigan. At the same time interviewed haulers consistently noted challenges with the high cost to education and outreach and the need for EGLE support to lead a statewide education campaign supporting food scraps and surplus food diversion.
- **Priority action:** Scaling up prevention, rescue, and organics processing infrastructure will be critical for Michigan to reduce edible food scraps and surplus food and achieve its diversion targets.

COMMERCIAL CORRUGATED CARDBOARD

431,500+ tons disposed in 2024, 3rd most prevalent material in the commercial waste stream; an additional 237,300+ tons must be diverted to reach the 45% recycling rate goal.

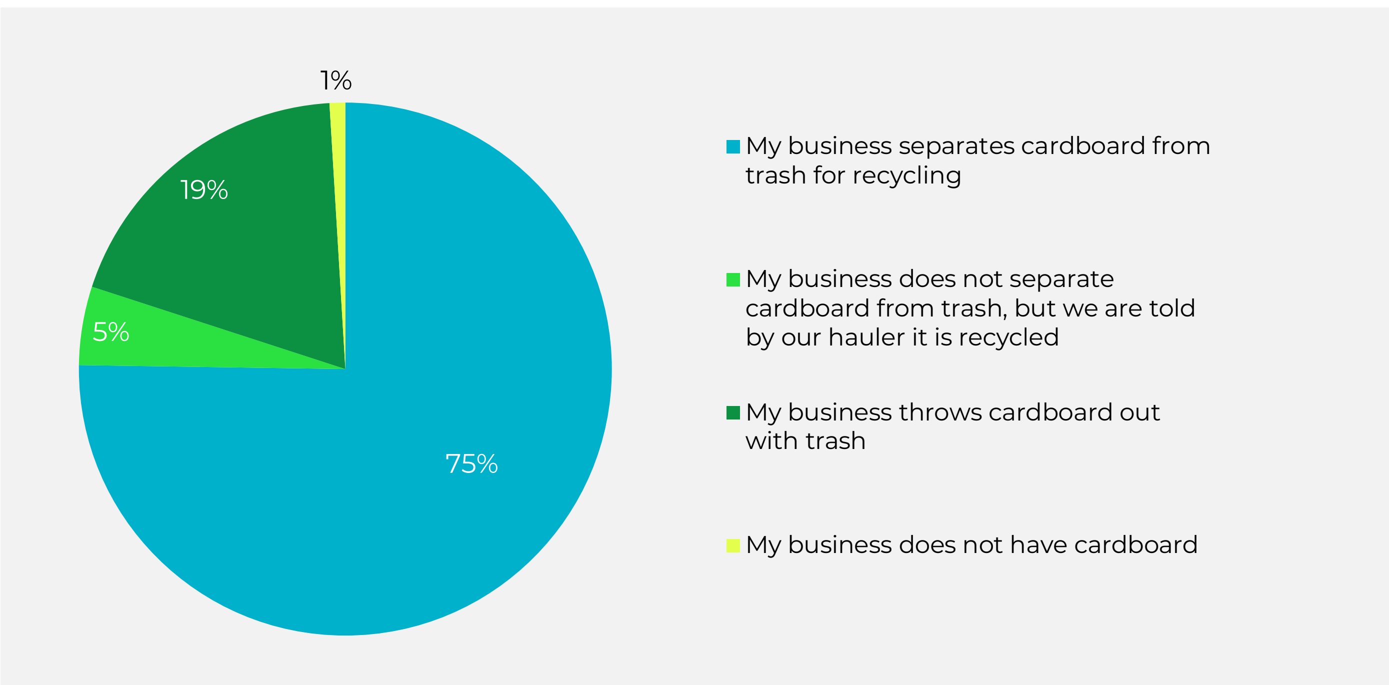
Corrugated cardboard is a highly recyclable commodity that is easily sorted at MRFs, often has strong market value, and has stable regional end markets in Michigan. Despite this, cardboard is frequently disposed, accounting for the third largest disposal category for the commercial sector in Michigan, as shown in the MSW Disposal Composition section of this report. In general, major retail chains recover cardboard through a business-to-business system, baling cardboard at store locations or backhauling to distribution centers before directly selling to end markets. However, small and medium businesses may not have the resources or volume generation to justify onsite baling operations or direct partnerships with end markets. As such, small and medium sized businesses represent a major opportunity for Michigan to capture recyclable cardboard out of the disposal stream.

Small Business Survey

To gain insight into corrugated cardboard recycling of small businesses in Michigan, a study was distributed to over 10,000 commercial entities across the state. Respondents were asked questions about business size and operations, the handling of waste, and overall feedback on how access and recycling operations could be improved. In total, 147 businesses responded to the survey, with 105 businesses providing valuable insight on their waste management operations for their flagship location. Survey responses represented 13 different COGs and 11 different NAICS codes, providing a comprehensive perspective. The survey questions are provided in the appendix.

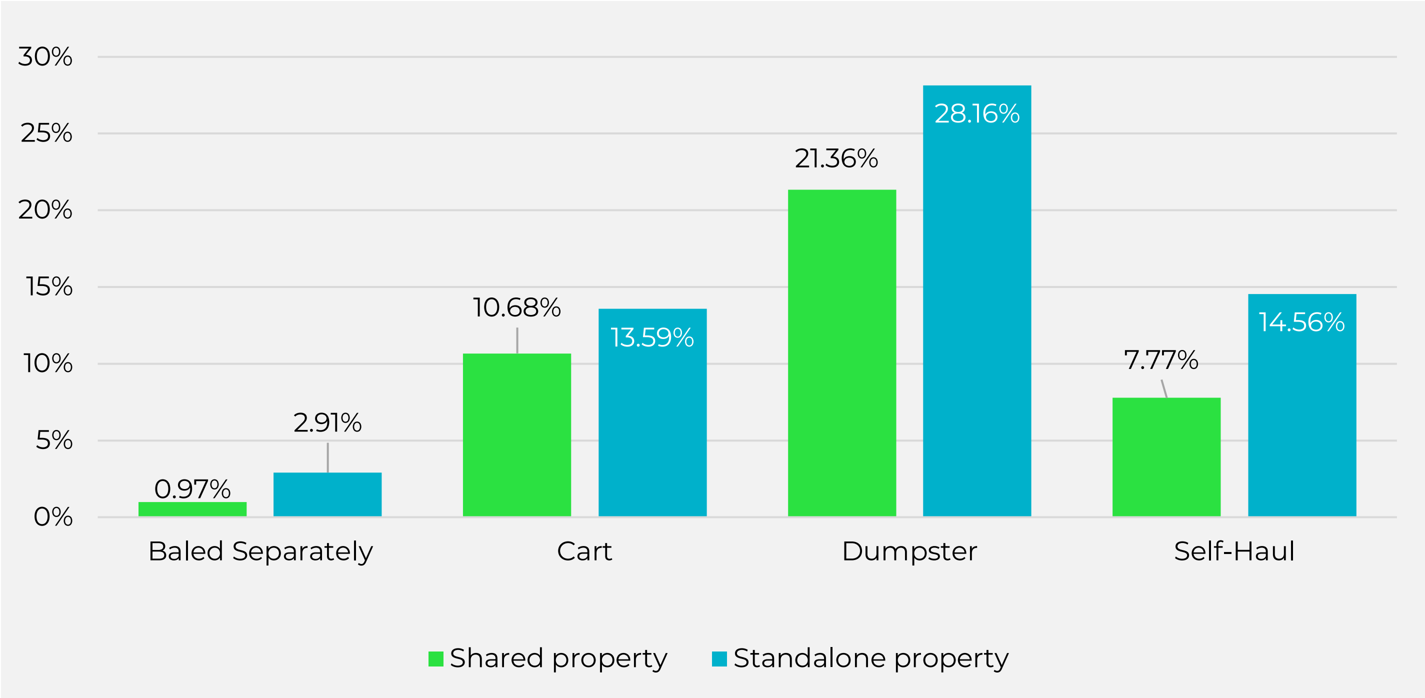
Overall, 75% of small business respondents reported that they separate their cardboard materials for recycling (Figure 17).

Figure 17: Proportion of Small Business Reporting of Cardboard Management



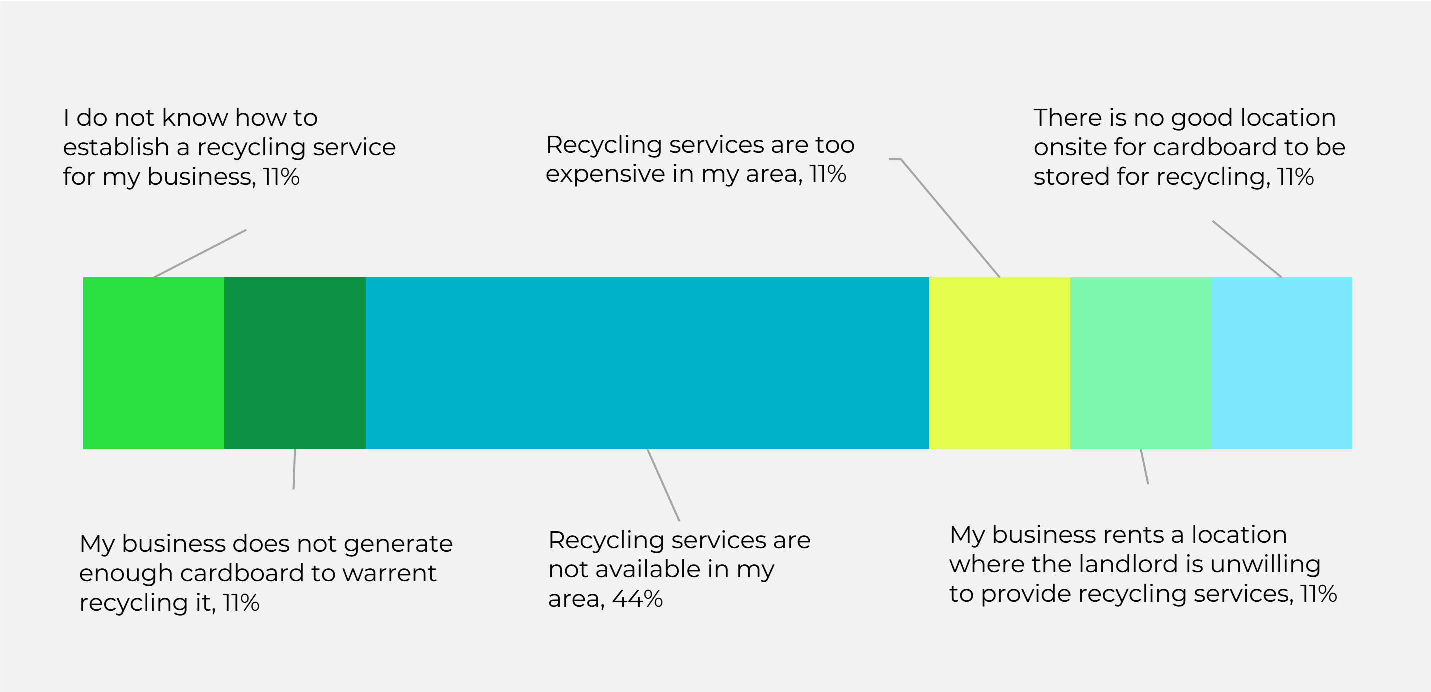
For both standalone and shared-property businesses, dumpsters were the most common collection method, followed by carts, self-hauling, and baling recycled materials separately (Figure 18). Weekly pickup was by far the most frequently reported collection schedule for both dumpsters (71%) and carts (84%). While most businesses indicated that cardboard is collected as part of a commingled recycling stream, 52% of those using carts reported that carts are completely full of cardboard at the time of collection. For dumpster-based collection, results were more varied, with 53% noting that dumpsters are typically one-quarter to half full of cardboard when collected.

Figure 18: Percent of Respondents Indicating Corrugated Cardboard Collection Container Type



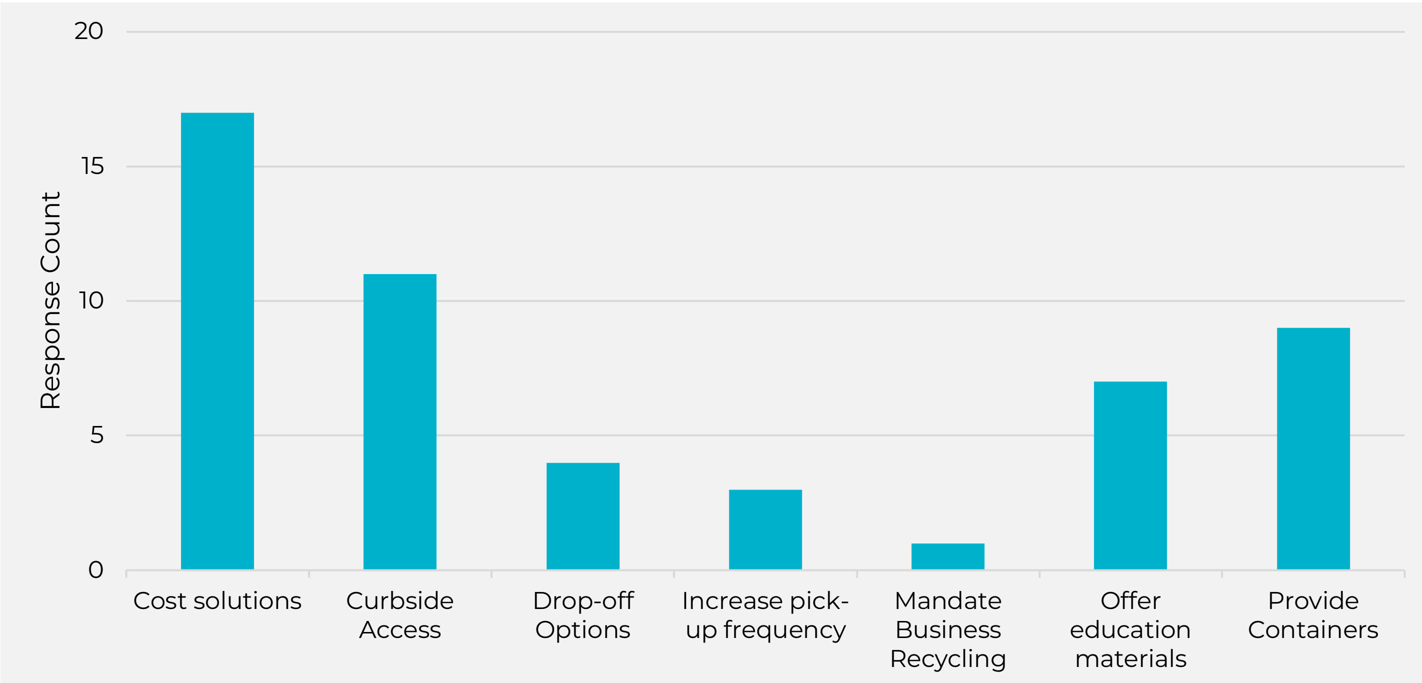
For the 24% of small businesses that do not separate their cardboard from their waste, additional questions were asked to gain a better understanding of why. The most common answer, accounting for 44% of these respondents, was that recycling services were not available in their area. Other reasons were evenly split amongst respondents and included high costs, lack of awareness about available programs, low volume of cardboard, limited onsite space, and that recycling services are decided by landlords (Figure 19).

Figure 19: Non-Recycling Participants Indication of Why They Did Not Recycle



The final question for all respondents was requesting feedback on how EGLE can make recycling easier for small businesses. Out of the 52 feedback responses provided, the most common theme was providing cost solutions for commercial recycling programs, accounting for 33% of the replies. The next common request was encouraging haulers to pick up curbside from commercial entities, followed by requests for onsite containers to be required on commercial properties (Figure 20).

Figure 20: Recycling Needs Reported by Small Businesses



Material Gap Takeaways

- **High disposal rates:** Corrugated cardboard makes up an estimated 431,577 tons of Michigan's commercial disposal stream.
- **Gap in participation by small businesses:** Nearly one-quarter of small business survey respondents reported not separating cardboard from trash.
- **Lack of service availability:** 44% of respondents said they do not recycle cardboard because on-site recycling services are not available in their area.
- **Additional barriers:** Other challenges include high program costs, limited space for containers, and property owner control over recycling services.
- **Opportunity for diversion:** To reach the state's 45% recycling rate goal, an additional 710,000 tons of paper materials must be recovered, with commercial corrugated cardboard representing one of the largest opportunities.
- **Priority action:** Develop pathways to expanding access to and participation in cardboard recycling programs among small businesses.
- **Supporting Policy Opportunities**: While mandated business recycling was represented by a small percentage of respondents, policy solutions to drive recovery have proven successful at the local, regional and statewide level, and can be encouraged to drive further diversion.'

SCRAP METAL

232,700+ tons disposed in 2024; diverting another 23,200+ tons are needed to reach the 45% recycling rate goal.

Residential Survey

The 2023 Michigan Gap Analysis estimated that approximately 717,300 tons of scrap metal are recycled annually in Michigan's residential and commercial sectors, much of which likely goes unreported to EGLE. This estimate was developed using scrap metal recycling data from Pennsylvania, scaled to Michigan's population.

As part of this phase of work, RRS conducted a statewide survey to collect direct feedback from residents and benchmark against the proxy-based estimate. The survey, distributed to over 1,000 residents and received 802 responses, aimed to assess how frequently residents replace metal items, how they removed those items from the household, and their overall attitudes toward scrap metal recycling and access in their communities. Respondents represented 368 ZIP codes across all 14 COGs, demonstrating broad geographic distribution and allowing for a representative analysis of Michigan residents.

The survey was organized into three material-focused sections: large appliances, small appliances and tools, and construction and building debris. Each section focused on a detailed list of items, and respondents were asked to indicate which items they had replaced in the past year, how many, and what they did with them. The three categories can be summarized as follows:

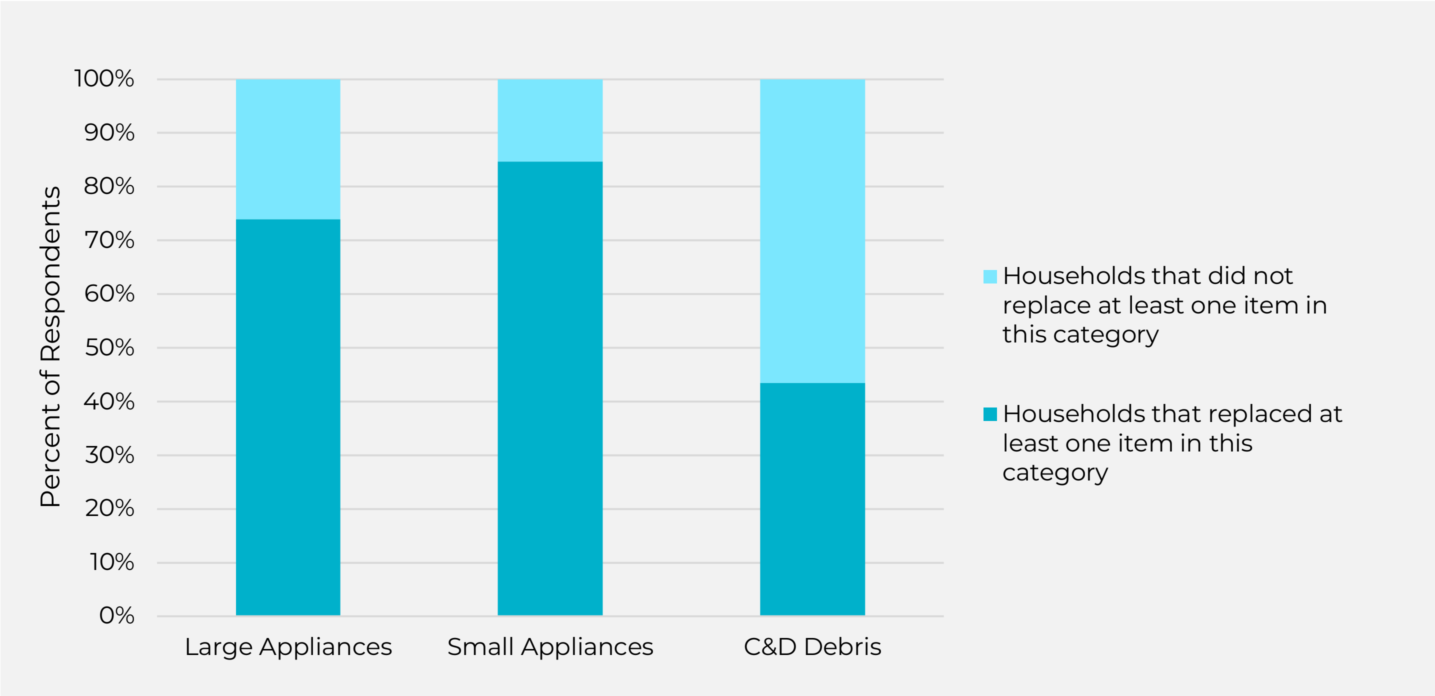
- **Large appliances** included items such as washing machines, stoves, and air conditioning units.
- **Small appliances, tools, and household items** comprised a larger number of items and included items such as filing cabinets, garden equipment, toasters, microwaves, and mobility-related items like scooters and bicycles.
- **Construction and demolition (C&D) debris** included aluminum siding, doors, pipes, railings, and smaller miscellaneous metal items such as brackets and nails.

In addition to reviewing survey responses, a quantitative analysis was conducted to estimate the total volume of materials replaced annually in Michigan. This estimate was derived from multiplying the number of reported replacements by the average weight of each material type. The material categories included in the survey and the average weights used in the analysis is presented in the Appendix.

Replacement Frequency

Overall, it was most common for residents to replace at least one item in the small appliance and tool category, with 85% of the 802 total survey respondents reporting a replacement in this category. For large appliances, 75% of respondents indicated they had replaced at least one item. In contrast, construction and building debris were replaced far less frequently, with only 43% of respondents noting that an item in this category was replaced (Figure 21).

Figure 21: Percent of households that reported replacing at least one item in each category within the past year.



Within the large appliance category, washing machines and air conditioners were the most frequently replaced items, with 28% of respondents replacing at least one washing machine and 24% replacing an air conditioning unit in the past year. In the small appliance and metal tool category, pots and pans were the most replaced item by 26% of respondents, followed closely by garden equipment by 24% of respondents. While construction and building debris were replaced less often overall, the most frequently noted items in this category were miscellaneous hardware (e.g., screws, nails), reported by 19% of respondents, followed by metal doorknobs and hinges at 14%.

In several cases, respondents indicated replacing the same type of item more than once within a single year. While this is unsurprising for smaller or consumable materials, such as pots and pans, a notable 29% of all respondents reported replacing the same large appliance multiple times within the past year. This trend was particularly evident for HVAC systems and air conditioners—of those who replaced these items, 28% reported doing so more than once during the year.

Management Method

If a respondent indicated they had disposed of a particular item, they were then asked to specify how it was managed to better understand the material's disposal pathway. The list of management method options provided in the survey accompanied by the corresponding destination categories is summarized in Table 13.

For example, if a respondent reported selling or giving away an item, it was assumed the item remained in use and was categorized as reuse. If the item was brought to a metal scrap yard, it was categorized as entering the recycling stream. When respondents indicated that a contractor or appliance service provider hauled the item away after delivery of a new product, the exact destination could not be confirmed. However, given that many of these items are composed of valuable scrap metal, and that disposal would typically incur a fee, it is likely that these materials were directed to a scrap yard for recycling rather than to landfill.

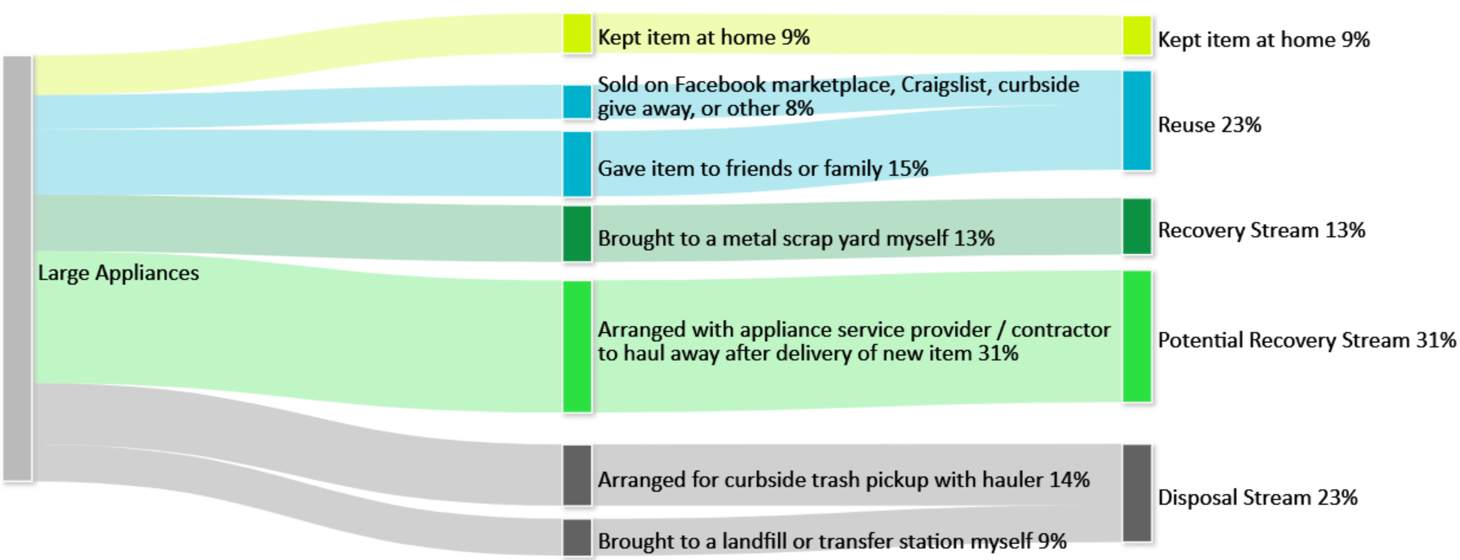
Table 13: Management methods reported by survey respondents and corresponding final material stream destination

FINAL DESTINATION	ASSOCIATED MANAGEMENT METHOD
Kept	• Kept Item at home either for continued use or because resident was unsure how to dispose of item
Reuse	• Sold on Facebook marketplace, Craigslist, curbside give away, or other • Gave item to friends or family
Recovery Stream	• Brought to a metal scrap yard myself
Potential Recovery Stream	• Arranged with appliance service provider / contractor to haul away after delivery of new item
Disposal Stream	• Placed item in recycling bin • Arranged for curbside trash pickup with hauler • Brought it to a landfill or transfer station myself

Large Appliances

For large appliances, the most reported management method was arranging for a service provider or contractor to haul away the old item during the delivery of a new one (31%). While the destination of these items is not always confirmed, they are categorized as entering a potential recovery stream due to the high scrap value of metal appliances. The second most common outcome was disposal: 23% of respondents either arranged for curbside trash pickup (14%) or brought the item to a landfill or transfer station themselves (9%). Another 23% of large appliances were either sold (8%) or given away to friends or family (15%) and were therefore assumed to remain in use and categorized under reuse. A total of 13% of respondents reported bringing the item directly to a metal scrap yard, a management method categorized under the recovery stream. Finally, 9% of respondents indicated that they kept the item at home, either for continued use or due to uncertainty about how to dispose of it (Figure 22).

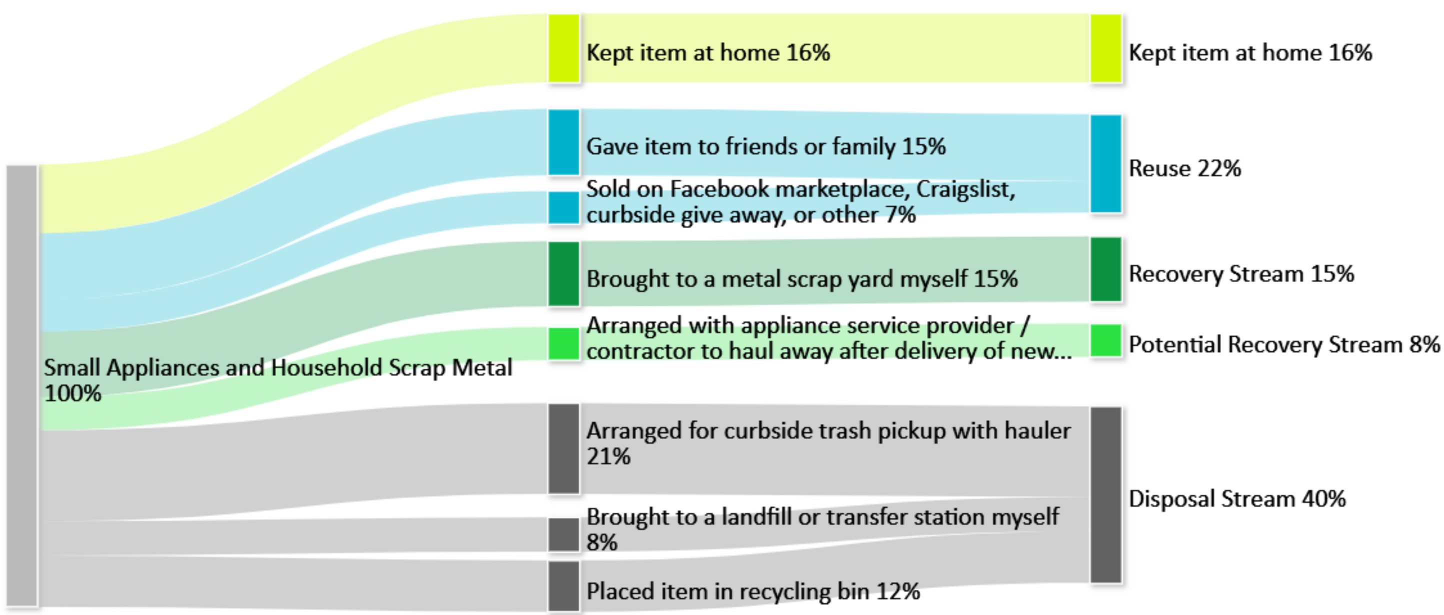
Figure 22: Residentially reported management method for large appliances



Small Appliances and Household Metal Items

For small appliances and household scrap metal, the most common destination was the disposal stream, with 40% of respondents either arranging for curbside trash pickup (21%) or bringing the item to a landfill or transfer station themselves (8%). An additional 12% indicated that they placed the item in a recycling bin, though in general scrap metal items cannot be recovered at material recovery facilities, and these items were assumed to be sorted into residue and ultimately disposed of. Sixteen percent of respondents reported keeping the item at home, either for continued use or due to uncertainty about how to dispose of it. A combined 22% of items were either given to friends or family (15%) or sold via platforms like Facebook Marketplace or Craigslist (7%) and were therefore assumed to remain in use and categorized under reuse. A total of 15% of respondents brought the item directly to a metal scrap yard, placing it into the recovery stream, while 8% arranged for an appliance service provider or contractor to haul it away, a method classified as potential recovery (Figure 23).

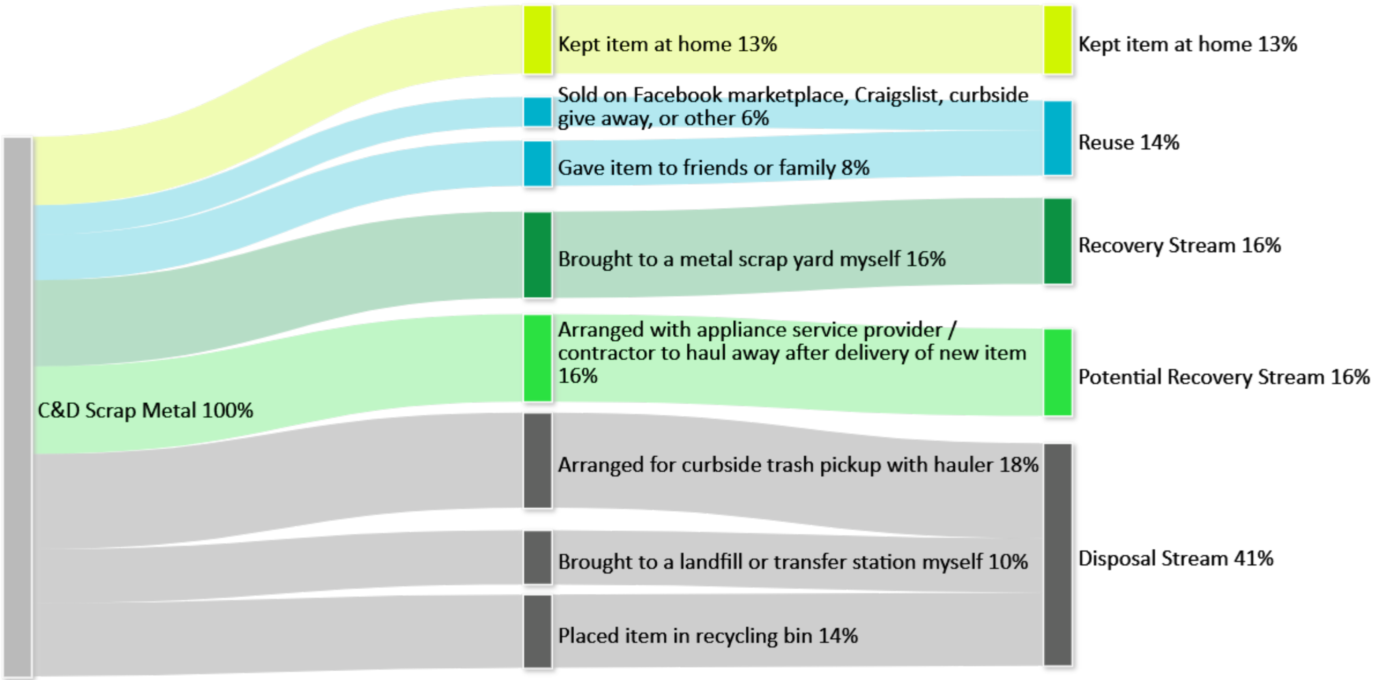
Figure 23: Residentially reported management method for small appliances and household metal items



Construction and Demolition of Scrap Metal Materials

For C&D scrap metal, disposal was the most common destination. A combined 41% of respondents either arranged for curbside trash pickup (18%), brought the material to a landfill or transfer station themselves (10%), or placed the item in a recycling bin (14%). While the recycling bin pathway may suggest an intent to divert, the actual outcome of that material is likely disposal. A total of 16% of respondents reported bringing the item directly to a metal scrap yard, which was categorized as entering the recovery stream. An additional 16% arranged for a contractor or service provider to haul away the item after replacement, a method classified as potential recovery—given the likelihood that the material's scrap value may incentivize recovery. In total, 14% of respondents managed items in a way consistent with reuse: 8% gave the item to friends or family, and 6% sold or gave it away through informal channels like Facebook Marketplace or Craigslist. Finally, 13% of respondents reported keeping the item at home, either for potential reuse or due to uncertainty about proper disposal (Figure 24).

Figure 24: Residentially reported management method for C&D scrap metal

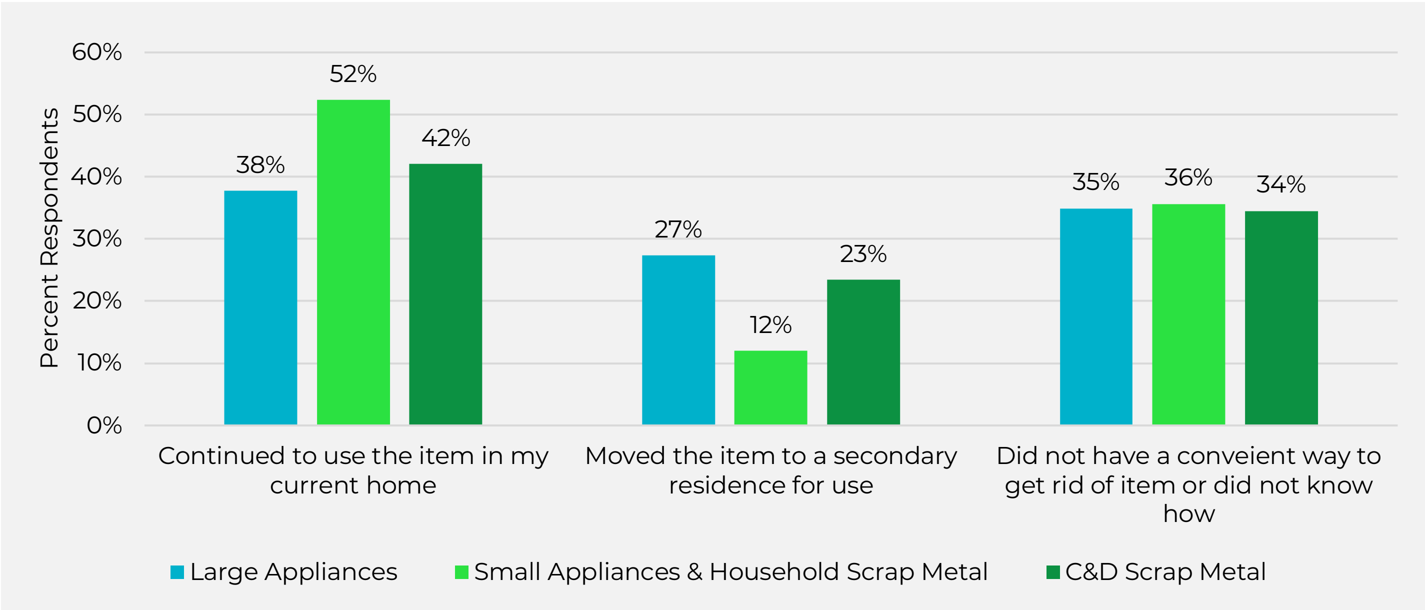


Kept Items

In all three material categories, a portion of respondents reported keeping replaced items at home. To better understand this behavior, those individuals were asked whether the item was being reused in a different capacity or retained due to uncertainty or limited access to disposal options.

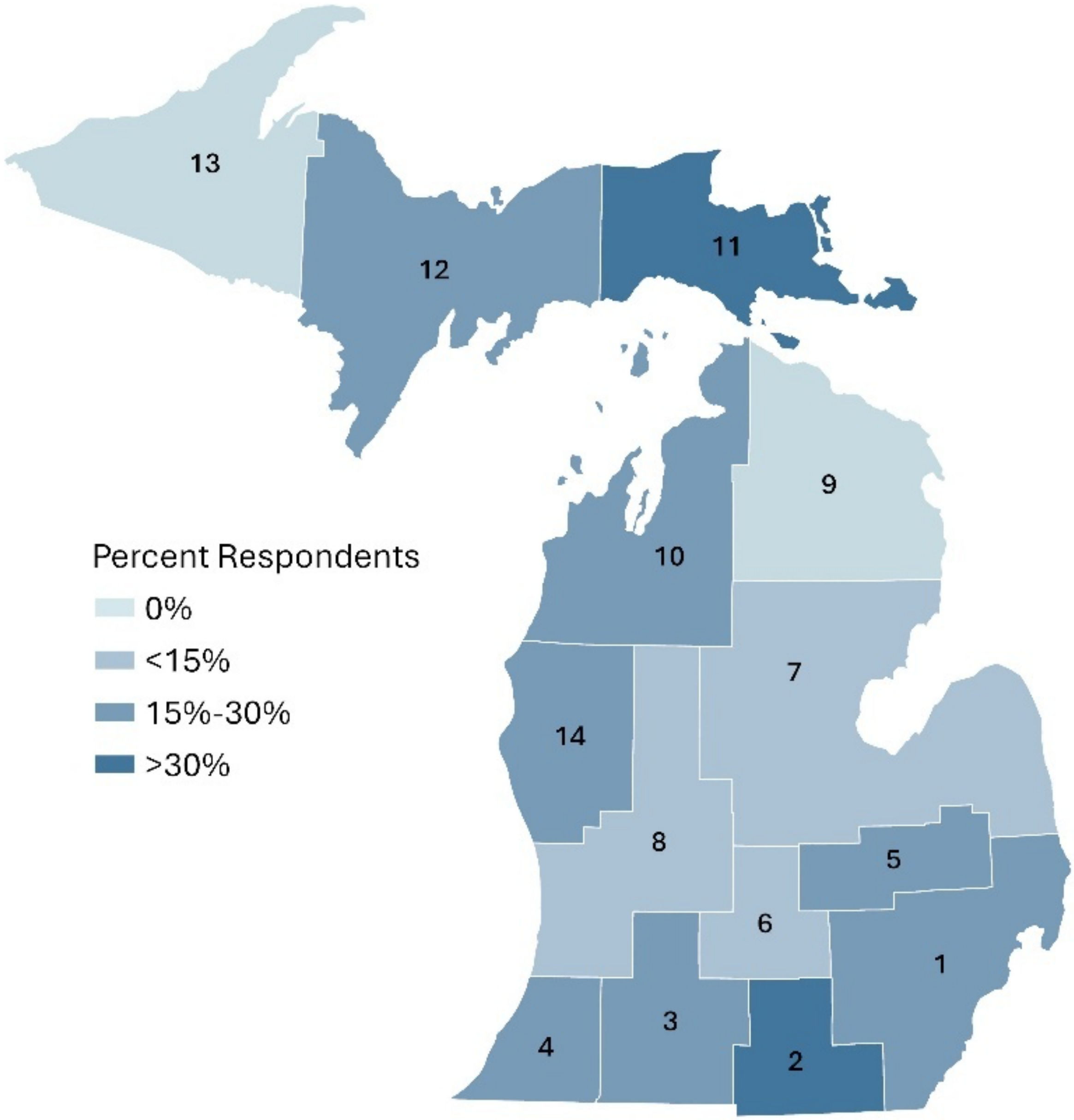
As shown in Figure 25, many respondents kept an item at home for continued use – 38% of kept large appliances, 52% of kept small appliances or household scrap metal, 42% of C&D scrap metal. However, a lack of convenient disposal options or confusion about how to dispose of items emerged as a consistent barrier across material types. Roughly one-third of retained items in each category were kept because respondents either did not know how to dispose of the item or lacked a practical means of doing so.

Figure 25: Reported reasons residents kept items at home after replacement



To explore whether these barriers varied geographically, respondents' answers were analyzed by Council of Government (COG) region. Apart from COGs 9 and 13, residents in every region reported challenges in disposing of at least one replaced item. Rates of confusion or limited access ranged from 11% (COG 7) to 50% (COG 11) (Figure 26).

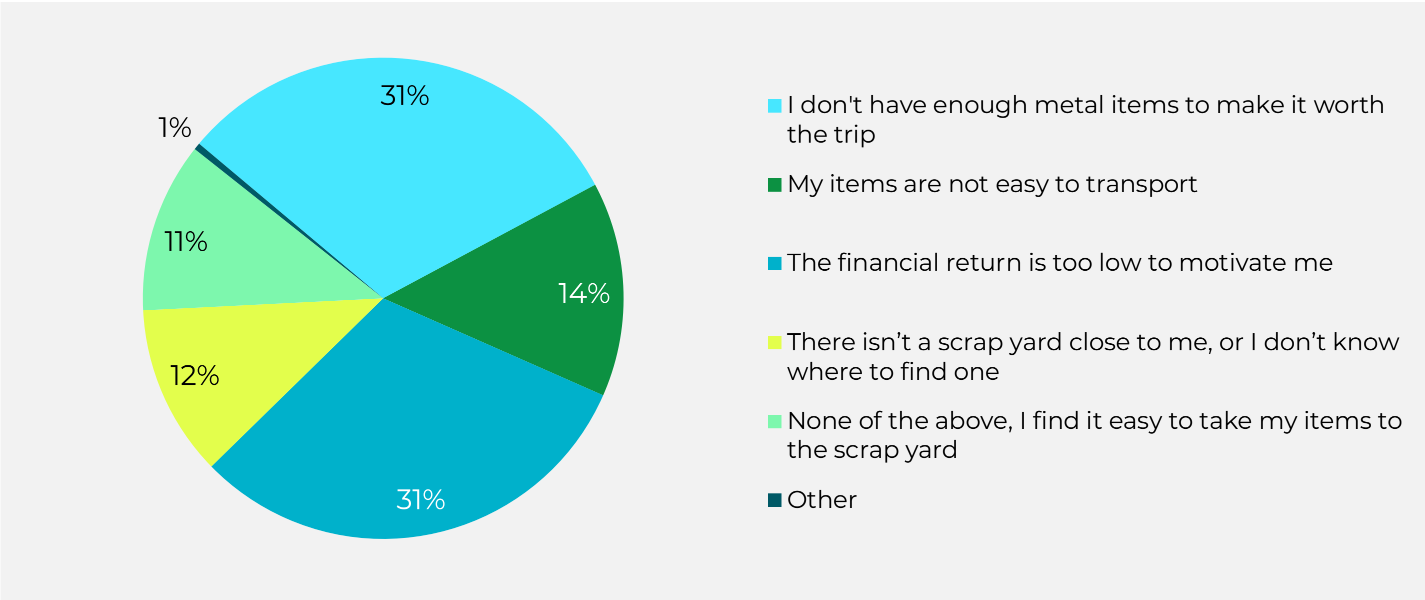
Figure 26: Percent respondents reporting confusion of or limited access to scrap metal recycling by COG



Scrap Metal Recycling Behaviors and Motivations

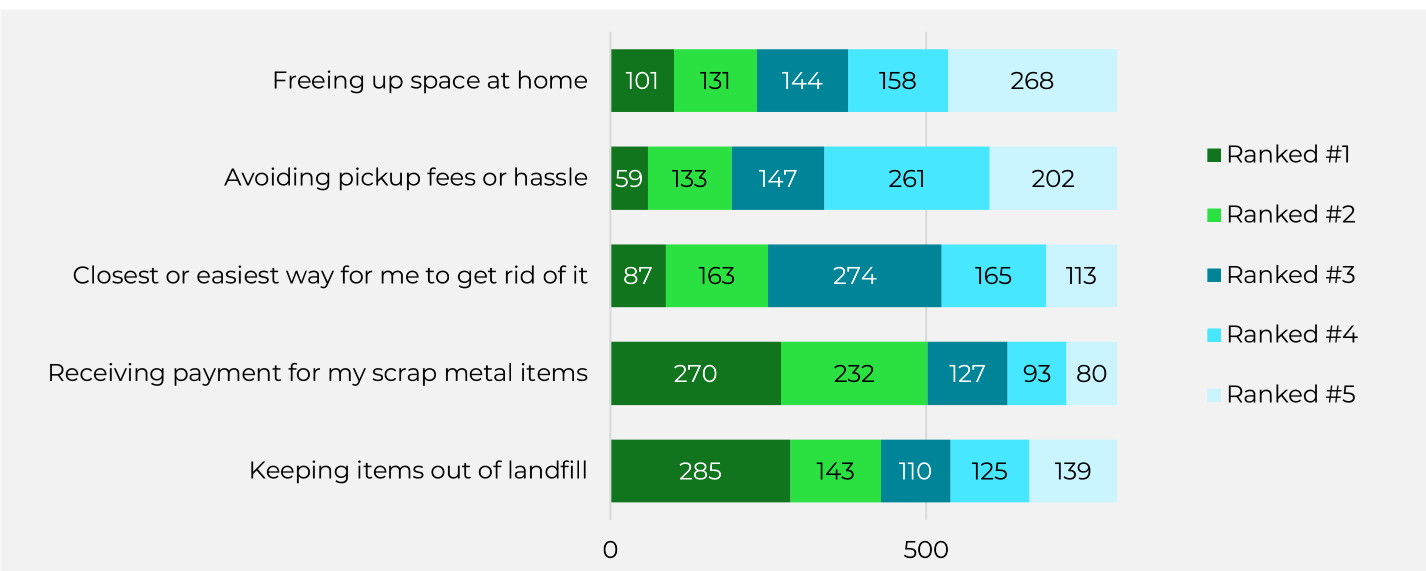
Before completing the survey, residents were asked a few questions about their views on current disposal options for metal items and what could make metal recycling more viable in the future. Notably, only 11% of respondents indicated that they find it easy to take their items to a scrap metal yard for recycling. When asked what factors currently prevent them from taking items to a scrap yard, the most common reasons were evenly split between not having enough material to justify the trip and the financial return not being worth the effort - each accounting for 31% of respondents. Transportation challenges, either lacking reliable transportation or finding the items difficult to transport, were also frequently mentioned (14%). Another 12% of respondents indicated they do not have a scrap metal yard close to their home or were unaware of how to find one (Figure 27).

Figure 27: Respondents reported view on scrap metal recycling in Michigan



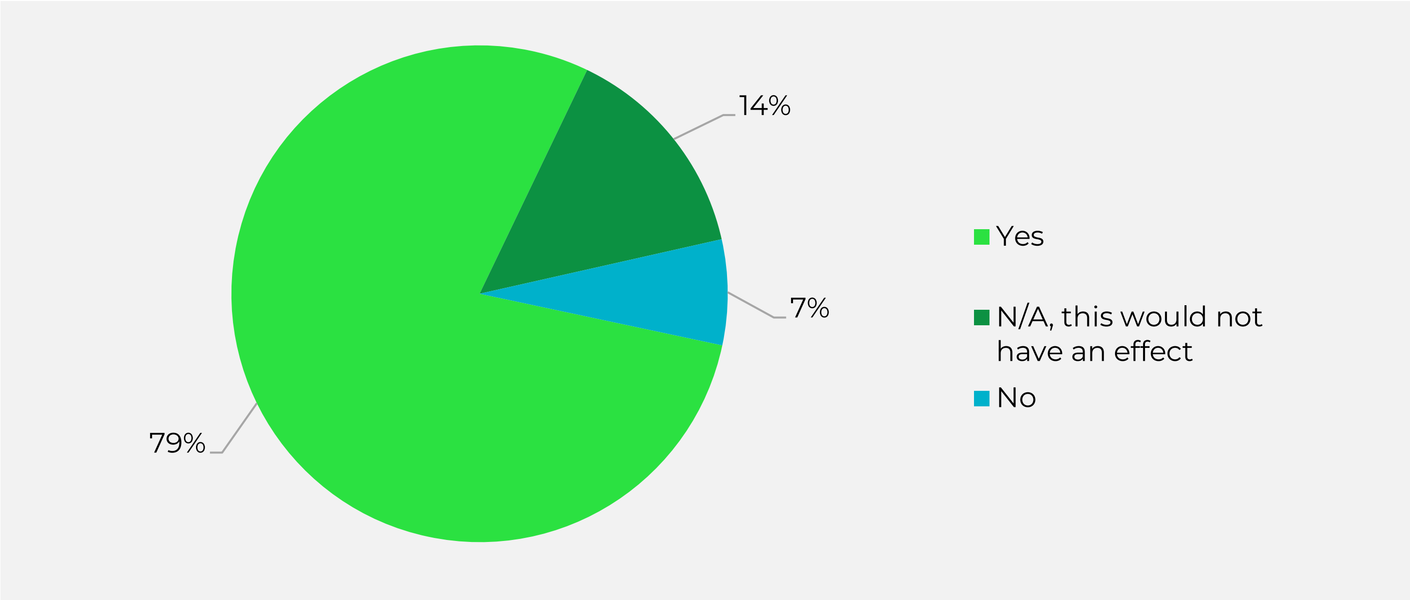
When asked about their primary motivation for bringing items to a scrap yard, 36% of residents cited keeping items out of landfills as their top priority, while 34% said they were primarily motivated by the payment received. In general, fewer residents were motivated by avoiding pickup fees or saving space at home (Figure 28).

Figure 28: Motivations for taking materials to a scrap yard.



Finally, residents were asked whether access to a "super drop-off" site, one that accepts scrap metal along with other materials, would increase their willingness to recycle scrap metal. Residents expressed interest in this, with 79% saying yes (Figure 29).

Figure 29: Residential response to being asked whether access to a super drop-off site that accepts scrap would increase their willingness to recycle scrap metal.



Residential Scrap Metal Recycling Projections

The survey asked residents to report the total quantity of scrap metal removed from their household, using either unit counts (e.g., number of air conditioning units, refrigerators, stoves, toasters, metal shelves, bicycles) or linear/square footage estimates (e.g., square feet of aluminum siding replaced, or linear feet of metal fencing removed). These reported quantities were then converted to estimated tonnage using average weight assumptions for each item type.

Based on this analysis, it is estimated that approximately 965,000 tons of materials containing scrap metal are replaced annually across the three surveyed categories.

Of that total, residents reported bringing about 147,500 tons of material directly to scrap metal yards themselves (Figure 30). This estimate is nearly double the 76,500 tons identified in the 2023 residential scrap metal recycling gap analysis, which was based on reported recycling in Pennsylvania scaled to Michigan's population. There are two likely reasons for this higher estimate in the Michigan survey data:

1. The tonnage calculations from this study are based on the total weight of whole items, which may include non-metal components, for example, the plastic housing of a dishwasher or other appliances.
2. The Pennsylvania-based estimate relies on recycling data formally reported to the state, rather than direct input from residents, and may underrepresent actual recycling behavior due to gaps in reporting. Not all scrap yards may report recycling data to the state, resulting in an incomplete picture.

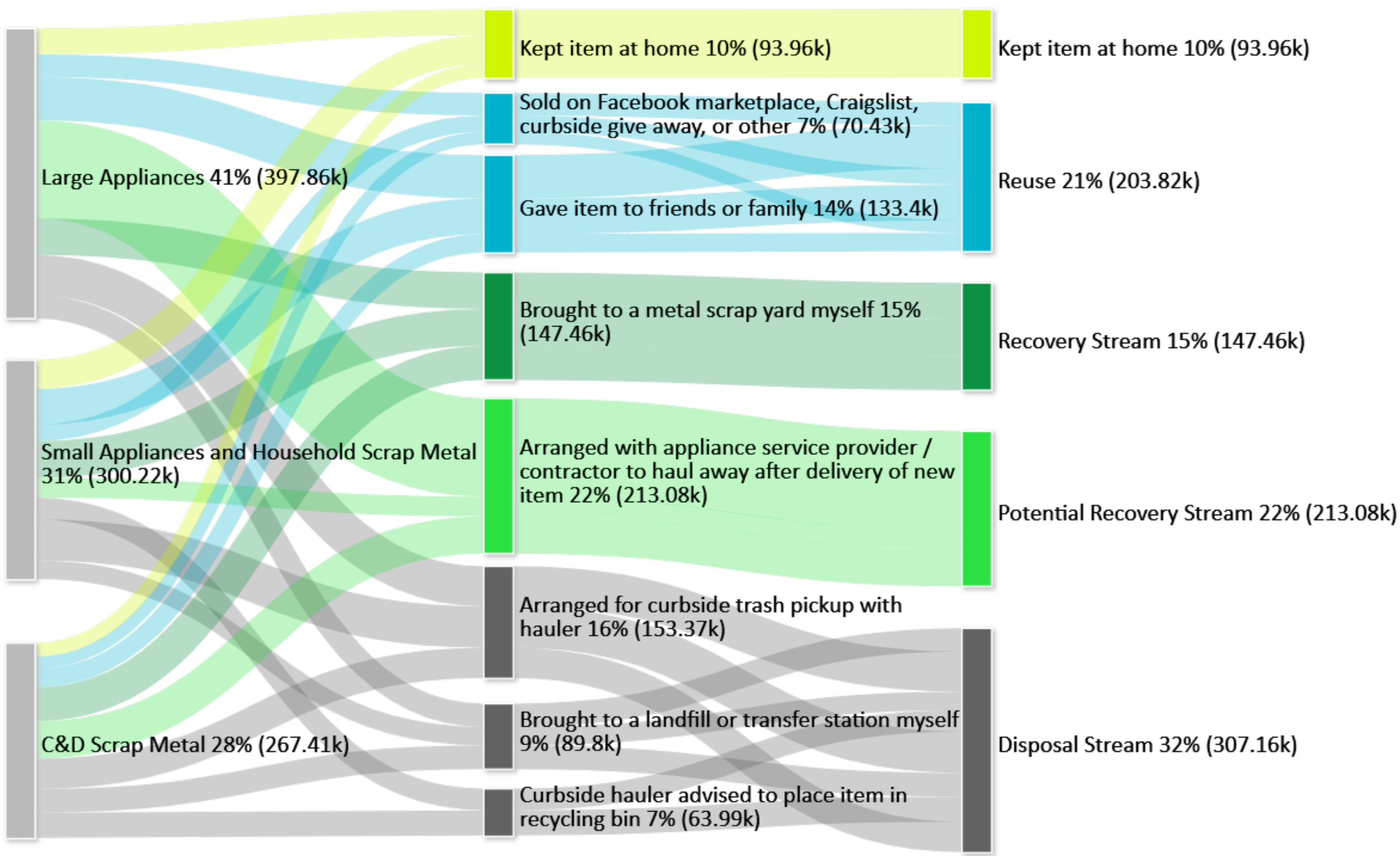
By asking residents directly about materials they brought to scrap yards in the past 12 months, this survey helps to address and fill that reporting gap.

The survey also identified 213,100 tons of scrap metal items that were reported as being hauled away by appliance service providers or contractors from residential homes. While the destination of these materials is unknown, it is likely that many of these items enter the commercial scrap metal stream due to their value. The 2023 Michigan gap analysis estimated 640,780 tons of scrap metal recovered annually from the commercial sector; the tonnage identified in this survey may represent a portion of that total.

Respondents also reported 203,100 tons of scrap metal items that were either given to family or friends or sold through platforms like Facebook Marketplace or Craigslist. Additionally, 94,000 tons of scrap metal were kept at home, with some items still being used by residents and others potentially retained due to disposal challenges.

Finally, the survey identified approximately 307,200 tons of scrap metal items sent to disposal, either brought to the landfill or transfer station directly by the residents, picked up at the curb by the trash hauler, or placed in the recycle bin by the resident where the material likely ended up in the residual stream and was eventually disposed of. In total, 32% of the replaced scrap metal tonnage was sent for disposal, highlighting a substantial opportunity for improved diversion through public education, infrastructure access, and clearer recycling guidance.

Figure 30: Estimate tons of residential scrap metal and management destination



Comparison to State Level Recycling and Disposal Data

In 2024, EGLE estimated that approximately 1.16 million tons of metal were recycled in Michigan. This total includes scrap metal, white goods, aluminum and steel cans recycled through MRFs, and redeemed beverage containers. Of this amount, about 36,300 tons were aluminum and steel cans recovered through MRFs and container redemption, leaving roughly 1.12 million tons of scrap metal (ferrous and non-ferrous metals and white goods).

Michigan does not currently track scrap metal recycling by residential versus commercial sources, but estimates can be developed using data from other states. In 2023, Pennsylvania reported that 16% of recycled scrap metal originated from the residential sector and 84% from the commercial sector. Applying this ratio to Michigan suggests that residents brought approximately 181,500 tons of scrap metal to scrap yards, while about 935,900 tons came from commercial sources such as retail appliance take-back programs, construction and demolition activities, and routine business operations.

Table 14: 2024 Estimated Scrap Metal Recycling in Michigan Broken Down by Generating Sector

GENERATING SECTOR	TOTAL SCRAP METAL RECYCLING ESTIMATE
Residential	181,500
Commercial	935,900
Total	1,120,000

Residential reporting also indicates that around 307,160 tons of scrap metal were disposed—either through curbside trash or recycling bins. However, waste characterization studies show a much lower amount of scrap metal in the disposal stream, with about 232,768 tons in total, of which 104,438 tons were attributed to the residential sector.

The discrepancy between these estimates is likely reflected in the high market value of scrap metal. Even when scrap is disposed of or improperly placed in curbside recycling, it is often recovered before final disposal. For example:

- MRFs may divert scrap metal to local scrap yards.
- Informal collectors may retrieve scrap set out at the curb.
- Metals sent to incineration may still be captured from incinerator ash.

Material Gap Takeaways

- **High replacement rates:** 74% of residents replaced a large appliance in the past year, 85% replaced a small appliance or other household metal item, and 43% replaced metal construction and demolition (C&D) debris.
- **Items retained at home:** Between 9–16% of replaced items were kept at home rather than recycled. About one-third of residents cited lack of knowledge or convenient removal options—totaling an estimated 94,000 tons of scrap metal remaining in households.
- **Barriers to recycling access:** Only 11% of residents felt it was easy to bring scrap metal to a scrap yard. The most common barriers were not having enough material to make a trip worthwhile, as well as a lack of nearby scrap yard options.
- **Interest in comprehensive drop-off sites:** 79% of residents said they would recycle more metal if comprehensive drop-off sites were available that accepted scrap metal alongside other materials.
- **Opportunity for diversion:** Michigan recycled approximately 1.12 million tons of scrap metal in 2024, yet an estimated 232,768 tons still entered the disposal stream.
- **Priority action:** Expanding access and convenience for scrap metal recycling could be a key focus for EGLE to capture this highly recoverable material and further increase recycling.
- **Supporting Policy Opportunities**: Driving supply to Michigan's strong scrap metal processing infrastructure can be a positive outcome of supportive diversion-focused local, regional and statewide policies, benefiting both industry and generators of these valuable materials.

WOOD WASTE AND STORM DEBRIS MANAGEMENT

554,700+ tons disposed in 2024; diverting an additional 55,400+ tons is needed to reach the 45% recycling rate goal.

Generation

Wood waste in Michigan includes pallets, dimensional lumber, engineered wood products (e.g., plywood, particle board, wafer board, strand board), painted and treated wood, and untreated wood (e.g., forest and mill residues such as bark, slab wood, sawdust, and materials like untreated fencing, wood roofing, and siding). It is generated across the residential, commercial, institutional, and industrial sectors.

Accurately quantifying wood waste generation is challenging due to data limitations. However, estimates can be developed using a range of proxy data sources. A detailed description of the wood waste generation and recycling estimate is presented in the Appendix.

Importantly, for the purpose of calculating Michigan's 45% recycling rate goal, only wood waste generated by the residential, commercial, and institutional (RCI) sectors are included. The State's diversion estimate does not account for industrial waste, and as such including industrial wood waste diversion would substantially distort the results. Industrial generators include sectors such as forestry and sawmills.

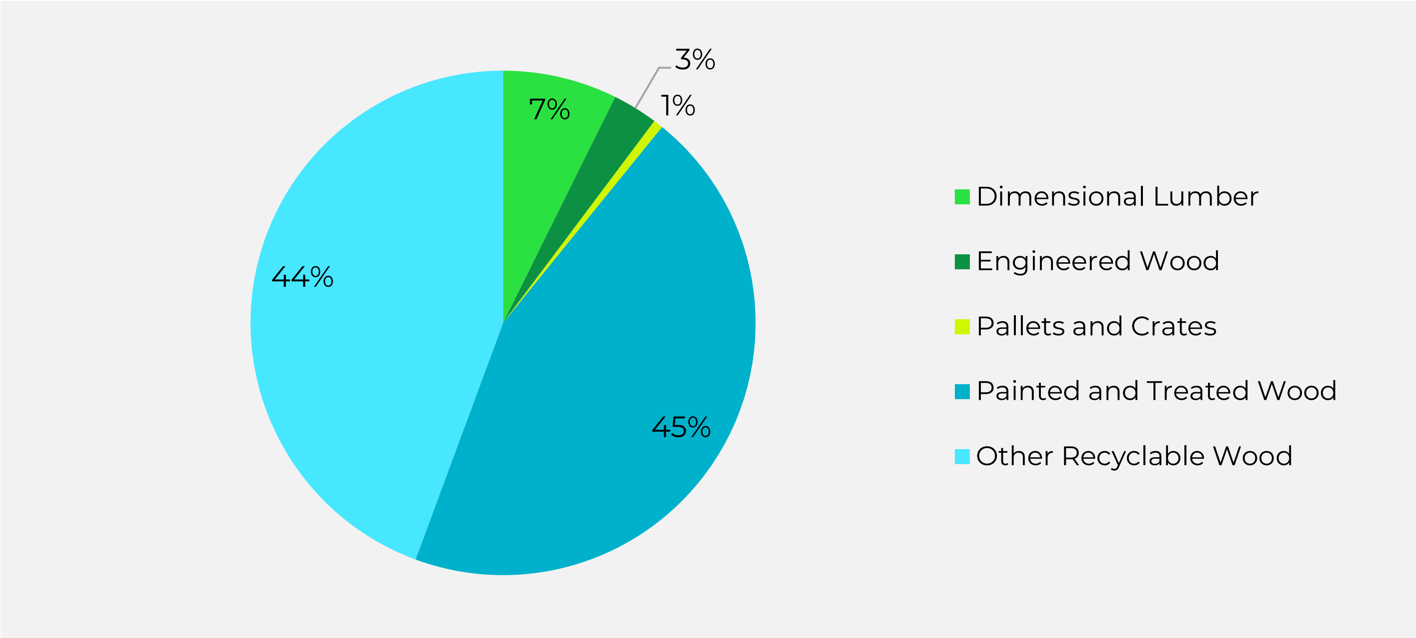
Table 15 presents an estimated breakdown of wood waste by material type including dimensional lumber, engineered wood, pallets and crates, painted and treated wood, and other recyclable wood such as untreated scrap from production prefabricated wood products, untreated/unpainted fencing, wood roofing and siding. Figure 31 provides the proportional breakdown. Approximately 45% of wood waste in Michigan's MSW stream is painted or treated wood waste, which has limited recycling options. Another 44% of wood waste is "other recyclable wood," which is untreated wood waste that could potentially be chipped or mulched.

Table 15: Estimated Wood Waste in MSW Disposal Stream (Tons)[6]

WOOD WASTE CATEGORY	SINGLE-FAMILY	MULTI-FAMILY	COMMERCIAL	TOTAL
Dimensional Lumber	7,800	1,700	31,100	40,600
Engineered Wood	2,500	600	13,100	16,100
Pallets and Crates	200	0	3,200	3,400
Painted and Treated Wood	96,400	21,700	130,400	248,500
Other Recyclable Wood	45,600	10,200	190,300	246,100
Total	**152,400**	**34,300**	**368,100**	**554,700**

[6] Due to the uncertainty of this data, numbers in this table were rounded to the nearest hundreds place such that individual rows may not add to the table total.

Figure 31: Proportion of Wood Waste by Type in MSW Stream



Wood waste is also disposed of in Michigan's C&D landfills; although this material is not included in the state's MSW recycling rate goal, accounting for this wood waste provides a holistic view of statewide generation. Table 16 provides a breakdown of the estimated wood waste disposed in Michigan's C&D landfills. Similarly to the MSW stream, the largest category of wood waste disposal is painted and treated wood.

Table 16: Estimated Wood Waste in C&D Disposal Stream (Tons)[6]

WOOD WASTE CATEGORIES	TOTAL
Painted and Treated Wood	150,300
Dimensional Lumber	41,600
Engineered Wood	52,500
Pallets and Crates	36,200
Total	**280,600**

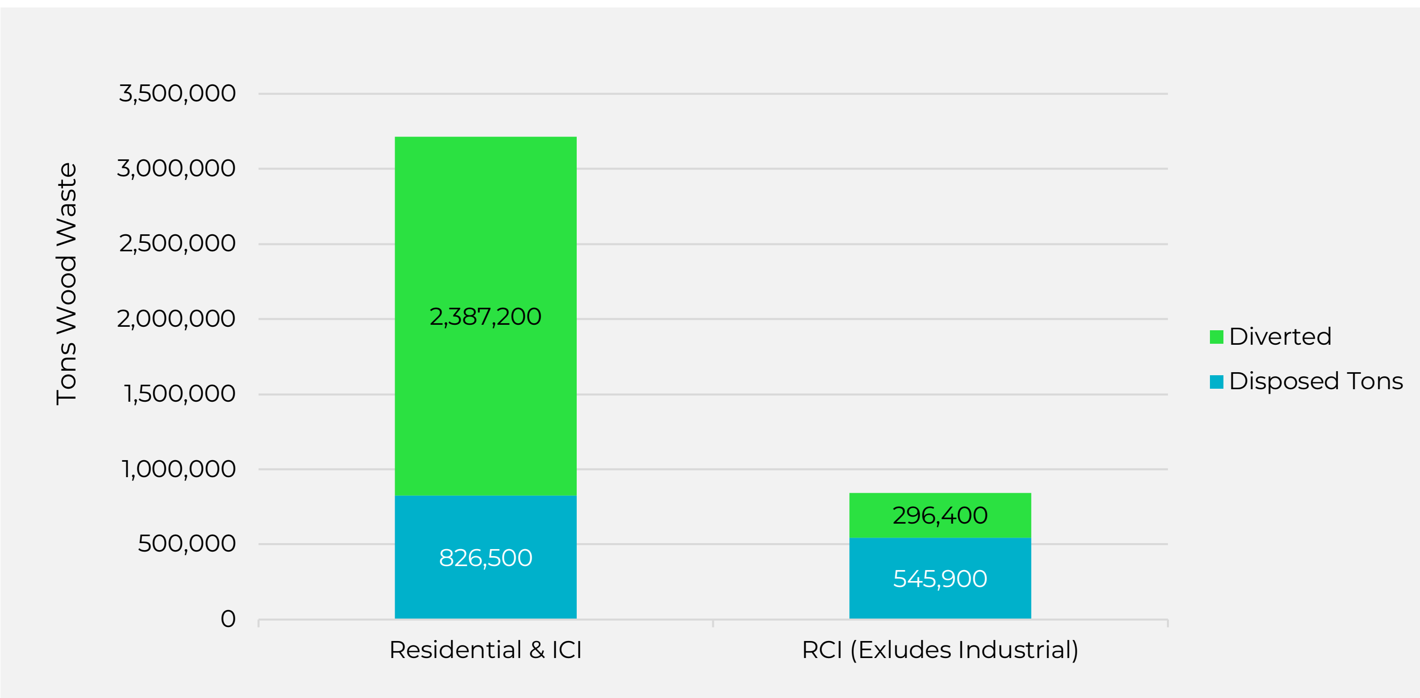
Table 16, Table 17, and Figure 32 provide a breakdown of modeled wood waste management pathways in Michigan. Overall, approximately 3.2 million tons of wood waste are managed each year across all sectors combined, with approximately 2.4 million tons diverted and 835,300 tons disposed. When industrial generation is included, about 74% of wood waste appears to be diverted through reuse, repair, recycling, composting, mulching/chipping, combustion for biofuel, and other uses such as animal bedding.

However, the 74% recycling rate is driven by industrial wood waste management, particularly through combustion (biofuel) and mulching/chipping. When industrial waste is excluded, the recycling rate for wood waste drops sharply to 35%. Note that reuse and repair of wood waste items such as furniture or reused lumber are not accurately captured in the data presented here due to data limitations.

Table 17: Modeled Wood Waste Generation in Michigan (Tons)[6]

MANAGEMENT METHOD	RESIDENTIAL AND ICI	RCI (EXCLUDES INDUSTRIAL)
Reused	10,500	6,200
Repaired	164,400	97,100
Recycled	23,200	13,700
Other Uses[7]	3,400	2,000
Composted	7,600	7,600
Mulched/Chipped	653,800	167,900
Combusted (Biomass Fuel)	1,524,300	1,900
Total Diverted	**2,387,200**	**296,400**
Disposed (MSW)	545,700	545,700
Disposed (C&D)	280,600	0[8]
Total Disposed	**835,300**	**545,700**
Total Generated	**3,222,500**	**851,100**
Recycling Rate	**74%**	**35%**

Figure 32: Wood Waste Generation in Michigan by Sector



[7] Other includes animal bedding and dismantling for parts used in other applications.
[8] While some commercial waste is disposed in Michigan's C&D landfills, the State's MSW diversion estimate does not account for disposal in these landfills.

Potential diversion opportunities for wood waste depend on the type of wood waste. Figure 33 and Figure 34 show the estimated flow of wood waste by type to different management strategies. All wood waste generation, residential, commercial, institutional, and industrial, is captured in Figure 33 while Figure 34 removes the industrial wood waste tons. Some key observations regarding the two figures are provided below:

Total wood waste

- Untreated wood waste is the largest category, with most of the untreated wood waste going to combustion and mulching. By and large, this category of wood waste is generated by industrial sectors.
- Pallets have the most widely implemented diversion management strategy, with a substantial portion being reused, repaired, and recycled. Approximately 81% of total wood pallets generated in Michigan are estimated to be diverted.
- Dimension lumber, engineered wood, and treated wood are typically sent to disposal. A small but known proportion of these materials are likely recovered in the state, and that data is not reflected in this analysis.

Residential, Commercial, and Institutional (RCI, Excluding Industrial)

- Combustion of untreated, non-industrial wood waste is assumed to be negligible, given the lack of economic feasibility for biomass fuel facilities to divert material from MSQ landfills. Only a small proportion of pallets generated in the commercial sector is directed to combustion.
- Pallets are diverted at a high rate (91%) from the RCI sector and sent to reuse, repair, recycling, mulching or chipping, and combustion.
- Dimensional lumber, engineered wood, and painted and treated wood is disposed of, along with a large proportion of untreated wood.

Figure 33: Total Wood Waste Generation Broken Down by Material and Management Method (Tons)

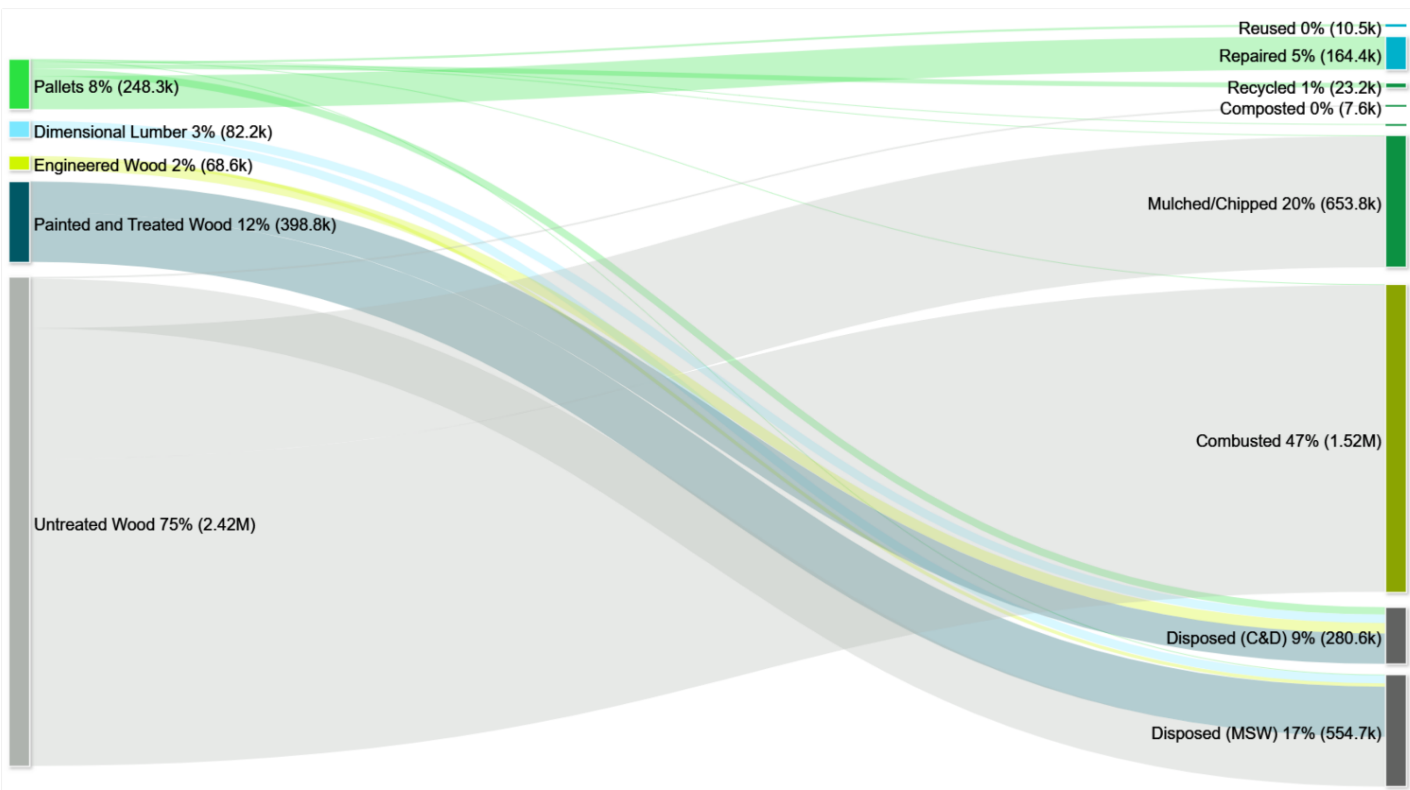
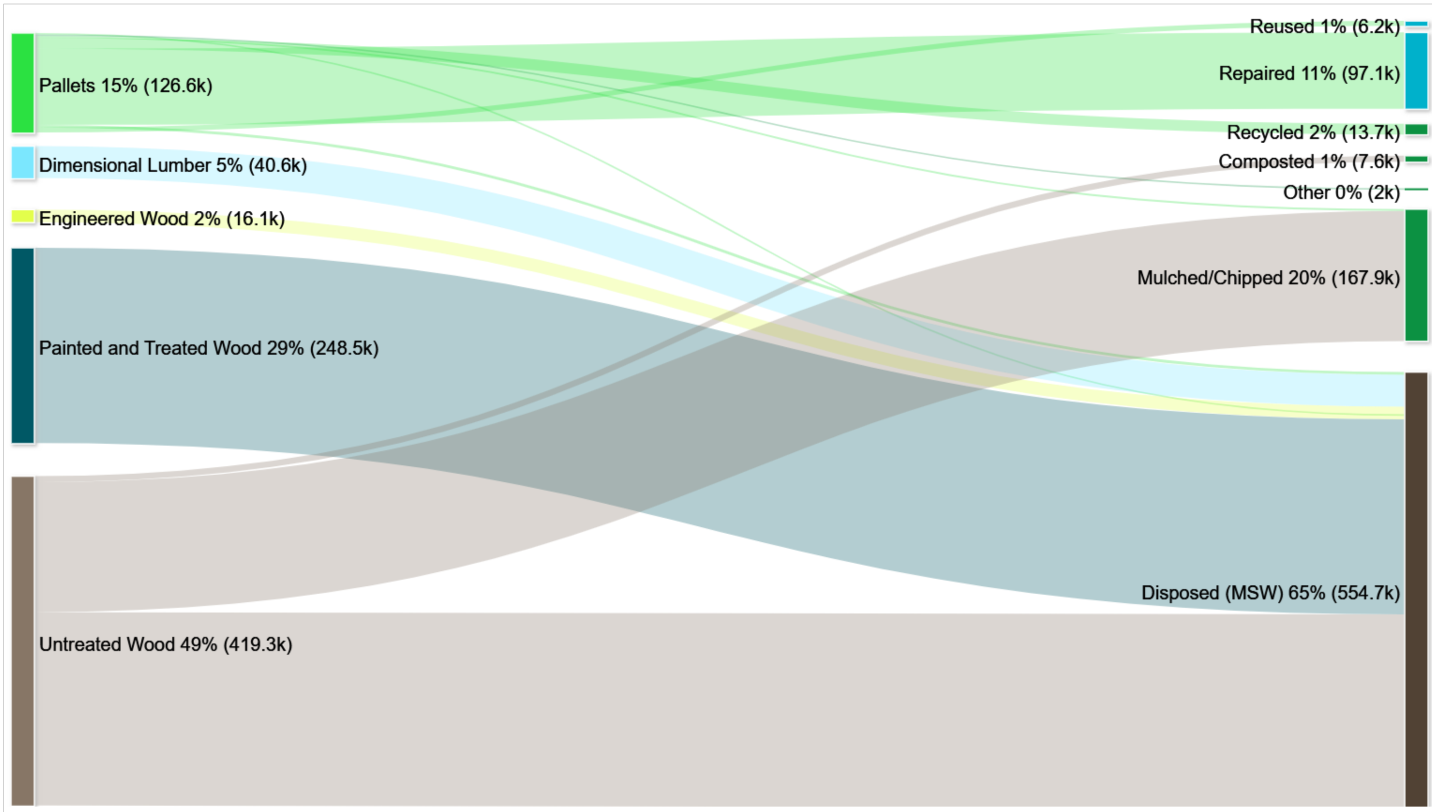


Figure 34: Residential, Commercial, and Institutional Wood Waste Generation Broken Down by Material and Management Method (Tons)



At present, EGLE's recycling data reported under Part 175 may be missing wood waste reuse and recovery. Additionally, the data reported by organics facilities under Michigan Waste Data System (WDS) provides almost no insights into wood waste mulching or chipping, which is likely a large outlet for untreated wood. Figure 34 shows that Michigan may be under counting the recycling market for commercial pallets, amounting to a total of 123,200 tons reused, repaired, recycled, combusted, mulched or chipped, or recycled in another way such as being used as animal bedding. Another estimated 165,600 tons of untreated wood generated in the residential and commercial sector may be going to mulching and chipping across Michigan without that recycling data being captured. The project team cannot say for certain if these materials are left uncaptured as part of the state's data collection efforts due to the proprietary nature of Michigan's recycling data.

Biomass to Fuel

As shown in the above figures, a major difference between the total wood waste management strategies in Michigan compared to management strategies for the RCI sectors (excluding industrial) is that the RCI sectors do not typically utilize combustion as a diversion strategy for wood waste, except for emergency management of storm debris.

Michigan is home to a well-established ecosystem of biomass fuel plants. These plants utilize wood waste as fuel, where the wood is combusted to produce heat to power a boiler, and the steam from the boiler operates a turbine that generates electricity. Typically, these biomass stations are part of the non-regulated electrical generation portfolio that sells its generated power to Michigan's grid through a system of power purchase agreements (PPA's). The power from these plants is dispatched as needed by the utilities to the grid and, as a result, is often on standby.

The financial terms of the PPA and the underlying powerplant structure require fuel to be purchased for prices at approximately $25–$35 per bone-dry ton of hardwood, limiting use to the most cost-effective woodchip supplies for power generation. Furthermore, supply chain costs also require that sources of wood remain within radii that do not exceed 100 miles and frequently target supplies that come within 50 miles of each plant. As a result, these plants focus almost exclusively on supply that is derived from forest residuals (the ~50% of the tree that remains after logging) and forest product residuals (sawdust and slab wood) supplied in the areas surrounding the plants. Most of these areas are in rural Michigan. The costs of sorting and supply chain MSW-derived wood waste flows are simply more than the current purchase price of wood waste fuel and are therefore financially unviable.

Storm Debris

Wood waste is typically generated gradually through routine tree maintenance, forestry activities, or construction and demolition. However, severe weather events, like the ice storm that hit northern Michigan in March 2025, can create a sudden, massive influx of wood debris. The 2025 ice storm caused massive damage over a 12-county region in northwest Michigan's lower peninsula, resulting in a disaster designation by Governor Gretchen Witmer. The aftermath of the storm required local and state agencies to mobilize in order to clear the debris, repurpose the wood as biomass fuel, and restore damaged infrastructure.

In response to the storm, the Michigan Department of Natural Resources (DNR) has opened 17 public debris disposal sites across the disaster area for resident to bring tree debris and vegetative debris to some sites (https://www.michigan.gov/dnr/about/newsroom/releases/2025/04/10/dnr-ice-storm-cleanup-updates?utm_source=chatgpt.com).

Emmet County

Emmet County has mounted a large-scale, multi-agency response to manage and repurpose an estimated 50,000 to 70,000 cubic yards of fallen trees and brush. The storm caused widespread damage, with downed trees across public and private land, prompting the rapid establishment of drop-off sites and partnerships for wood grinding and energy recovery.

To put the amount of wood debris that needs to be managed into perspective, The Emmet County Transfer Station typically processes 2,500 cubic yards of wood waste annually and grinds once every two years. Due to the storm, 5,000 cubic yards are now stockpiled from this one event. In addition, the Little Traverse Conservancy's Offield site holds 7,000–10,000 cubic yards of wood debris, with additional volume expected from smaller satellite sites. Finally, the Emmet County Road Commission has reported between 35,000 and 50,000 cubic yards of debris and continues to receive more.

To manage the additional influx, a contractor has been hired to grind and haul the wood waste under a FEMA-supported agreement. The material will then be sent to biomass facilities including Consumers Power's co-gen plant in Grayling, and potentially as far as Cadillac, as well as alternative outlets like the Packaging Corporation of America in Filer City and a cement plant in Charlevoix.

While Emmet expects reimbursement for the cost of chipping and transporting the wood waste, efforts to secure FEMA reimbursement are reportedly slowed by political complications at the state level.

Looking to the future, Emmet County's response reflects a growing emphasis on resilience and resource recovery. By quickly mobilizing local contractors, leveraging biomass energy infrastructure, and tracking debris volumes, the county has created a model for post-disaster wood waste recovery that emphasizes both sustainability and emergency preparedness.

Table 18 shows an estimated 192,000 tons of wood waste was generated from the storm alone, accounting for 8% of the estimated statewide total untreated wood waste generated annually (see appendix for methodology).

Table 18: Estimated storm generated wood waste from the March 2025 Michigan ice storm

AFFECTED COUNTIES	CUBIC YARDS	TONS
Alcona	72,100	18,000
Alpena	61,100	15,300
Antrim	50,900	12,700
Charlevoix	44,400	11,100
Cheboygan	76,400	19,100
Crawford	59,400	14,900
Emmet	50,000	12,500
Mackinac	109,200	27,300
Montmorency	58,400	14,600
Oscoda	60,500	15,100
Otsego	55,000	13,800
Presque Isle	70,400	17,600
Total	**767,800**	**192,000**

In addition to the trees damaged and downed in the storm, many utility poles were damaged. Over 2,800 utility poles have had to be replaced by Presque Isle Electric and Gas, more than 5 times the normal replacement figure of 500 poles per year. Utility poles are typically treated with creosote to protect against the rotting; however, this treatment renders the poles difficult to recover. As a result, most of the damaged poles are sent to landfills permitted to handle this type of waste.

Material Gap Takeaways

- **High generation:** Approximately 851,100 tons of wood waste are generated annually in the residential, commercial, and institutional (RCI) sectors across Michigan, representing about 8% of total MSW generation.
- **Low recycling rate:** Compared to industrial wood waste recycling, the RCI sector has a much lower recycling rate of 35%, due to fewer collection opportunities and greater stream complexity.
- **Underreporting of diversion:** An estimated 296,400 tons of wood waste are recovered annually from the RCI sector through reuse, repair, recycling, composting, and mulching/chipping. However, much of this recycling may not be captured in Part 175 data or fully reported by composters.
- **High disposal volume:** Roughly 554,700 tons of RCI wood waste are disposed of each year, with the largest volumes coming from painted and treated wood, as well as untreated materials such as scrap lumber, fencing, roofing, and siding.
- **Recovery challenges:** Wood waste is difficult to capture because it typically requires specialized collection efforts, such as comprehensive drop-off programs.
- **Priority action:** Expanding access to wood recovery opportunities through comprehensive drop-offs and strengthening reuse markets could be a key focus for EGLE.
- **Supporting Policy Opportunities**: Driving waste wood supply to a robust Michigan wood recovery and end-market infrastructure would be a positive outcome of supportive diversion-focused local, regional and statewide policies, benefiting both industry and generators of these valuable materials.

DIVERSION PROGRAM ACCESS

SECTION SUMMARY

This section of the report summarizes the single-family and multi-family curbside/onsite and drop-off recycling access update. Additionally, feedback from local governments is summarized highlighting needs and gaps for recycling programs across the state.

Key findings include:

- **Curbside and onsite gaps:** While most single-family residents in Michigan have curbside recycling, major gaps remain. Many rural communities lack curbside service altogether or rely on subscription providers that require extra steps, added cost, and may discontinue service. Fewer than 20% of multi-family residents live in communities that mention onsite recycling with likely much lower actual uptake of services at these properties, highlighting a significant access gap compared to single-family curbside service. Onsite recycling is critical to ensuring recycling services are as convenient for multi-family residents as trash services such that recycling containers are available on the property similar to trash containers.
- **Drop-off recycling shortfalls:** Drop-off sites are essential for ensuring all residents can recycle and for collecting materials not accepted at the curb (e.g., plastic film, electronics, HHW, C&D, bulky waste, scrap metal). Currently, only 63% of residents have access to drop-off sites operated by their municipality or county that accept any recyclables, and only 26% have access to a municipal or county operated drop-off that accepts other recyclables such as scrap metal, bulky plastic, plastic film, or expanded polystyrene.

RECYCLING PROGRAM RESEARCH METHODOLOGY

The Project Team gathered updated recycling program data via research of community web pages. The research methodology was designed to determine access via the experience of a typical resident looking for information on available programs. This access search targeted updates to single and multi-family curbside recycling, recycling drop-off, and food scraps and surplus food curbside and drop-off (represented in the Specific Materials Gaps section).

The researched recycling programs represent approximately 80% of the population statewide. Program factors researched include the program website link, program name, the materials acceptance list, the program service (user) type, single-family or multi-family access. and drop-off service type for drop-off programs. Once the data was collected, the data was reviewed and flagged for any inconsistencies or contradictory information. Flagged programs were re-searched. In the final analysis portion of access searching, the data is reviewed again to ensure that there are no discrepancies between attributes such as the service type and access type, and programs with high-population counts were again reviewed for accuracy. Additionally, programs that were found to have multi-family access or food scraps and surplus food access were manually reviewed to ensure that their program webpages or feedback notes accurately reflected their coverage area and service type.

SINGLE FAMILY CURBSIDE RECYCLING ACCESS & GAPS

Single Family Curbside Access

Curbside recycling access in Michigan is provided through three main approaches:

- **Municipal service:** Recycling services managed directly by a city or county public works department.
- **Contracted service:** Provided by a franchised hauler under agreement with the local government. Recycling services may be automatically provided to residents or residents may still be required to sign up or opt-in to receive a recycling cart or bin.
- **Subscription service:** When residents/property owners in the community must select a hauler operating in the area. The residents/property owners choose to enroll in a subscription curbside recycling program.

Table 19 represents the residential service type by population throughout the state, and Figure 35 displays the curbside service type of each community throughout the state. The most common method of curbside recycling, which covers 54% of the state's population, is provided by contracted or franchised services. About 19% of residents access curbside recycling through subscription-based programs, where households individually opt-in to programs. A total of 7% of the population receives curbside recycling directly from their municipality or local government, while less than 1% fall under other service types which may be hybrid or unspecified program service types. Approximately 20% of the population do not have access to single-family curbside recycling programs.

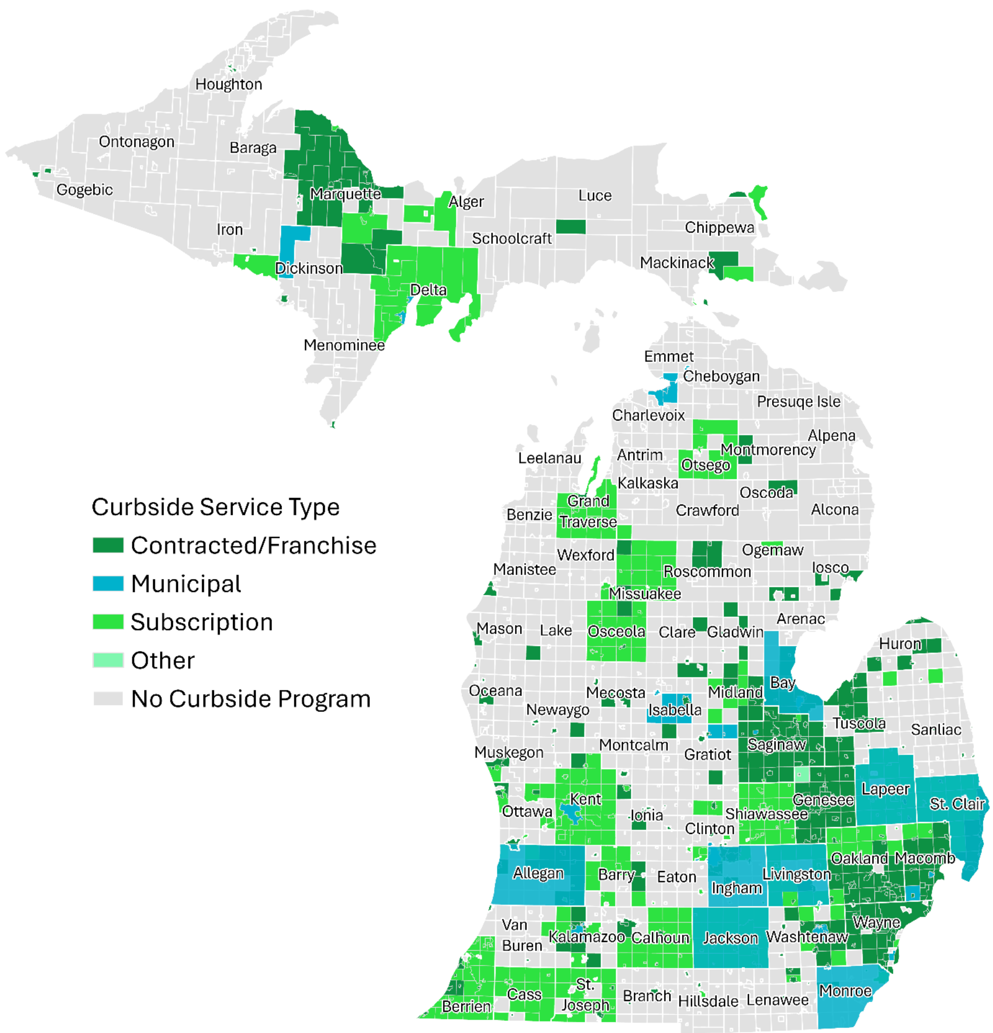
The curbside recycling access levels presented in Table 19 are consistent with the findings of the 2023 Gap Analysis.

Table 19: Curbside Service Type by Population

CURBSIDE SERVICE TYPE	SINGLE FAMILY POPULATION	PERCENT OF TOTAL STATE SINGLE FAMILY POPULATION
Municipal Service	616,159	7%
Contract/Franchise Service	4,420,330	54%
Subscription Service	1,530,250	19%
Other	10,759	<1%
No Program	1,669,277	20%

Note: Population with access includes only single-family residents

Figure 35: Curbside Recycling Service Type



Single Family Curbside Gaps

- **Access remains constant:** Curbside recycling access has made steady progress in Michigan with 80% of Michigan's single-family population having access, but a large share of that access (19%) is provided through subscription services which requires residents to take extra steps to sign up and may come at an additional cost. Investments have been made to improve access including infrastructure grant funding towards projects such as carts provided to 9,600 households in Madison Heights and Frankenlust Township shifting from 18-gallon bins to carts. These represent a mix of new access or improvements to existing access but leave a need for additional curbside access.
- **Low rural curbside:** Curbside access is not available in many rural communities, and where available, it is often a subscription service. However, subscription services are considered a barrier to entry for many residents, as they typically cost significantly more than municipally contracted service, require action by residents to initiate service, and result in lower participation and recycling rates.

MULTI-FAMILY ONSITE ACCESS & GAPS

Multi-Family Onsite Recycling Access

Onsite multi-family programs refer to recycling programs that provide access to multi-family residents that are as convenient to use as trash programs (e.g., onsite recycling carts place near trash dumpsters) and are typically defined as being accessible to properties that have four or more units at each property.

A detailed review of municipal communications to residents about recycling found that 31 communities across Michigan explicitly indicate that onsite recycling access is available for multi-family residents (Table 20), representing 17% of the state's multi-family population. However, municipal communication about program availability is different from verified, on-the-ground access to convenient and clearly communicated recycling services. This analysis did not confirm the extent to which onsite recycling is actually provided at multi-family properties within these communities. In addition, effective onsite access depends not only on the presence of service, but also on factors such as container placement, container capacity, collection frequency, and regular, consistent communication to residents about how to participate. The number of multi-family properties in Michigan that meet these conditions is unknown, but it is almost certainly lower than 17%, demonstrating that multi-family recycling access and uptake of services is a gap in the state of Michigan.

Table 20: Multi-family Curbside Recycling Access

COMMUNITY COUNT	MULTI-FAMILY POPULATION OF COMMUNITIES (5+ UNITS)	PERCENT OF STATE MULTI-FAMILY POPULATION
31	316,861	17%

Multi-Family Onsite Recycling Access Gaps

- **Low multi-family access:** Only 17% of multi-family residents live in municipalities that address multi-family recycling services on their community recycling pages. Actual uptake of onsite recycling services at multi-family properties in Michigan is unknown, but it is most certainly lower than 17%.
- **Limited information:** Many community recycling pages do not mention multi-family recycling, leaving residents with little access to program details.

DROP-OFF RECYCLING ACCESS & GAPS

Drop-Off Recycling Access

When onsite recycling programs are unavailable or underdeveloped at multi-family properties due to limited infrastructure or lack of program support, off-site drop-off recycling programs may become the only practical pathway for residents to participate in recycling. Although drop-off programs require residents to take additional steps to collect and transport materials, they still provide an important opportunity for residents of multi-family properties to engage in recycling and contribute to statewide recycling goals.

Drop-off recycling programs also serve as a critical access point for single-family households that technically have curbside recycling available, but only through an added cost of subscription service. In these cases, drop-off programs can provide a more accessible alternative for households that might otherwise go without recycling service.

In addition, drop-off programs can expand the range of materials residents can recycle beyond what curbside programs typically accept. Many drop-off programs in Michigan are designed to collect traditional recyclables such as corrugated cardboard, newspaper, magazines, glass bottles and jars, aluminum and steel cans, plastic bottles, jugs, and tubs. However, drop-off programs can also provide a higher level of service by accepting materials that are often excluded from curbside collection because they are not compatible with MRF processing, such as plastic film, bulky plastics, scrap metal, furniture, mattresses, and C&D debris. As a result, drop-off access is important not only for filling gaps where curbside or onsite service is unavailable, but also for providing recycling opportunities that would otherwise not exist. Without pathways to divert these harder-to-recycle materials, Michigan will be unlikely to achieve its 45% recycling goal.

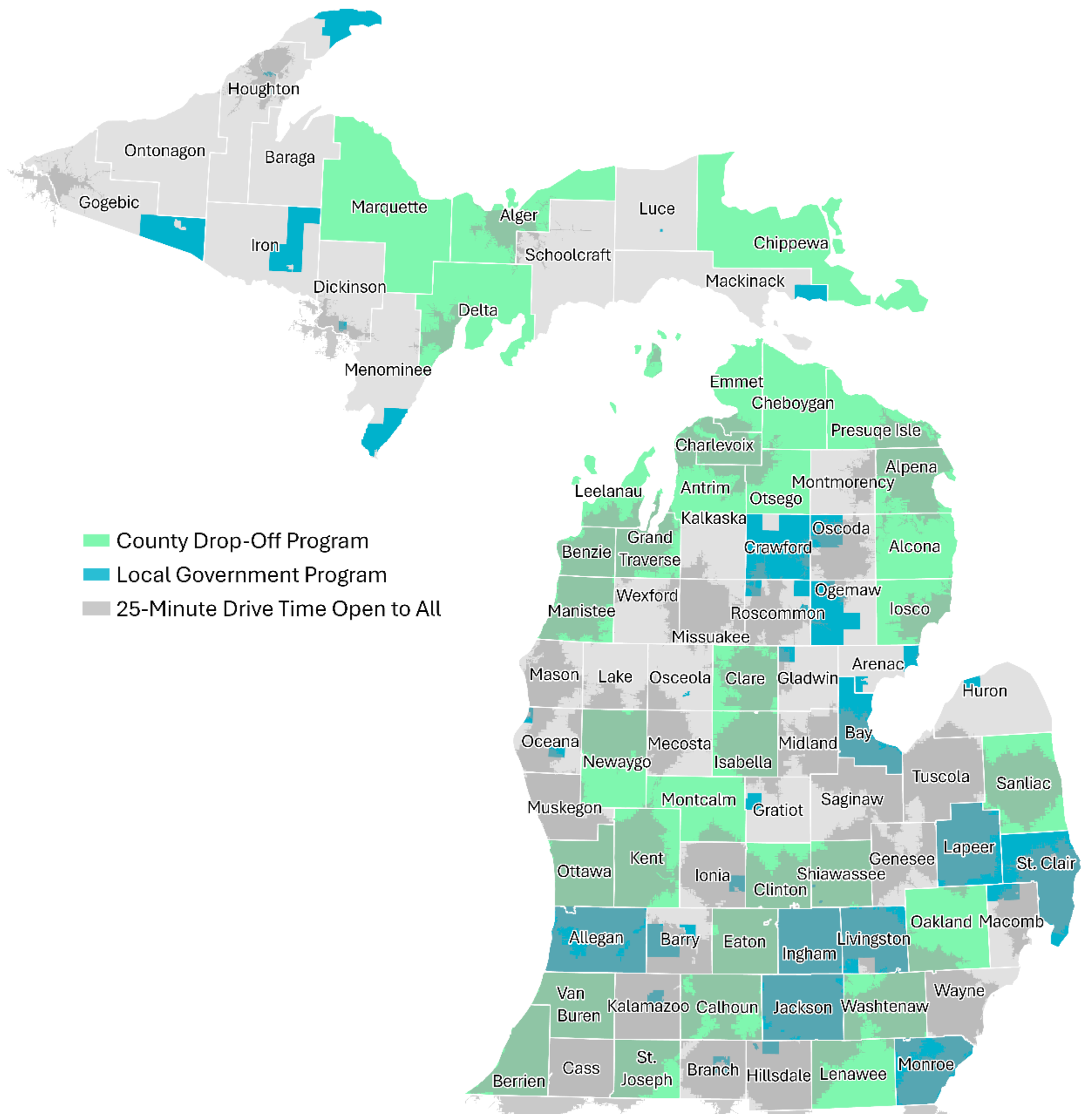
Drop-off programs can be structured and operated in several different ways. Some are operated at the county level and are open to all county residents, while others are run by local governments, such as townships or municipalities, and are limited to their own residents. Drop-off programs may also be privately owned and operated, in which case they are often open to a broader range of users. Publicly funded sites, supported through taxes or local fees, are typically restricted to residents who pay into the system. In contrast, privately operated facilities are more likely to charge user fees by vehicle, load, or item, allowing access to anyone willing to pay for the service.

Table 21 and Figure 36 present drop-off access to county-wide programs, local government run programs, and “open to all” facilities where any user of any community can access the site. A total of 42% of Michigan residents have access to a county-run drop-off program, and another 28% have access to a local government-operated program. In addition, 72% of Michigan residents live within a 25-minute drive of a drop-off site that is open to the public, though these sites are likely to charge a fee for some or all materials that may hinder participation. While this access level appears strong, it still leaves 18% of Michigan residents without access to any drop-off recycling program.

Table 21: Drop-off Program by Population

PROGRAM TYPE	POPULATION WITH ACCESS	PERCENT OF STATE POPULATION
County Program	4,279,772	42%
Local Government Program	2,815,115	28%
Total with county or local government program access	6,394,276	63%
Total within 25 minutes open to all area (private sector)	7,218,969	72%
Total with no drop-off access	1,790,320	18%

Figure 36: Residential Access to Drop-Off Recycling Program



Drop-off Program Material Acceptance

Like access to curbside service, access to a drop-off recycling site represents only one component of a robust recycling system in Michigan. The materials accepted at the drop-off sites described above vary widely. Some sites are limited to high-value commodities such as corrugated cardboard, office paper, aluminum and steel cans, and #1 and #2 plastic bottles and jugs, while others provide far more comprehensive recycling services that go beyond the materials typically accepted through curbside programs.

Some drop-off sites also accept other recyclables that are generally excluded from curbside programs as they are not MRF-compatible, including scrap metal, bulky rigid plastics, plastic film and bags, and expanded polystyrene (commonly known as Styrofoam). Access to drop-off collection for these materials is particularly valuable because there is no other pathway to divert these materials from disposal. Where available, drop-off acceptance of these materials helps to fill critical gaps in the recycling system and provides residents with more complete options for non-traditional recyclables.

Only 26% of Michigan residents can recycle at least one other material, as listed above, at a county or municipally operated drop-off (Table 22). This table does not account for privately run drop-off opportunities such as scrap yards or retail plastic bag takeback programs. However, private sector drop-offs cannot reliably fill this gap. Market conditions dictate private participation, meaning these options can emerge or disappear quickly as markets fluctuate. For other materials such as bulky plastics and expanded polystyrene, private investment in recycling is unlikely to ever be widespread as commodity values are lacking. Hard-to-recycle materials such as electronics, HHW, and C&D debris face similar limitations. Moreover, participation in recycling programs is driven by convenience. If recycling these materials requires multiple trips to different locations, fewer residents will participate. With adequate support and funding, municipal and county programs could fill this critical gap by offering comprehensive, one-stop drop-off opportunities for both non-traditional recyclables and hard-to-recycle materials, increasing convenience and participation statewide.

Table 22: Drop-Off Access for Hard-to-Recycle Materials

POPULATION WITH COUNTY OR LOCAL GOVERNMENT PROGRAM	POPULATION	PERCENT OF STATE POPULATION
Drop-Off Accepts Other Recyclables	1,660,441	26%

Drop-Off Recycling Access Gaps

- **Missing access:** Drop-off sites are critical for providing recycling options to all residents and for collecting materials not accepted at the curb (e.g., electronics, HHW, C&D debris, bulky items, scrap metal). Many Michigan counties still lack county-run or locally run drop-off programs. With only 26% of the state's population having access to a drop-off center taking additional material types, the establishment of additional drop off centers accepting a full range of recyclable materials continues to be a high priority to reach a 45% recycling rate.

LOCAL GOVERNMENT INSIGHTS

Through targeted interviews with EGLE regional representatives, key counties representing varying population types across the state, haulers and processers, the Project Team gathered updated information on recycling and organics recycling programs as well as recycling, organics, scrap metal, batteries, and wood waste processing sites across Michigan. The purpose of this effort was to build a more accurate and current understanding of access to material recycling infrastructure across the state, and to gather feedback and insights in helping identify data gaps, validate existing information, and explore opportunities to improve access to recycling food scraps and surplus food collection services in Michigan communities.

The counties and authorities representing varying population types across the state interviewed included Delta County, Emmet County, Kent County, Oakland County, RRRASOC, Van Buren County, and Washtenaw County. Other counties were contacted but did not respond to the interview request.

Local Government Feedback on Curbside Programs

Counties and authorities were asked about community curbside recycling and organics programs available in their communities including type of service provider (e.g., municipal, franchised, subscription), eligible households (e.g., single-family and/or multi-family), container type, and accepted materials. Updates to curbside program access data are included in the Access section, such as the addition of curbside recycling, food scraps, and surplus food programs by communities or haulers; changes in commonly accepted materials; and transitions from municipally provided or contracted programs to subscription-based services.

Key takeaways:

- **Service by population:** Larger cities often operate municipal single-stream programs, while smaller or rural areas rely on subscription services. Subscription services require residents to take extra steps to sign up, can come at an additional cost, and can be dropped by the provider at short notice.
- **Multi-family limits:** Some municipal curbside programs serve multi-family units only by request (e.g., Emmet County), while others do not offer service at all (e.g., RRRASOC) leaving a large multi-family gap reflected in the multi-family onsite analysis presented above.
- **Cardboard only:** Subscription haulers may serve multi-family and commercial customers but often provide recycling only for cardboard.
- **Service cuts:** Some municipalities have discontinued curbside recycling (e.g., Pontiac), leaving residents with only subscription services, and in some cases haulers now provide trash service only.

Local Government Feedback on Drop-Off Programs

During the interviews, counties and authorities were asked about community recycling and food scraps and surplus food drop-off programs available in their communities including program type (county program, residents only, specified residents, private open to all, other), location, and accepted materials. The responses regarding food scraps and surplus food access are incorporated in the food scraps and surplus food section of this report.

Key takeaways:

- **Accurate listings:** Communities stressed the importance of listing all available drop-off centers to meet benchmark recycling standards, ensuring access for residents without curbside service (e.g., apartment dwellers, rural areas).
- **Location matters:** Facility location within a region is critical, as distant sites may be inaccessible due to transportation constraints. Additionally, locations along common transportation routes or near shopping centers provide greater convenience.
- **Free to residents:** County-operated drop-offs funded by millage that are free for residents; businesses and non-residents may pay small fees (e.g., Delta County) are preferred to ensure equitable access.
- **Material variation:** Accepted materials vary across county, township, and privately operated sites (e.g., plastic bags, EPS, mixed rigid plastics) causing confusion to residents.
- **Special events:** Some communities offer rotating or event-based drop-offs (e.g., Pontiac's weekly Saturday collections), which could count as access.
- **Lost sites:** One Oakland County drop-off location was permanently closed due to contamination issues.

INVESTMENT IMPACTS AND CONTINUED SUPPORT

SECTION SUMMARY

This section highlights EGLE's investments in Michigan's recycling and circular economy since the establishing of the Renew Michigan Fund in 2018. From 2019-2025, over $50 million in EGLE grants supported recycling market development, recycling infrastructure, and circular economy projects. Additional investment tracked towards Michigan's circular economy demonstrates $443 million in combined state, private and federal investment.

Key findings include:

- **2022 Investment Surge:** A notable acceleration of investment in 2022 with $12.7M in EGLE grants awarded, double the prior year, with subsequent years adding new funding pathways to the NextCycle program offerings including Circular Economy Grants, Harvest Grants, and Seed Awards.
- **Priority Gaps:** EGLE investments have focused on priority gaps such as market development, food rescue and diversion, expanded access including support of comprehensive drop-off recycling programs, and a growing number of reuse solutions.
- **Tracked Investment:** Across Michigan nearly 400 projects are tracked totaling $443 million in investment towards the circular economy from both public and private entities.
- **Case Studies:** Success stories from communities across Michigan where EGLE grant investments have translated into real diversion gains, from rural mobile recycling programs and urban drop-off sites to food scraps composting in the Upper Peninsula and AI-powered sorting technology at Michigan MRFs.

SUMMARY OF EGLE GRANT FUNDING 2019 THROUGH 2024

From 2019 to 2024, nearly 300 EGLE-supported grants and projects focused on recycling and circular economy received $209 million in combined public and private investments, of which EGLE contributed over $60 million. Projects supported by EGLE grant funding are among nearly 400 tracked projects across the state totaling $441.8 in tracked investment. This funding includes tracking from recycling market development, grants, recycling infrastructure grants, circular economy grants, funding for NextCycle MICROS projects, NextCycle Seed Awards, organics infrastructure grants, critical minerals grants, and NextCycle Harvest Grants. These EGLE funded programs are all tracked based on targeting specific gaps in recycling and the circular economy. EGLE funds other important programs such as scrap tire grants and electronics collection, as well as recycling grants which are not currently tracked as part of the total investment.

EGLE funded projects address a broad range of priorities aimed at advancing Michigan's circular economy, improving waste diversion infrastructure, and increasing the overall efficiency and sustainability of materials management systems. The largest portion of the funding, accounting for 35% of total funds, was directed toward Recycling Processing and End Markets, largely driven by prioritized gaps in these areas considered critical for increasing volumes recovered. Increasing investments in organics projects represent EGLE's growing commitment to fund solutions for organics recovery.

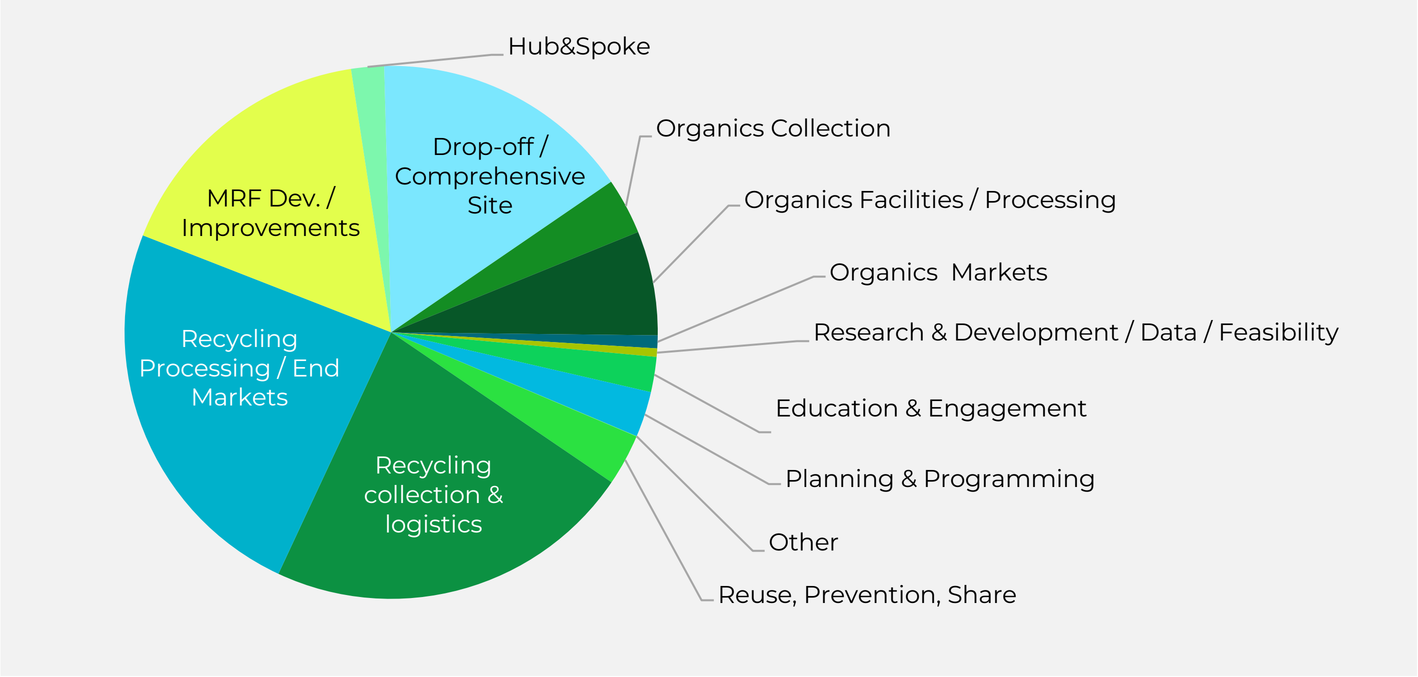
Tracked projects across the state can be explored from the NextCycle Michigan website project investment data visualization. From Southeast Michigan to the Upper Peninsula, projects to build a stronger circular economy demonstrate the variety of solutions which are possible, collaborative approaches, and real investment going towards solutions. EGLE specific grant investment tracking can be found through the outgoing grants and loans dashboard.

Table 23: Summary of Tracked Projects 2019 through 2024

PROJECT FOCUS	# OF TRACKED PROECTS	EGLE INVESTMENT	TOTAL INVESTMENT
Reuse, Prevention, Share	32	$ 1,630,373.00	$ 2,525,373.00
Recycling Collection & Logistics	75	$ 11,629,072.00	$ 23,080,186.00
Recycling Processing / End Markets	64	$ 21,445,277.00	$ 304,530,468.00
MRF Dev. / Improvements	32	$ 8,627,855.00	$ 68,649,633.00
Hub&Spoke	9	$ 1,032,447.00	$ 1,213,885.00
Drop-off / Comprehensive Site	34	$ 8,212,961.00	$ 10,415,070.00
Organics Collection	40	$ 1,790,868.00	$ 2,881,713.00
Organics Facilities / Processing	39	$ 3,283,659.00	$ 4,265,591.00
Organics Markets	10	$ 424,189.00	$ 657,036.00
Research & Development / Data / Feasibility	19	$ 275,660.00	$ 10,627,676.00
Education & Engagement	16	$ 1,108,187.00	$ 11,336,187.00
Planning & Programming	17	$ 1,445,956.00	$ 1,622,020.00
Grand Total	388	$60,906,504.00	$441,804,838.00

Note: 2025 Grant funding is not currently represented in this table, but [$11.8M was announced by EGLE](#).

Figure 37: Summary of EGLE Funding



INDUSTRY INVESTMENT

Beyond EGLE investment, organizations around Michigan are prioritizing investment in solutions. $380.9 million of investment from private and public entities around the State is tracked directly tied to tangible projects which address gaps in Michigan's recovery and recycling system.

Notable private investments include:

- Graphic Packaging's' announcement in 2022 of a new paperboard production line at their facility in Kalamazoo able to utilize recycled material. This project allows for more material to be repurposed in Michigan, adding to the circular economy.
- WM Detroit MRF announced in 2022 is a $35 million investment into a state-of-the-art facility to process 40 tons per hour of recyclable material. Supported by an EGLE Recycling Market Development grant, this investment adds capacity in Southeast Michigan.
- ACI Plastics Recycling Facility in Flint is a $10 million investment, also supported by an EGLE Recycling Market Development Grant, to create jobs and recover plastics to feed back into Michigan manufacturing.

Investment tracking includes accounting for grant awards, known budgets for projects, sponsored awards, reported private investments from NextCycle teams and publicly announced investments. Tracking for all projects is limited to what is reported and does not account for all match, changes in projects, or unreported investments. From initial reporting in 2023, overall investment tracking has been adjusted to account for abandoned investments such as the $280 million planned Anergia commitment towards the Kent County Sustainable Business Park and unpursued expansion of the now closed Great Lakes Tissue facility. While gaps in tracking exist, the overall investments tracked demonstrate the momentum towards investment in a circular economy and progress towards a $1 billion capex investment calculated initially in 2017 as part of the Governors Recycling Council Report and refreshed in 2021 to place a target investment need aligned with Michigan's 45% recycling rate goal. Tracking has expanded to account for broader solutions beyond just traditional recycling efforts to include reuse, upstream, and circular economy solutions.

Additionally, support for circular economy projects includes funding from the Economic Development Administration Build to Scale program which awarded $813,330 to Centrepolis Accelerator at Lawrence Technological University matched 1:1 by the State of Michigan to support $1,626,660 going towards technical support of both public and private circular economy projects. Over the course of the three-year grant period, 24 public sector projects represented 162 communities were completed, supporting the advancement of their circularity solutions, and 64 private sector projects received technical assistance and entrepreneurial support to advance their work. These efforts add to the $441.8 in tracked project investment.

INVESTMENT SUCCESS CASE STUDIES

The investment figures presented above represent real programs taking shape in communities across Michigan. Many of the grant projects that have secured funding over the last few years are still ongoing, but these initiatives are already producing positive impacts. From rural counties expanding recycling access through mobile collection, to cities building comprehensive drop-off sites that accept materials well beyond what curbside programs can collect, to Upper Peninsula composters diverting food scraps from landfill, EGLE's investments are translating into tangible infrastructure, expanded access, and measurable diversion gains. The case studies below highlight a sample of grantees whose work reflects both the diversity of Michigan's recycling and organics landscape and the impact that targeted public investment can have when paired with strong local partnerships.

City of Holland: Increasing Access to Curbside Recycling

2020 Recycling Infrastructure Grant | Grant Award: $267,646 | Total Project: $512,514

Summary

In 2020, the City of Holland used a Recycling Infrastructure Grant from EGLE to transition its residential curbside recycling program from a 32-gallon bag system to 96-gallon single-stream carts. All 9,300 households in the city received new carts capable of holding more than 300 pounds of recyclables. The Recycling Partnership provided educational materials, resident communications, and program metrics to support the transition. The program is funded through a residential user fee collected by the City's Department of Public Works, with an enterprise fund of $1.2 million established to cover any unforeseen expenses.

Outcome

Recyclables collected per household nearly doubled, rising from 71 pounds to 169 pounds annually, while solid waste shipped to landfill dropped from 7,000 tons to 3,800 tons per year. Holland's success with this grant has continued to generate momentum. Through EGLE's Recycling Quality Improvement Grant, the city partnered with The Recycling Partnership to reduce contamination rates in its recycling stream. Most recently, Holland received a 2024 Recycling Infrastructure Grant through NextCycle Michigan to develop a comprehensive drop-off site and earned both the "Best in Show" and "People's Choice" awards at the NextCycle Michigan Pitch Showcase.

Calhoun County: Mobile Drop-Off Recycling for Rural Access

2020 Recycling Infrastructure Grant | Grant Award: $37,944 | Total Project: $47,430

Summary

Calhoun County used a 2020 Recycling Infrastructure Grant to purchase a truck and trailer to bring recycling access directly to rural households and public events across the county. The program was designed to address gaps in curbside service in less densely populated areas where traditional collection infrastructure is not economically viable. The program is supported through a community host agreement with the C&C Landfill. Implementation was complicated by the COVID-19 pandemic, which introduced labor shortages, supply chain delays, and reduced participation during the program's early stages.

Outcome

Despite these challenges, the program expanded drop-off service to three mobile collection centers operating three hours per day, three days per month. Between late 2022 and 2023, the mobile unit also participated in 11 community events, collecting water bottles, cardboard, and polystyrene. In total, the program diverted an additional 86 tons of material from disposal, demonstrating the potential of mobile collection to meaningfully fill access gaps in rural communities.

Partridge Creek Compost: Food Scraps Diversion in the Upper Peninsula

2022 Recycling and Organics Infrastructure Grant | Grant Award: $180,000 | Total Project: $225,750

Summary

Partridge Creek Compost (PCC), a general-permitted composting facility in Ishpeming in Michigan's Upper Peninsula, received a 2022 Recycling and Organics Infrastructure Grant to purchase food scraps diversion equipment and supplies, including hydraulic tipping trailers, a tractor, a dump truck, and 100 curbside bins. The grant enabled PCC to establish a food scraps collection partnership with Northern Michigan University's dining hall and to launch a curbside food scraps pickup program for residents, local restaurants, and educational institutions. PCC's funding model draws on hauling contract fees, tipping fees, compost sales, and community partnerships, supplemented by additional support from the USDA, the Marquette County Conservation District, and the Americana Foundation.

Outcome

By March 2025, the program was serving more than 320 households and an estimated 9,500 individuals across the region. PCC has continued to grow through engagement with NextCycle Michigan, where the team is working to address three interconnected regional challenges: growing more food locally, diverting compostables from landfill, and producing compost to build farmland in an area where the Upper Peninsula imports 90% of its food and 60% of Marquette County's waste stream is compostable. In 2024, EGLE awarded PCC an additional $144,000 to purchase a front-end loader and trommel screener to increase processing capacity and support continued expansion of curbside and drop-off services.

City of Lansing: Urban Comprehensive Recycling Drop-Off Site

2022 Recycling Infrastructure Grant | Grant Award: $490,917 | Total Project: $618,647

Summary

The City of Lansing used a 2022 Recycling Infrastructure Grant to develop a comprehensive recycling and food scraps drop-off site, featuring roll-off containers, signage, and a covered storage area. The City funds its recycling and yard waste programs through a fee of $117 per year assessed to single-family households.

Outcome

The site opened in September 2025 and accepts cardboard, boxboard, paper, plastic, metal, glass, polystyrene, and food scraps, operating daily from dawn to dusk. The Lansing site demonstrates how grant funding can help municipalities establish comprehensive drop-off infrastructure that goes well beyond traditional curbside recyclables, providing residents with a single, convenient location to manage a broad range of materials that would otherwise enter the disposal stream.

Glacier Technology: AI-Powered Robotic Sorting at Michigan MRFs

2023 Recycling Market Development Grant | Grant Award: $367,000 | Total Project: $735,800

Summary

Glacier Technology received a 2023 Recycling Market Development Grant to install AI-powered robotic sorting systems at two Michigan material recovery facilities — SOCRRA and RRRASOC, both in Oakland County, which collectively serve 1.2 million people. The project was designed to demonstrate the performance and scalability of Glacier's robotic sorters and AI analytics systems on live recycling lines. Glacier's engagement with Michigan began through the NextCycle Michigan Recycling Innovation and Technology Accelerator Track in 2022, where the company worked with industry experts to test and refine its technology, emerging from stealth mode following the cohort and receiving national attention and investment.

Outcome

At SOCRRA, the robot was deployed on the container line, while at RRRASOC it operated on the residue line. AI analytics identified operational efficiency improvements at both facilities, and Glacier trained site staff to independently service the equipment going forward. Glacier was recognized with a Pitch Award at the 2022 NextCycle Michigan Pitch Showcase, reflecting the potential of AI-enabled sorting technology to improve recycling quality and efficiency at scale across Michigan's MRF network. Since, Glacier has expanded their reach in the state, working with other working with other Michigan MRFs.

POLICY REVIEW AND CONSIDERATIONS

SECTION SUMMARY

This section of the report provides an overview of current policy on organics, EPR policy as it relates to compostable packaging and electronics, and the bottle deposit system. The policy recommendations address high-priority policy areas identified by EGLE and encompass a wide breadth of divertible materials, including food scraps and surplus food, compostable packaging, electronics, and consumer beverage packaging. The policy considerations target more than a quarter of divertible MSW material with opportunities focused on building successful models from other states and communities with proven success. There is a trend towards harmonizing policies under EPR with recent enactments to standardize nationwide requirements.

Key findings include:

- **Organics Policy Considerations**
    - » **High generation:** Organics including food scraps and surplus food is the largest portion of Michigan's MSW stream, making up 37% of residential waste and 38% of commercial waste.
    - » **Zoning and permitting barriers:** Small- and mid-sized composting facilities are often not explicitly permitted land use under many local city zoning laws and ordinances and are not covered by the Right to Farm Act.
    - » **Zoning actions:** Communities could update ordinances using tools like the U.S. Composting Council's Model Zoning Template and Guidelines to reduce permitting burdens and support local composting.
    - » **Funding mechanism & education needs:** Resident education is critical for food scraps and surplus food reduction but requires sustained funding to remain effective. Adjusting landfill tipping fees can both discourage landfilling and generate dedicated revenue for education and outreach. Alameda County, CA provides a successful example.
    - » **Access and collection:** Municipalities could be encouraged or required to provide food scraps and surplus food collection, either commingled with yard waste or through drop-off programs, following examples like Washington's HB 1799.
    - » **Policy alignment:** If considering a diversion mandate or landfill ban, Michigan could align with California and Vermont, where source separation is required for most generators without tonnage thresholds.
    - » **Practical tools:** NRDC's Tackling Food Waste in Cities: A Policy & Program Toolkit offers a framework of foundational, ambitious, and transformative strategies that communities can adopt based on their capacity and progress.
- **EPR Policy Considerations – Compostable Packaging**
    - » **Current policy gap:** Compostable packaging is recognized under all seven U.S. packaging EPR laws, but many states (e.g., Maine and Oregon) focus narrowly on recycling, leaving compostable packaging categorized as non-recyclable and unsupported by infrastructure funding.
    - » **Foundational requirement & needs assessments:** Needs assessments identify strengths and gaps in state materials management systems, guiding data-driven EPR design and resource allocation. To effectively integrate compostable packaging into EPR frameworks, both packaging and composting infrastructure must be included within program scope and evaluated through statewide needs assessments.
    - » **Opportunities for Action:** Conduct a comprehensive EPR needs assessment before considering implementation to ensure compostable packaging and supporting infrastructure are fully integrated and aligned with state-specific recycling goals.
- **EPR Policy Considerations – Electronics**
    - » **Current landscape:** 23 states and DC have electronics EPR laws, using either convenience standards (broad access requirements) or performance standards (weight-based targets).

- » **Program models:** Convenience standards (Oregon, Washington, Vermont) ensure widespread collection access and pooled manufacturer funding; performance standards (Illinois, Minnesota, New York, etc.) assign weight-based obligations to manufacturers. Michigan's law is based on a take-back model, where manufacturers are responsible for offering convenient, cost-free returns.
- » **Effectiveness:** Convenience models are considered more stable, with stronger infrastructure, transparent cost allocation, and reliable year-round service.
- » **State examples:** Oregon and Washington demonstrate successful convenience programs, each providing free statewide access and collecting millions of pounds of electronics annually.
- » **Opportunities for Action:** Consider expanding covered devices and transition from non-binding performance targets to a convenience standard model with a coordinating body to ensure equitable, statewide collection and sustainable funding.

- **Bottle Deposit System Considerations**
  - **Expand Covered Containers:** Consider supporting the expansion of redemption eligibility to more beverage containers to boost participation by increasing users' opportunities for redemption.
    - » **Adopt Universal Redemption:** Consider implementing universal redemption that requires retailers to redeem all deposit containers, regardless of whether that brand is sold there.
    - » **Reallocate Unredeemed Deposit Use:** Consider redirecting a portion of unredeemed deposits toward recycling system improvements, restructuring funding in this way could ease the operational burden on redemption points and open the possibility for redemption points to increase access and convenience for residents. Consider including MRFs as eligible for redemption to recognize their increasing share and burden of handling unredeemed containers since disruption of services during Covid.
    - » **Boost Convenience & Infrastructure:** Consider providing residents with multiple options for redemption such as reverse vending machines, hand counting, and bag-drop systems to lower barriers to participation and increase redemption rates.
  - **Update the Deposit Value:** Michigan's deposit has remained at 10¢ for decades and has eroded substantially in purchasing power. Evaluate whether the deposit could restore its motivational impact. A careful evaluation of inflation effects is needed.
    - » **Improve Transparency & Public Engagement:** Public trust is essential to participation in the process. Continual education and outreach on the benefits of Michigan's redemption system, enhanced system transparency, and exploring community-based redemption initiatives such as donation bins and local champions to support redemption.

ORGANICS POLICY REVIEW AND CONSIDERATIONS

Organics including food scraps and surplus food, yard waste, and compostable paper and packaging, make up 2.5M tons of disposal; diverting another 1.0M tons is needed to reach the 45% goal.

Impact of Removing Zoning and Permitting Barriers for Food scraps and Surplus Food Composting

Under Michigan's Part 115, Solid Waste Management Act, counties and regions are required to develop and maintain approved Material Management Plans (MMPs) that prioritize waste reduction, reuse, recycling, and organics recovery before disposal. MMPs can identify food scraps composting as a critical strategy to increase materials recycled and reduce landfill reliance. However, outdated or restrictive local zoning and permitting requirements can often prevent the development of composting infrastructure needed to implement these plans, creating a disconnect between state-approved planning goals and local land-use regulations. Removing these barriers is essential to increasing food scraps recovery and achieving the state's 45% recycling rate goal.

Local Zoning Challenges and Opportunities

Detroit's Urban Agriculture Ordinance, adopted in 2013, serves as a leading example of how local governments can proactively modernize zoning codes to support composting and food scraps recovery as part of broader sustainability, food system, and public health goals. The ordinance clearly defines land uses for urban farms and gardens, and importantly, permits composting as an accessory use. Urban farms are allowed to accept organic waste from off-site sources, compost it on-site, and use or redistribute the finished product at no cost.

This framework has enabled community organizations, nonprofits, and entrepreneurs in Detroit to process food scraps and yard waste locally, support soil health, and reduce reliance on landfills, all without being subjected to the same zoning or permitting requirements as large-scale solid waste facilities.

By contrast, many other Michigan cities and townships lack specific zoning language that permits composting or urban agriculture. In these communities:

- Composting may fall under undefined or industrial land uses, requiring special use permits or even being prohibited by default.
- Local ordinances may not allow off-site materials to be used in composting, eliminating access to food scraps—a critical nitrogen source for effective composting of yard waste and leaves.
- Setback, lot size, or zoning district restrictions often make small-scale composting impractical, especially in urban areas.

Many local zoning codes do not explicitly recognize composting as a permitted land use. Without this designation, composting operations can be regulated under the same standards as landfills or incinerators, making them difficult or impossible to site. Part 115, and its associated administrative rules, recognize differences in scale, feedstocks, and environmental risk among composting operations. However, many local zoning ordinances do not reflect these distinctions, instead applying standards designed for large-scale solid waste disposal facilities to small- and medium-scale composting operations. This regulatory mismatch can result in unnecessary permitting hurdles, increased costs, or outright prohibitions for community-based composting activities that pose minimal environmental risk and align with state materials management priorities.

Why Local Zoning Still Matters—Even with the Right to Farm Act

It's a common misconception that Michigan's Right to Farm Act (RTFA) removes local barriers to composting. While RTFA offers strong protections for on-farm composting, its scope is limited:

- It only applies to farm operations operating under Generally Accepted Agricultural and Management Practices (GAAMPs).
- Composting must be part of the farming operation, typically for on-farm use.
- Most community-scale, nonprofit, or urban composting projects—such as school gardens, neighborhood drop-off sites, or small businesses—do not qualify for RTFA protections.
- Even operations that do qualify can still be delayed or challenged under local zoning ordinances if composting is not clearly defined or permitted.

While the RTFA provides important protections for certain agricultural composting activities, it does not address the broader range of community-scale and urban composting operations possible under many Material Management Plans developed pursuant to Part 115. As a result, local zoning ordinances remain the primary determinant of whether composting infrastructure can be sited and operated in much of Michigan. Modernizing zoning and permitting frameworks is therefore a critical step toward achieving the state's materials recycling goals and fully implementing Michigan's materials management planning system. Some Michigan municipalities have taken proactive steps to address these barriers. For example:

- **Cascade Township, Michigan**
    - » The zoning code was amended in 2024 (with support from EGLE and NextCycle Michigan) to allow food scraps composting and vermiculture, with clear guidance on site planning and operations. This enabled the development of Wormies, a local composting business, and demonstrated how zoning updates can support environmental entrepreneurship.
- **Detroit, Michigan**
    - » As noted above, Detroit continues to lead by example with its ordinance allowing composting as an accessory use under its Urban Agriculture framework and currently working (with support from NextCycle Michigan) in amending its City Zoning code to explicitly allow very small "exempt" (under 500 cubic yards) compost sites to operate in the City as long they do not create a nuisance. Greater than 500 cubic yard compost sites will follow siting and operating requirements that align with EGLE's Part 115 rules.

Additionally, communities outside of Michigan provide further examples of reducing barriers to food scraps composting:

- **San Diego County, California**
    - » Implemented a streamlined zoning ordinance (Section 6977) in 2022 that reduced permitting burdens for small-scale composting, replacing an expensive and lengthy use-permit process with a low-cost Zoning Verification Permit for facilities processing up to 100 cubic yards.
- **Montgomery County, Maryland**
    - » Amended its code to allow farmers to compost more off-site material.
- **Baltimore, Maryland**
    - » Created new zoning categories in 2018 for urban agriculture and allowed on-site composting of off-site materials with free redistribution, much like Detroit's approach.

These examples illustrate that zoning reform is a critical enabler of food scraps recycling. They show how composting can be decoupled from industrial waste infrastructure and integrated into local food systems, climate goals, and job creation strategies.

Moving Forward: Tools and Recommendations for Michigan Communities

To meet Michigan's 45% recycling rate goal, an additional 100 medium and 40 large-sized organics facilities are needed to process 1 million tons of additional material. To achieve this, communities can:

- Define composting as permitted or accessory use in agricultural, residential, and mixed-use zones
- Allow the acceptance of off-site organic materials under reasonable operational standards
- Update setbacks, size limits, and other code provisions to reflect the realities of small, medium and large-scale composting
- Use available tools like the U.S. Composting Council's Model Zoning Template and Guidelines to develop locally tailored solutions
    - » https://www.compostingcouncil.org/page/Model-Zoning-Template-and-Guidelines

By updating local zoning codes to support composting, Michigan communities can unlock critical infrastructure needed to divert food scraps, improve soils, create local jobs, and build more resilient, circular economies.

State Permitting Constraints

At the state level, Part 115 of the Natural Resources and Environmental Protection Act limits the amount of food scraps that can be processed at small (5%) and medium (10%) composting facilities. Only large general-permitted facilities may exceed this threshold, but the costs and regulatory burden associated with general permits pose barriers for many operators. Part 115 also requires a 100-foot setback from property lines for medium and large facilities regardless of surrounding land use, further limiting viable sites, especially in cities.

To meet the state's recycling target, these facilities will require:

- Increased food scraps capacity (beyond the 10% limit for medium facilities)
- More flexible setbacks based on surrounding land use

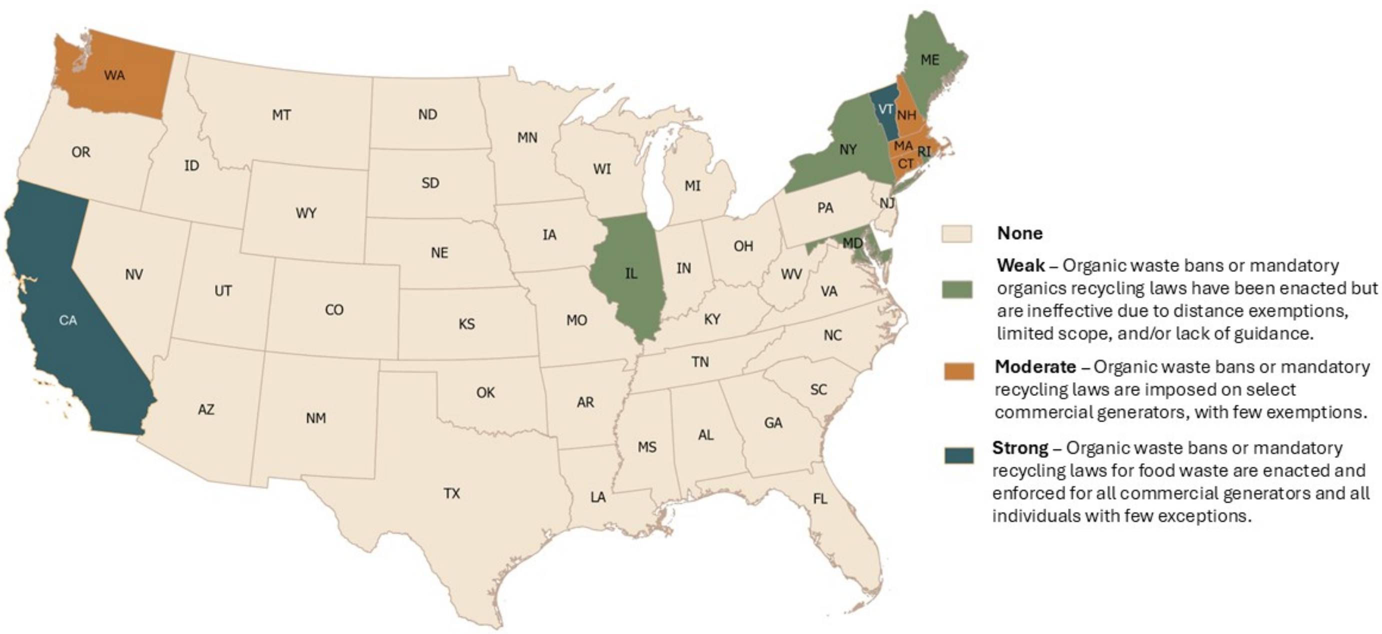
Both adjustments could be addressed through revisions to the general permit framework.

State of Food Scraps and Surplus Food Recycling Mandates in the US

Currently, eleven states and the District of Columbia (DC) have food scraps and surplus food diversion requirements and landfill bans (Figure 38). Of these twelve jurisdictions, ReFED considers two states (California and Vermont) to have strong policies; meaning, they are "enacted and enforced for all commercial generators and all individuals with few exceptions." Four states (Connecticut, Massachusetts, New Hampshire, Washington) and DC have moderate strength policies—the laws are imposed on select commercial generators with few exception—and five states (Illinois, Maryland, New Jersey, New York, Rhode Island) have weak policies—the laws are enacted but are ineffective "due to distance exemptions, limited scope, and/or lack of guidance" (Figure 38).[9]

Eleven states have active food scraps and surplus food landfill bans with varying levels of coverage depending on regulated entities (Table 24), with California being the only state with a statewide landfill ban for residential, industrial, and commercial entities.

Figure 38: U.S. Food Scraps and Surplus Food Policy Strength Ratings[10]



[9] ReFED U.S. Food Waste Policy Finder

[10] ReFED; Food Law and Policy Clinic – Harvard Law School. "State Food Waste Policy Ratings." https://policyfinder.refed.org/uploads/policy-finder-backend-state-score-sheet-0225.pdf

Table 24: Summary of State Food scraps and Surplus Food Regulations[11]

STATE	LANDFILL BAN	TAX INCENTIVE FOR DONATIONS	DIVERSION REQUIREMENT
AZ		Deduction **Eligible donor:** Restaurants, farmers	
CA	✓	Credit **Eligible donor:** Farmers, non-retailer packers/processors, taxpayers engaged in processing/distributing ag products	All waste generators are required to source separate and recycle organic waste **Generation Threshold:** • 8 cubic yards/week by 2016 • 4 cubic yards/week by 2017 • 2 cubic yards/week by 2020 • Any generation amount by 2022
CO		Credit **Eligible donor:** Taxpayers, C-corporations	
CT	✓		Eligible industrial and commercial generators required to source separate and recycle food scrap **Generation Threshold:** • 104 tons/year by 2014 • 52 tons/year by 2020 • 26 tons/year by 2022
DC	✓	Credit **Eligible donor:** Taxpayers, businesses	Retail food stores, high-education campuses, arenas/stadiums, and hospitals/nursing homes must donate excess food and source separate organic waste for composting, AD, or other processing **Generation Threshold:** • 10,000 sq. ft. by 2023 • Subject to mayoral requirements by 2024
IA		Credit **Eligible donor:** Taxpayers who produce food	
IL	✓		Permanent large event facilities located in counties with composting operating are required to offer food and organic waste source separation disposal and composting
KY		Credit **Eligible donor:** Taxpayers who receive income from ag products	
ME			Beginning July 1, 2030, entities generating more than 2 tons/wk. within 20 miles of an organics recycler will be required to divert according to the state's waste management hierarchy (prioritizing reduction and donation). By July 1, 2032, the threshold will drop to 1 ton/wk. within a 25-mile radius. **Generation Threshold:** • 2 tons/week by 2030 • 1 ton/week by 2032 • Subject to department rules by 2035
MA	✓		Commercial entities producing >0.5 ton/wk. of organic waste required to source separate and ensure proper recycling **Generation Threshold** • 1 ton/week by 2014 • 0.5 ton/week by 2022
MD	✓	Credit **Eligible donor:** Farm businesses, Montgonery County residents	Persons, businesses, schools, supermarkets, convenience stores, and cafeterias producing >1 ton/wk. of food scraps and surplus food required to source separate organic waste **Generation Threshold:** • 2 tons/week by 2023 • 1 ton/week by 2024
MO		Credit **Eligible donor:** Taxpayers	

[11] ReFED; Food Law and Policy Clinic – Harvard Law School. "Policy Finder Policy Matrix." https://policyfinder.refed.org/uploads/policy-finder-policy-matrix-0225-1.pdf

STATE	LANDFILL BAN	TAX INCENTIVE FOR DONATIONS	DIVERSION REQUIREMENT
NE		Credit **Eligible donor:** Grocery stores, restaurants, ag producers	
NH	✓		Any entities generating >1 ton/wk. of food scraps and surplus food are required to divert the waste from a landfill if there is an alternative capacity within 20mi (e.g., composting, AD, energy recovery)
NJ	✓		Entities generating >52 TPY of organic waste are required to source separate and recycle the waste (unless they are >25mi from a processing facility)
NY	✓	Credit **Eligible donor:** Farmers	Businesses, organizations, and institutions generating >2 tons/wk. of food scraps and surplus food must donate edible food and recycle remaining scraps if they are within 25mi of a processing facility. From 2026-2027, the threshold lowers to >1 ton/wk. and 50mi, and beginning 2028, the threshold lowers to >0.5 ton/wk. and 50mi.
OR		Credit **Eligible donor:** Farmers, farm businesses	
PA		Credit **Eligible donor:** Food rescue projects serving low-income neighborhoods	
RI	✓		Higher educational and research institutions producing >52 TPY of organic waste, all other educational entities producing >30 TPY of organic waste, and all other covered commercial entities producing >104 TPY of organic waste must divert the waste from landfill and incineration. Applies to those generators located within 15 mi of a composting or AD facility. **Generation Threshold:** • 104 tons/year by 2016 • 52 tons/year by 2018
SC		Credit **Eligible donor:** Meat packers, butchers, processing plants	
VA		Credit **Eligible donor:** Farmers	
VT	✓		All persons, organizations, business entities, municipalities, Vermont state governmental agencies, departments, and subdivisions, and federal agencies required to source separate and recycle organic waste **Generation Threshold:** • 104 tons/year by 2014 • 52 tons /year by 2015 • 26 tons/year by 2016 • 18 tons/year by 2017 • Any tons/year by 2020 (applies to all generators)
WA	✓		As of 2025, any business producing >4 cubic yards/wk. of organic waste are required to arrange on-site composting or organics collection. Businesses are encouraged to donate edible food. In 2026, the threshold decreases to include any business that produces >96 gallons/wk. of organic waste. Municipalities with > 25,000 residents are required to adopt a compost procurement ordinance. By April 1, 2027, municipalities are required to provide curbside composting to single-family homes and businesses in urban areas. **Generation Threshold:** • 8 cubic yards/week by 2024 • 4 cubic yards/week by 2025 • 96 gallons/week by 2026
WV		Credit **Eligible donor:** Farmers	

Note: the following states have no known food scraps and surplus food regulation or tax incentive: Alabama, Alaska, Arkansas, Delaware, Florida, Georgia, Hawaii, Idaho, Indiana, Kansas, Louisiana, Maine, Michigan, Minnesota, Mississippi, Montana, North Carolina, North Dakota, New Mexico, Nevada, Ohio, Oklahoma, South Dakota, Tennessee, Texas, Utah, Wisconsin, Wyoming

Food scraps and Surplus Food Recycling Considerations

Organics, specifically food scraps and surplus food, is currently the largest portion of Michigan's MSW stream from single-family, multi-family, and commercial settings, accounting for 37% of the residential stream and 38% of the commercial stream. One of the largest barriers to improving organics recycling programs is the state's current zoning and permitting of organics facilities. The scope of Right to Farm Act (RTFA) is too limited to adequately support small- and medium- sized composters, leaving local ordinances to determine whether such facilities are allowed. As a result, many cities and townships across the state do not explicitly classify composting as a permitted land use, making it challenging to establish new sites.

It is recommended that communities amend local zoning ordinances to reduce permitting burdens for local composters by utilizing tools like the U.S. Composting Council's Model Zoning Template and Guidelines. The model zoning framework, developed by the U.S. Organics Composting Council, the NRDC, and the Environmental Law Institute (ELI) is designed to support community composting by establishing it as a permissible land use under municipal zoning code. The model provides ready-to-use legal language across five basic zoning districts, includes standards for conditional use permits—which are required in specific zone types—and resources to help communities adapt the framework to their local context.

Resident education is a critical component of reducing food scraps and surplus food, but it requires sustained funding to be effective. Adjusting landfill tipping fees can serve a dual role: generating dedicated revenue for statewide food scraps diversion education and outreach, while also creating a financial disincentive for landfilling in comparison to composting. For example, Alameda County, California, has implemented a tiered tipping fee model for self-haulers, requiring landfill operators to charge 10% to 50% higher fees for source-separated food scraps and plant material. This approach both encourages diversion and provides a potential funding mechanism for educational initiatives. In areas where curbside collection is not available, municipal drop-off programs could be explored.

Effective education must be paired with access, as awareness alone is insufficient, residents need convenient and reliable pathways to properly divert their food scraps. It is recommended that the state encourage or, if possible, require that municipalities collect food scraps, with the option of collection commingled with yard waste. The state of Washington, for example, under HB 1799, requires cities of a certain population to provide food scraps collection programs. Bolstering the collection options within cities can also benefit surrounding suburban areas by creating regional collection opportunities.

If the state considers implementing a food scraps and surplus food diversion mandate or landfill ban, it is recommended to align with the policies of California and Vermont, two of the most effective U.S. jurisdictions with such policies, both of which have enforcement for commercial generators and individuals alike. In California and Vermont, unlike other jurisdictions, source separation is required for most waste generators, without specifying a generation tonnage threshold.

For communities looking to strengthen their food scraps and surplus food recycling, it is recommended they consider NRDC's Tackling Food Waste in Cities: A Policy & Program Toolkit. The toolkit offers foundational strategies for easier to obtain targets—such as establishing local baseline food scraps and surplus food generation and setting actionable goals—as well as more ambitious approaches, including updating municipal waste collection services and educating the public on food waste reduction and prevention, all of which offer significant reduction potential. Lastly, the toolkit presents transformative strategies, intended to build on the improvements of the foundational and ambitious strategies with targets such as transitioning to a pay-as-you-throw waste pricing system or adopting a mandatory organics recycling policy. Municipalities at earlier stages of their food scraps and surplus food recycling efforts can adopt foundational strategies, while those further along can pursue more transformative or ambitious actions. The framework is meant to align with each community's goals, capacity, and stage of progress.

Compostable packaging is a growing part of the disposal stream

Compostable Packaging EPR Review

There are currently 7 states in the US that have enacted Extended Producer Responsibility (EPR) policies for packaging. These states are California, Colorado, Maine, Oregon, Minnesota, Maryland and Washington. Compostable packaging is included as a covered product in all packaging EPR laws, but not all states include composting systems as part of the funding. Maine and Oregon's packaging EPR programs are focused on recycling only, and as such, compostable packaging in these states is treated like other non-recyclable packaging; producers will contribute financially to the program but will not receive the benefit of improved product management. The range of compostable packaging considered a covered product varies based on the definitions and exemptions for covered products in these programs.

Compostable Packaging EPR Example: Minnesota

Under the Packaging Waste and Cost Reduction Act (Minn. Stat. § 115A.144-115A.1463), producers are required to join a producer responsibility organization to fund the statewide program for packaging. Packaging is defined as materials that are used to transport, market, protect, or handle a product. The law includes compostable materials, which are defined as materials that meet American Society for Testing and Materials (ASTM) Standard Specification for Labeling of Plastics Designed to be Aerobically Composted in Municipal or Industrial Facilities (D6400), materials that meet the ASTM Standard Specification for Labeling of End Items that Incorporate Plastics and Polymers as Coatings or Additives with Paper and Other Substrates Designed to be Aerobically Composted in Municipal or Industrial Facilities (D6868) and materials comprised only of wood or paper with no coatings or additives.

By 2032, all packaging, food packaging, and paper products must be refillable, reusable, recyclable, or compostable. The Minnesota Pollution Control agency will establish a collection list of the covered compostable products that summarizes the statewide collection requirements for packaging. The producer responsibility organization, Circular Action Alliance (CAA), will set annual eco-modulated producer fees to cover program costs. As this law was passed in early 2024, and the program is still in the early phases of development, data is not yet available on the impact of EPR on divergence of compostable packaging in Minnesota.

Compostable Packaging Example: Colorado

Colorado's statewide EPR law (HB 22-1355) requires producers of all packaging materials, regardless of whether the packaging is recyclable, compostable, or non-recoverable, to pay producer dues to the producer responsibility organization to fund the program (State of Colorado). Certified compostable packaging is defined in the law as products that have received certification for undergoing aerobic biological decomposition in a controlled composting system meeting ASTM standards for plastics and polymer coatings with paper; however, Colorado's compostable labeling act only applies to food service products and plastic products. The EPR program will provide funding or other assistance to compost facilities to reduce the costs and increase the effectiveness of efforts to process and recover compostable packaging materials. On August 14, 2025, Colorado's Producer Responsibility Advisory Board recommended that Circular Action Alliance's amended Program Plan be approved by the Department. Program implementation is scheduled to begin in early 2026, so the impacts of this law on divergence are not yet available (Colorado Department of Public Health and Environment).

Compostable Packaging EPR Example: California

California's Plastic Pollution Prevention and Packaging Producer Responsibility Act (SB 54) was passed in 2022 to address packaging waste through extended producer responsibility, requiring that all packing and plastic food ware sold in California is recyclable or compostable by 2032 (CalRecycle). The Covered Material Category list for California's Packaging EPR includes compostability determinations. Under the law, to be considered "compostable," a material must meet all the requirements of AB 1201. It must not contain PFAS in an amount greater than 100 ppm, be third-party certified, designed for compostability with food and yard waste, labeled to be visibly distinguishable from non-compostable products, and an allowable organic input under the US Department of Agriculture's National Organic Program Regulations, meaning that: paper must be newspaper or recycled paper such as cardboard without glossy coatings or colored inks, woody material must be untreated without added synthetic substances, and fiber materials must not have any added synthetic substances

(CalRecycle 2025). Examples of packaging materials deemed compostable on the covered material category list include kraft paper, molded fiber, cardboard without plastic components, paperboard, mixed paper, and untreated wood without plastic components (CalRecycle 2025). Plastic and polymers designed for compostability are not deemed compostable under the current CalRecycle covered material categories list (CalRecycle 2025). Circular Action Alliance has been approved as the producer responsibility organization for the packaging program, but the statewide needs assessment is still underway to determine the program plan and budget for packaging EPR (CalRecycle).

Compostable Packaging EPR Example: Maryland

Under Maryland's EPR law, SB 901, producers must join a state-approved producer responsibility organization that will coordinate compliance and submit a plan to the Maryland Department of the Environment (MDE) and ultimately fund the statewide packaging program. Maryland differs from other states, as the legislation allows for the possibility of multiple producer responsibility organizations to operate, if proven necessary to the success of the program. The law is inclusive of recyclable, reusable, and compostable materials and calls for an evaluation through a statewide needs assessment by July, 2034 of the current infrastructure, capacity, and associated costs to manage them as well as a statewide covered materials list of materials deemed recyclable or compostable. No definition of compostable material is currently published; however, several compostable materials collectors serve on the EPR advisory council.

Compostable Packaging EPR Example: Washington State

The state of Washington signed the Recycling Reform Act (SB 5284) into law in May of 2025, which aims to expand recycling access across the state and modernize the state's materials management systems. Producers of covered packaging and paper products must join or form a producer responsibility organization by July 2026, which will coordinate the program's operations, fees, and develop the program plan. The Washington Department of Ecology will oversee the program and guide implementation through facilitating an advisory council, developing statewide collection lists, conducting two statewide assessments, and reviewing program plans and annual reports. The law includes compostable materials, defined as products capable of composting in a composting system and in compliance with at least one equivalent standard specification (ASTM D6400, D6868, D8410; ISO 17088; EN 13432) or is comprised only of wood or fiber substrates of greater than 98% fiber by dry weight and exclusive of specific additives.

Compostable Packaging EPR Recommendations

Compostable packaging is recognized under all seven U.S. state packaging EPR laws; however, not all states allocate program funds to support composting infrastructure for its effective management. States like Maine and Oregon focus specifically on recycling, and as a result, compostable packaging is categorized as a non-recyclable material. Producers are still required to contribute to the program financially, but the packaging does not receive the benefit of an improved system, effectively limiting its integration within the scope of these EPR frameworks. To meaningfully address compostable packaging through broader packaging EPR, it is a foundational requirement that both compostable packaging and composting infrastructure be included within program scope and their capacity and prevalence be evaluated under a statewide needs assessment. Doing so ensures organics management facilities are adequately supported through infrastructure upgrades and EPR initiatives.

Several states have taken a phased approach to EPR by first conducting comprehensive needs assessment before enacting full EPR legislation. Needs assessments are useful tools for evaluating the current state of materials management: specifically, what is working well within a state's current system and what needs improvement. By conducting statewide evaluations, states can develop tailored data-driven solutions. The data and stakeholder opinions gathered during the assessment can be used to properly allocate resources during EPR implementation.

Maryland, for example, completed a statewide needs assessment that directly informed its subsequent passage of a full EPR law (SB 901) enabling the state to be reimbursed by the PRO for the costs of administration related to the needs assessment. The findings project that a well-designed EPR program could increase the State's recycling rate from 34% to 50% or greater, capturing $202 million of material value, an increase of $53 million from baseline estimates, reduce over 1 million metric tons CO_2 equivalents, and create over 2,000 additional jobs. The assessment also identified gaps, such as the need for upgraded infrastructure to manage compostable packaging to prevent contamination rates from increasing. Infrastructure upgrades at organics processing facilities could cost between $1.8 million and $25.5 million per site, depending on facility size, excluding the costs associated with anaerobic digestion improvements.

Several other states, including Illinois, Hawaii, Minnesota, and Rhode Island, have enacted legislation requiring needs assessments to evaluate their existing materials management systems. Illinois is conducting a comprehensive assessment of its current collection systems, infrastructure, and capacity for managing packaging and paper products across reuse, recycling, composting, and disposal pathways. Hawaii's assessment focuses on identifying what is needed to reduce waste generation, increase reuse, improve collection services, and expand local processing capacity. It also includes a county-level analysis of the resources required to minimize the amount of packaging and paper waste sent to landfills or waste-to-energy facilities. Minnesota's law mandates that by December 2026 (and every five years thereafter), the state must complete a comprehensive needs assessment covering materials flow, recycling, composting, infrastructure capacity, and service gaps to inform its EPR packaging strategy. Rhode Island's assessment aims to determine the infrastructure, policy, and programmatic needs to support a statewide redemption and recycling system. This includes a baseline analysis of recyclable and compostable materials by jurisdiction, type, collection method, and end market destination, along with an evaluation of the capacity of materials recovery facilities (MRFs), composting operations, and transfer stations.

Electronics Recycling Review and Policy Considerations

57,500+ tons of electronics were disposed in 2024; diverting another 5,700+ tons of electronics are needed to reach the 45% recycling rate goal, more than double current recycling rates.

In 2024, Michigan disposed of more than 57,500 tons of electronics. Achieving the state's 45% recycling rate goal will require diverting an additional 5,700 tons of electronics annually, more than doubling current recycling levels.

As of 2025, 23 states and the District of Columbia have implemented extended producer responsibility (EPR) laws for electronics. EPR programs for electronics generally fall into two broad categories: convenience standards or performance standards. Convenience standards require manufacturers to meet specific requirements related to consumer access to electronics recycling, such as minimum numbers or geographic distribution of collection sites. In these programs, electronics recycling is typically coordinated through a centralized entity that contracts for collection and recycling services and allocates costs among participating manufacturers. These programs generally provide recycling services at no cost to collectors, such as municipalities or contracted organizations. Examples of states using convenience standards include Oregon, Washington, and Vermont.

Performance standards assign individual manufacturers a collection obligation, typically expressed as a weight-based target calculated using market share or statewide recycling goals. Manufacturers are responsible for financing the collection and recycling necessary to meet those obligations. Examples of states that have performance standard-based policy for electronics EPR are Illinois, Minnesota, New Jersey, New York, Pennsylvania, and Wisconsin.

Michigan's Electronic Waste Takeback Law (Part 173 of the Natural Resources and Environmental Protection Act), enacted in 2008, established an electronics manufacturer responsibility framework based primarily on convenience standards. The law requires manufacturers of covered electronic devices to provide takeback programs for residents and small businesses and requires electronics recyclers to register annually and comply with applicable operational, environmental, health, safety, and data security requirements. Under Part 173, manufacturers that do not maintain compliant takeback programs may be subject to restrictions on the sale of covered devices in Michigan. The law also authorizes EGLE to promulgate rules related to manufacturer takeback programs, recycler standards, reporting, and other program requirements.

Some states with convenience standard programs report more consistent access to electronics recycling and more stable funding for year-round collection operations, as costs are distributed across manufacturers and managed through a centralized system. In Oregon and Vermont, e-waste recycling is managed by contracted organizations, while in Washington it is run by the Washington Materials Management and Financing Authority. These entities establish agreements with recyclers, manage collection networks, and allocate costs among manufacturers.

Below are examples of two states that have electronics recycling programs that use convenience standards, which are regarded by experts as the most stable programs in the US. Oregon is an example of a convenience standard program that uses contracting through a producer responsibility organization to operate electronics collection and recycling, and recently approved expansions to the list of covered devices and the funding structure in the law. Washington is an example of a state with a convenience standard program that is managed by a government entity.

Electronics EPR Example: Oregon

Oregon operates a statewide electronics recycling program that provides no-cost recycling of covered electronic devices through a network of collection sites distributed across counties and population centers. The program is funded by electronics manufacturers and administered by the Oregon Department of Environmental Quality (Oregon Department of Environmental Quality).

In 2022, the program collected 12.4 million pounds of electronic devices for recycling and over 39,000 devices for reuse (Oregon Department of Environmental Quality).

Electronics EPR Example: Washington

Washington's E-Cycle program provides statewide, no-cost recycling of covered electronic devices through more than 200 collection sites (required in counties and cities with populations greater than 10,000). The program is financed by participating manufacturers and administered by a central authority (Washington Materials Management and Financing Authority 2025).

In 2024, the program collected over 12.8 million pounds of electronics. Since program inception, over 479 million pounds of electronics have been collected statewide.

Electronics EPR Recommendation

Despite Michigan's existing manufacturer takeback framework, statewide electronics recycling rates remain low relative to disposal volumes, and access to recycling services continues to vary across the state, particularly in rural communities.

To improve electronics recovery and increase statewide access to convenient recycling opportunities, Michigan could consider the following implementation and policy strategies:

  - Evaluate opportunities to better align state grant investments with manufacturer takeback obligations established under Part 173.
- Strengthen oversight and accountability mechanisms for manufacturer takeback programs, recycler registration, certification, and reporting requirements.
- Improve statewide data collection and reporting consistency for manufacturers and recyclers to better measure program performance and material recovery.
- Expand public awareness efforts related to manufacturer takeback programs and available electronics recycling opportunities.
- Encourage partnerships among manufacturers, recyclers, local governments, retailers, and community organizations to expand convenient collection opportunities statewide.
  - Evaluate whether additional convenience standards, geographic access requirements, or program coordination mechanisms could improve statewide service consistency.
  - Consider whether additional rulemaking or statutory clarification may support more consistent implementation, accountability, and statewide access to electronics recycling services.

Bottle Deposit System Review and Recommendations

48,400+ tons of redeemable containers were disposed in 2024; 167,400+ tons of additional glass bottles and jars, aluminum cans, and plastic PET bottles must be diverted to reach the 45% recycling rate goal.

Bottle redemption rates have fallen significantly in Michigan in recent years, with 2024 recording the lowest redemption rate since the 1990s at only 70% (Schneider, 2025). Given the falling rates, there is renewed interest from across the state and impacted stakeholders to investigate ways to improve Michigan's bottle redemption system. Several research projects are underway, including a research effort at the University of Michigan and a ballot initiation from the Michigan Environmental Council (MEC). The research that is being undertaken, detailed below, will provide an assessment of the current system, policy recommendations, and diversion impacts of bottle bill expansion.

University of Michigan Bottle Redemption System Research

Researchers at the University of Michigan are currently working on a study of the bottle deposit system and exploring opportunities for improvement. The project is structured around three core objectives:

- **Benchmarking:** A comparative analysis of bottle deposit systems across the ten U.S. states with similar laws was conducted to identify best practices and variations in system design.
- **Stakeholder Engagement:** In-depth interviews were carried out with key actors across the value chain, including beverage distributors, retailers, recyclers, environmental organizations, and policymakers, to gather diverse perspectives and identify systemic challenges and opportunities.

1. **Collaborative Workshop:** A facilitated in-person workshop brought together a broader group of stakeholders to discuss and deliberate the key themes and potential solutions identified during the interviews.

The study synthesizes stakeholder insights and available data to evaluate the performance of the current system and assess options for reform. The findings will be compiled into a comprehensive report aimed at informing policy discussions and future improvements to Michigan's bottle deposit program.

Michigan Environmental Council Bottle Redemption Actions

The Michigan Environmental Council (MEC) is pursuing an initiative campaign to expand Michigan's bottle bill to include all beverages except milk and nutritional supplements[12]. The MEC is also in favor of expanded redemption options for improved convenience and funding for overall system improvement and pollution protection. In 2025, the MEC worked with RRS's Signalfire Group to assess the potential impacts of an expanded bottle bill on recycling rates and environmental benefits.

Bottle Bill Recommendation

Bottle deposit systems consistently achieve the highest recycling rates for covered materials, particularly glass, which is otherwise difficult to recycle economically due to contamination. Deposit-based redemption significantly reduces contamination and supports true bottle-to-bottle recycling. However, Michigan's redemption rates have declined sharply in recent years, reaching a historic low of 70% in 2024. This decline is attributed to several factors, including the diminished real value of the ten-cent deposit, limited and inconvenient redemption infrastructure, and an increasingly cumbersome redemption experience.

To reverse this trend, Michigan could consider a set of targeted reforms designed to strengthen and modernize the program. First, expanding the range of covered containers—such as through the Michigan Environmental Council's ballot initiative to include nearly all beverage types—would increase participation by providing residents with more opportunities to redeem deposits. Universal redemption, where retailers are required to accept all deposit containers regardless of brand, would remove a key barrier to convenience and access.

The state could also reallocate the use of unredeemed deposits. At present, 25% are returned to retailers and 75% are directed to the Cleanup & Redevelopment Trust Fund. Legislative proposals, such as Senate Bill 1112, suggest redirecting a portion of these funds toward improving redemption infrastructure, supporting retailers and distributors, and enhancing water security. At the same time, dedicating a share of unredeemed deposits to support material recovery facilities (MRFs) would help offset the loss of high-value aluminum and PET bottles diverted through the deposit system, ensuring broader recycling system stability.

Improving convenience is equally important. Michigan could expand redemption options by supporting multiple access points, including reverse vending machines, hand-count stations, and bag-drop systems, making participation easier and more consistent across the state. Finally, updating the deposit value to reflect inflation would restore its motivational effect, while investments in public education, transparency, and community-based redemption initiatives would strengthen trust and reinforce the program's environmental and economic benefits.

[12] https://environmentalcouncil.org/better-bottle-bill/petition/

BEYOND MICHIGAN'S 45% RECYCLING RATE GOAL: MICHIGAN MATERIALS MANAGEMENT PLANNING AND ALIGNMENT WITH MICHIGAN'S CLIMATE ACTION INITIATIVES

SECTION SUMMARY

This section presents opportunities for EGLE to align local governments (e.g., counties and solid waste management districts) with the state's climate action initiatives of which a core part is waste reduction and diversion via the Materials Management Planning (MMP) process

Key findings include:

- **MMP shift:** In 2024, EGLE launched MMP to replace county solid-waste plans, prioritizing reduction, reuse, recycling, and composting before disposal. All 83 counties must develop MMPs toward the 45% recycling rate goal; Part 115 broadens oversight (MRFs, digesters, diversion facilities) with centralized reporting.
- **Climate alignment:** Ensure plans align with the MI Healthy Climate Plan that sets a 50% food scraps and surplus food reduction goal, disaster-debris planning, and standardized carbon metrics (e.g., EPA WARM, ReFED, NRDC). This alignment would be a boost to reaching the state's recycling goal because food scraps and surplus food is the largest single category in disposal and would strengthen Michigan counties preparedness for climate change.

ALIGNMENT OPPORTUNITIES IN MICHIGAN'S MATERIALS MANAGEMENT PLANNING PROCESS

In 2024, EGLE initiated the statewide MMP process, which replaces traditional county Solid Waste Management Plans and shifts focus to a hierarchy of reduction, reuse, recycling, and composting before disposal. Counties are required to develop MMPs (individually or jointly), supported by EGLE grants and guidance, with current solid-waste plans remaining in effect until each new MMP is approved. The updated Part 115 law ties planning to Michigan's recycling goals of 45% and expands oversight to new facility types (e.g., MRFs, waste-diversion facilities, anaerobic digesters) with centralized online reporting via Re-TRAC.

The MMP process represents a major shift in materials management in Michigan and is a critical step to reaching the state's recycling goals and prioritizing reduction, reuse, and recycling of materials long term. At the same time there are significant opportunities to better align the MMP process with the state's broader climate mitigation and adaptation planning strategies, particularly those outlined in the MI Healthy Climate Plan (MHCP). The MI Healthy Climate Plan and MMP process are directly interconnected at a policy level, but neither plan explicitly recognizes the other. Strengthening the connection between the two plans presents a valuable opportunity to improve policy effectiveness, support local governments and stakeholders in planning efforts, and align the efforts required for both planning processes.

An opportunity for aligning materials management planning and climate planning lies in food scraps and surplus food reduction. The MHCP codifies a 50% food scraps and surplus food reduction goal for the state. The carbon reduction benefits of food rescue and keeping food out of the landfill could be recognized in MMP plans and inform MHCP inventories, however, none of these plans are currently required to calculate the carbon benefits associated with these strategies or the cost of doing nothing and maintaining the status quo disposal rate. All 83 counties are currently developing plans to better manage organic and inorganic recyclables, presenting a timely opportunity to embed emission reduction metrics into their planning processes. Food scraps and surplus food reduction, more than other initiatives like greening-the-grid initiatives, are in direct control and oversight of local governments.

Recent work with a food rescue agency showed a 45,000 $MTCO_2eq$ benefit of food rescue. Despite this large potential carbon reduction, counties are not currently required to estimate the carbon benefits of these strategies or the consequences of inaction. Tools such as the EPA's Waste Reduction Model (WARM), ReFED impact metrics, and Natural Resources Defense Council's (NRDC) food scraps and surplus food calculators can provide standardized, data-driven methodologies for estimating these impacts at the county level. Providing webinar training or direct technical assistance to Designated Planning Agencies (DPAs) would help ensure that food scraps and surplus food reduction strategies are consistently measured and incorporated into local plans.

Another area of opportunity lies in disaster debris management, which becomes increasingly relevant in the context of a changing climate. Michigan is experiencing an increase in severe weather events, including more extreme storms and precipitation, with periods of drought in between. These events generate significant volumes of waste, from downed tree debris in rural areas, to flooding in urban areas resulting in significant volumes of household and bulky waste from flooded basements. Assisting counties with Disaster Debris management templates and direct technical assistance would support aligning the MMP process with climate adaptation outcomes identified in the MHCP.

Counties can play a key role in managing the lifecycle of renewable energy infrastructure. As Michigan continues its transition to cleaner energy sources, there is a growing need to address the reuse and recycling of solar panels, electric vehicle (EV) batteries, and wind turbine blades. Recognizing these emerging material streams within MMPs can help support both environmental protection and the development of recycling markets.

Sustainable purchasing policies also present a promising area for integration within county MMPs. Government procurement has a substantial carbon footprint, and incorporating sustainability into purchasing decisions can drive broader changes in consumption patterns, waste reduction, and more circular options. Sustainable purchasing policies promote reuse and repair, the use of recycled content, and the elimination of toxic materials. The Michigan Green Communities program has already supported a cohort of municipalities in developing sustainable purchasing practices. Building on this progress, EGLE could provide additional guidance to counties to ensure that sustainable procurement becomes a common feature of local MMPs.

Consistent, high-quality data is foundational to achieving data-driven solutions in both MMP and MHCP planning. Currently, counties have little access to reported recycling and diversion data needed for effective planning and waste reduction. In response, some counties are considering policies that would mandate direct data reporting from haulers. However, this approach risks creating administrative burdens for haulers and operators who work across counties, as well as data inconsistencies across the state. To address these challenges, EGLE could consider establishing a standardized, statewide data reporting platform, potentially by expanding existing tools such as Re-TRAC. Such a platform would streamline data collection, improve overall data quality, and enable more effective planning at both the local and state levels. Public access to the platform would further increase transparency and help stakeholders across government and industry better understand their roles in advancing Michigan's recycling and climate goals. Ideally, the system would also be capable of calculating and reporting the climate impacts of materials management activities, thereby reinforcing the integration of waste reduction strategies with the state's carbon accounting framework.

Better aligning the MMP process with the MHCP presents a clear opportunity to advance shared climate and sustainability goals. As all counties develop updated MMPs, integrating strategies like food scraps and surplus food reduction, disaster debris planning, sustainable purchasing, and others can drive measurable climate benefits. Providing counties with consistent data tools, technical support, and policy guidance will strengthen both local planning efforts and statewide progress toward climate and statewide diversion targets.

CONCLUSION

The 2025 Michigan Gap Analysis reflects meaningful progress toward the state's 45% recycling rate goal since the 2023 update, while making clear that substantial work remains. Michigan has achieved a record 25% recycling rate, supported by more than $60 million in EGLE investments from 2019 through 2024, contributing to $443 million in total circular economy investment statewide. In 2025, EGLE announced an additional $11.8 million in new grant awards targeting the priority material gaps addressed in this report.

This report identifies several priority areas where focused effort will be essential for continued progress:

Food Scraps and Surplus Food. Organics remain the single largest opportunity, representing 18% of all disposal. Despite growth in curbside and drop-off programs, 67% of Michigan residents still lack convenient access to food scrap diversion. Expanding affordable, equitable access to collection programs and removing barriers to small and medium-sized organics processing capacity are critical next steps.

Commercial Corrugated Cardboard. Cardboard is highly recoverable and retains strong market value, yet 25% of small businesses surveyed do not separate it from trash. Expanding hauler access, reducing costs, and supporting staff training and dumpster right-sizing can help close this gap.

Scrap Metal, Electronics, and Wood Waste. Residents have indicated they would recycle more if convenient access to collection sites were available. Expanding regional comprehensive drop-off sites that accept a broad range of hard-to-recycle materials, including scrap metal, electronics, and wood waste, represents one of the most actionable opportunities for increasing diversion.

Single-Family Access: While 80% of Michigan residents have access to curbside recycling programs, many residents predominantly in rural areas rely on subscription programs that require residents to sign up for service, often at an additional fee. These program structures create barriers to curbside recycling, lower participation, and ultimately result in more recyclables headed to disposal.

Drop-Off Access: Only 26% of Michigan residents have access to a municipal or county run drop-off program that accepts material beyond traditional curbside recyclables, highlighting a key gap in diverting other recyclable material which is necessary for the state to reach the 45% diversion goal. Expanding comprehensive drop offs statewide can result in significant recovery volumes.

Multi-Family Access. Only 20% of multi-family households live in a municipality that even addresses the need for onsite recycling. Expanding convenient, on-site recycling access for apartment residents is an underserved and high-priority gap.

Policy. Michigan has significant policy levers available to accelerate progress. EPR frameworks for packaging and electronics, reforms to the bottle deposit system, and alignment of the Materials Management Planning process with the MI Healthy Climate Plan all represent opportunities to build more durable, system-wide support for recycling and organics diversion.

Michigan is making significant strides, and the infrastructure, investment, and policy foundations needed to reach 45% are increasingly being put in place. Reaching that goal will require sustained commitment across all levels of government, the private sector, and Michigan communities.

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APPENDIX

REGIONAL STATE REPORTING AND FUNDING SNAPSHOTS

Indiana

Goals and Requirements

Indiana has a statewide recycling goal for municipal solid waste of 50% (Indiana Department of Environmental Management 2025). There is no set target date to reach this goal. Indiana does not have any state level recycling or diversion requirements for residents or commercial businesses. There are some local ordinances within the state to require recycling.

Reporting

To track progress towards this goal, the state requires recyclers to report quantity and type of processed recyclable material to the Indiana Department of Environmental Management (IDEM) annually. Indiana defines a recycler as anyone who acts as a recyclable materials broker, solid waste facility that handles recyclables, material recovery facilities, and solid waste management districts. The state uses the Re-TRAC Connect reporting system to collect the following information:

- Name and location of recycler's establishment
- Description of primary business activities
- Quantity in tons of recyclable material by commodity type
- The city and state of the recycling facility
- The name and principal address of the broker
- The state or foreign country in which the manufacturer (end user) is located

The state uses this information to publish their annual recycling index report (https://www.in.gov/idem/recycle/files/reporting_recycling_2023_index_report.pdf)

Funding

The state works to support recycling programs through granting opportunities to communities and end markets. The Indiana Community Recycling Grant Program provides grant assistance to counties, municipalities, schools, universities and nonprofit organizations within Indiana to support recycling education and outreach, waste reduction activities, organics management, and household hazardous waste collection and disposal programs. Grants start at $1,000 and go up to $100,000 with a 25% cash match (https://www.in.gov/idem/recycle/resources/indiana-community-recycling-grant-program/).

Indiana also offers the Indiana Recycling Market Development Program (RMDP) to support recycling end markets. The grant is open to public and private businesses, solid waste management districts, local governments, and nonprofit organizations doing business in Indiana, with awards ranging from $50,000 to $500,000 with a 50% cash match (https://www.in.gov/idem/recycle/recycling-market-development-program/).

In 2025, the IDEM awarded $3.7M of RMDP grant funding to five organizations to address recycling access across the state (https://events.in.gov/event/five-central-indiana-organizations-receive-grants-to-expand-recycling-programs).

- **American metals**
  - Awarded $911,519 toward purchase of equipment to disassemble and process magnets, batteries, and ferrous metals from items including EV motors, power tools, and windmill turbines with the goal to process 3,000 tons of battery material and 12,000 tons of magnet material annually.
- **Electronic Recyclers International Indiana, Inc.**
  - Awarded $1,000,000 to purchase equipment for solar panel dismantling and material recycling with the goal of recycling 250 tons annually.

- **Heidelberg Materials, Inc.**
  - » Awarded $500,000 to purchase closed-circuit plant and conveyors for recycling concrete and rebar from construction debris with the goal of diverting 75,000 tons annually.
- **Indiana Oxygen Company**
  - » Awarded $500,000 to purchase excavating, power handling, and drying equipment to process calcium hydroxide (lime) waste by-product material from the company's holding pond with the goal of recycling 74,000 tons of waste by-product annually.
- **Republic Services**
  - » Awarded $788,481 to purchase optical sorter for PET for their 96th street MRF to with the goal of diverting 410 tons of additional plastics. The grant recipient Republic Services also intends to use the funding to expand curbside recycling access in the City of Fishers (Gonzalez 2025).

Ohio

Goals and Requirements

Ohio's solid waste management planning process has established 10 goals for solid waste management districts (SWMD) to achieve. The second goal of these ten goals requires SWMD to show at least 25% of MSW reduction or diversion (Ohio Environmental Protection Agency 2021).

SWMD must ensure that at least 80% of the residential population in each county has access to recycling opportunities – either through curbside or drop-off programs. SWMD are also required to provide commercial recycling opportunities to ensure businesses have access to facilities or services that allow them to recycle. Ohio sets a minimum required recycling list for collection programs that includes corrugated cardboard, mixed paper, paperboard, glass bottles and jars, aluminum cans, steel containers, and plastic bottles and jugs (https://codes.ohio.gov/ohio-administrative-code/rule-3745-27-90).

Reporting

Ohio has established recycling reporting and solid waste management planning requirements for all SWMD, overseen by the Ohio Environmental Protection Agency (OEPA). Each district must submit an Annual District Report to OEPA that includes data on solid waste generation, recycling and composting tonnages, disposal and transfer activities, and progress toward program implementations such as curbside and drop-off recycling services. Districts also conduct annual commercial and industrial surveys to capture recycling data from businesses that is not otherwise reported by haulers. To supplement local data collection, OEPA directly surveys materials recovery facilities (MRFs) and major retail chains, then provides that data to districts for inclusion in their annual reports. This collaborative approach enables Ohio to maintain one of the nation's most comprehensive datasets on recycling by MRFs and private-sector entities. In addition, permitted solid waste facilities are required to submit quarterly and annual reports to either OEPA or their respective district, detailing the types and volumes of waste managed. OEPA uses the Re-TRAC Connect system to facilitate this reporting. SWMDs must also update their solid waste management plans every five years, each covering a 15-year planning horizon (https://epa.ohio.gov/divisions-and-offices/environmental-innovation/oei-programs/solid-waste-management-planning).

Funding

The OEPA offers several grants to solid waste management districts, private businesses, and nonprofits and institutions to support recycling programs including:

- **Academic Institution Grants**
  - » Funding for recycling efforts such as outreach and education, recycling equipment, and conference sponsorships.
- **Community and Litter Grants**
  - » Funding the purchase of equipment for collection and processing of recyclables and construction and demolition debris.
- **Market Development Grants**
  - » Funding for equipment to support end market development and remanufacturing recycled commodities.

- **Scrap Tire Grants**
  - » Funding to create or expand scrap tire processing capacity and product manufacturing.
- **Source Reduction Grants**
  - » Funding the purchase of equipment to promote reuse.

In 2024, OEPA granted $1.6M in total through the above listed grant programs (Gonzalez 2025).

Minnesota

Goals and Requirements

Minnesota set a recycling goal for the Twin Cities metro area to achieve a recycling rate of 75% or higher and Greater Minnesota to achieve a recycling rate of 35% or higher by 2030 (*Minnesota Statue 115A.551 Recycling*. 2024).

The Minnesota Statues § 115A.552 requires that all counties in Minnesota must:

- Ensure residents – including those living in single and multi-family housing units - have an opportunity to recycle.
- For cities with a population of 5,000 or more, curbside recycling services or a centralized drop-off site for recycling must be available that accepts at least four broad types of recyclable material.
- Maintain at least one recycling center in each county.
- Provide recycling education information describing how, when, and where materials may be recycled.

In addition to the above, the metro county region (Anoka, Carver, Dakota, Hennepin, Ramsey, Scott, and Washington) requires businesses generating at least 4 cubic yards of trash per week to recycle at least three materials.

Finally, Hennepin County has additional requirements:

- Residents are required to recycle paper, cardboard, glass bottles and jars, metal cans, plastic bottles, jugs, cups, containers, and cartons.
- Multi-family property owners are required to provide recycling services co-located with trash containers, appropriately labeled containers, and educate residents about recycling practices.
- Businesses generating more than one ton of trash per week or contract for eight cubic yards or more trash service per week must implement food scraps and surplus food recycling.

Reporting

Minnesota supports its recycling and diversion goals through robust, statewide reporting requirements for both public and private entities involved in solid waste management. All permitted solid waste facilities, including landfills, transfer stations, composting facilities, recycling facilities, waste-to-energy facilities, and refuse-derived fuel (RDF) facilities, are required to submit annual reports to the Minnesota Pollution Control Agency (MPCA). These reports must include volumes and types of materials managed, origin of the waste (e.g., county), and final disposition location (e.g., recycled, landfilled, transferred).

For MRFs, additional information MPCA collects includes residue rates and total actual recycled materials and location of end markets, including what was sent to those end markets and quantity. The MPCA allows facilities to submit two forms: a complete form that will not be made public, and a second form with redacted information that will be made public.

Haulers are also subject to annual reporting requirements, including the volume and composition of waste and recyclables collected, service area coverage, and information on which facilities material is hauled to. This ensures comprehensive tracking of material flows across the state's waste management system.

Reporting is conducted through Re-TRAC Connect.

Additional reporting requirements may apply at the local or regional level. For example, solid waste management districts and counties must also compile and submit recycling and diversion data, often requiring local governments and service providers to maintain detailed records.

Funding

The MPCA provides grant opportunities statewide and specific to Greater Minnesota (outside Twin Cities metro region).

- **Statewide Waste Reduction and Reuse Grants**
  - » Nonprofits, private businesses, institutions, state agencies, tribal governments, and local governments are eligible for grants ranging from $75,000 to $750,000 with a 25% match. The focus of the grants is to support initiatives transitioning Minnesota to reusable systems, community repair and training, and sustainable reuse infrastructure (https://www.pca.state.mn.us/grants-and-loans/statewide-waste-reduction-and-reuse-grants).
- **Greater Minnesota Waste Reduction, Reuse, Recycling, and Composting Grants**
  - » Counties, cities, townships, and tribal governments with populations under 45,000 and located outside the Twin Cities metro area are eligible for grants ranging from $50,000 to $250,000 with a required 25% local match. The focus of the grant funding is to enhance programs aimed at waste reduction, reuse, recycling, or composting (https://www.pca.state.mn.us/grants-and-loans/greater-mn-waste-reduction-reuse-recycling-and-composting-grants).

In 2024, MPCA announced more than $1M in grants awarded to five recipients (Minnesota Pollution Control Agency (MPCA) 2024b).

- **Todd County**
  - » Awarded $186,625 to increase the number of recycling material drop-off sites.
- **Red Lake Band of Chippewa**
  - » Awarded $249,268 to expand the Obaashiing Transfer Station to process recyclables, household hazardous waste, electronics, and reuse.
- **Otter Tail County Solid Waste**
  - » Awarded $250,000 to purchase 3,000 household recycling carts for single stream recycling.
- **Chisago County Environmental Services**
  - » Awarded $249,766 to improve the safety and efficiency of the current household hazardous waste facility and expand floor space so that reuse, reduction, and recycling efforts can be added.
- **Pope Douglas Solid Waste Management**
  - » Awarded $160,591 to purchase optical sorters for their MRF.

Also in 2024, MPCA announced $5.3M invested to grow recycling markets in Minnesota (Minnesota Pollution Control Agency (MPCA) 2024a).

- **Eureka Recycling**
  - » Awarded $850,000 to purchase an optical sorter to separate glass, paper, and cardboard from plastics, improving recycling at their Minneapolis facility.
- **Cosmic Recycling**
  - » Awarded $850,000 to purchase equipment to recycle solar panels, increasing capacity and creating seven new jobs in Fairfax.
- **GypCycle**
  - » Awarded $850,000 to purchase equipment to convert drywall into agricultural gypsum, supporting local farming and reducing landfill waste in Princeton.
- **LEI Packaging**
  - » Awarded $850,000 to establish a molded fiber production line using 500–700 tons of paper and cardboard annually in Chisago City.
- **Prime Extrusion**
  - » Awarded $200,000 to purchase an extruder and shredder to recycle difficult HDPE scrap, diverting 1 million pounds of plastic from landfills annually in Greenfield.

- **Polk County**
  - » Awarded $850,000 to purchase a flip-flow separator and bag opener to recover more recyclables and improve glass recycling at their facility.
- **Employment Enterprises**
  - » Awarded $119,479 to purchase balers and handling equipment to expand recycling services for cardboard, paper, metals, and glass in Little Falls.
- **FoamCraft Packaging**
  - » Awarded $149,519 to purchase a shredder and pelletizer to recycle foam waste into new foam and create animal bedding from pallets in Owatonna.
- **Slagle's Demolition Landfill and Rolloff**
  - » Awarded $131,607 to purchase equipment for mechanized recycling sorting and processing, increasing recycling and reducing physical strain on workers in Cass County.
- **Atlas Games**
  - » Awarded $99,359 to increase Replay Workshop's capacity to recycle and market small quantities of post-consumer HDPE and polypropylene in Proctor.
- **True North Goodwill Industries**
  - » Awarded $350,000 to upgrade equipment to recycle 6,000 mattresses annually and recover steel, foam, felt, and wood in Duluth, adding 1.5 staff positions.

Wisconsin

Goals and Requirements

Wisconsin does not have a recycling goal; however, the state does mandate that all residents, businesses, and institutions participate in recycling programs under the Waste Reduction and Recycling Law (Wisconsin Department of Natural Resources 2022). The mandate requires:

- **Universal Access**
  - » All citizens must have access to recycling services, either through curbside collection or drop-off centers.
- **Material Disposal Bans**
  - » Landfilling or incinerating of aluminum containers, glass containers, steel containers, plastic containers (1 and #2), newspapers, magazines, office paper, and corrugated cardboard is prohibited statewide.
- **Effective Recycling Programs (ERPs)**
  - » Local governments, designated as Responsible Units (RUs), are required to implement ERPs that encompass:
    - – Provision of recycling services to residents, either curbside or drop-off.
    - – Education and outreach initiatives to inform the public about recycling practices.
    - – Enforcement of recycling requirements for businesses, institutions, and special events.

Reporting

To ensure compliance and track progress, Wisconsin mandates annual reporting from both RUs and MRF through the Wisconsin Department of Natural Resources Switchboard platform.
(https://dnr.wisconsin.gov/topic/Recycling/reports.html?utm_source=chatgpt.com).

- **Responsible Units**
  - » Must submit an annual report by April 30 each year, detailing:
    - – Quantities of recyclable materials collected. This data is collected by RUs from haulers that are required to report to them.
    - – Actual costs associated with the recycling program.
    - – Compliance with program criteria.
  - » This reporting is essential for maintaining ERP approval and eligibility for state recycling grants.

- **MRFs**
  - Required to submit an annual self-certification by March 30, confirming:
    - Compliance with operational standards.
    - Tonnage of materials processed.

Funding

Wisconsin supports its recycling initiatives through grants administered by the Department of Natural Resources (DNR):

- **Basic Recycling Grant**
  - Available to RUs (cities, towns, villages, counties, tribes, or solid waste management systems) to offset costs associated with residential recycling and yard waste programs (https://dnr.wisconsin.gov/aid/Recycling.html)
- **Recycling Consolidation Grant**
  - Provides additional funds to RUs that consolidate recycling services with other units, promoting efficiency and cost-effectiveness (https://dnr.wisconsin.gov/aid/Consolidation.html)

Eligibility for these grants requires RUs to maintain an approved ERP and meet specific criteria outlined in Wis. Stat. § 287.24. Grant applications typically open in August for the following fiscal year.

Additionally, the DNR offers specialized grants, such as the E-Cycle Wisconsin Electronics Collection Grant, to support the collection and proper disposal of electronic waste, with priority given to areas lacking permanent collection sites.

WASTE CHARACTERIZATION MODELING METHODOLOGY

To develop a desktop model of Michigan's MSW composition, the project team aggregated data from four recent and regionally relevant waste characterization studies:

- **Iowa**
  - 2022 Iowa Statewide Material Characterization Study
- **Michigan**
  - 2024 Economic Impact Potential and Characterization of Municipal Solid Watse in Michigan
- **Pennsylvania**
  - 2022 Pennsylvania Waste Characterization study
- **Wisconsin**
  - 2020-2021 Wisconsin Statewide Waste Characterization Study

The Michigan study provided direct data from the state; however, the study was limited to one season and lower overall total sorting volumes[13]. To provide additional robustness to estimating Michigan's MSW stream, the project team incorporated the Iowa, Pennsylvania, and Wisconsin study results.

Study Selection Criteria

The studies were evaluated using the following criteria:

- **Geographic Relevance:** Preference was given to studies conducted in Michigan or neighboring states to ensure regional comparability.
- **Waste Stream Coverage:** Each study included data from both residential and commercial sources.
- **Regulatory Context:** Studies from states with similar regulatory framework, namely yard waste landfill bans and deposits were prioritized. However, given that Iowa and Michigan are the only deposit states in the Midwest region, Pennsylvania, Wisconsin, and Minnesota were also considered due to their yard waste landfill ban.

[13] For example, the Michigan study sorted a total of 15,200 pounds of material whereas the Iowa, Wisconsin, and Pennsylvania studies sorted 108,700, 46,800, and 314,500 pounds respectively.

- **Post-Covid Waste Compositions:** Waste composition studies conducted after 2020 were reviewed to account for potential changes in waste generation patterns due to the impacts of COVID-19.
- **Methodological Rigor:** The project team verified that all selected studies followed ASTM-aligned methodologies, including randomized sampling, a minimum sample size of 200 pounds, and an adequate number of samples to capture variability in the MSW stream.

Data Aggregation and Composition Estimation

To develop a representative MSW composition profile for Michigan, residential and commercial data from the three studies were aggregated. This required standardizing waste categories across studies, using a universal set of definitions and consolidating subcategories (e.g., grouping newsprint, magazines, and office paper under a single "paper" category) to ensure consistency.

The aggregated composition data were then applied to Michigan using the following process:

1. Per Capita Disposal Rate:
    a. 8,508,293 tons[14] / 4,041,760 households[15] = 4,210 pounds per household per year MSW disposed
2. County-Level MSW Disposal:
    a. For each Michigan County: County 2020 households * 4,210 pounds per household per year MSW disposed / 2000 pounds per ton = total county MSW disposed tons
3. Allocation by Generator Type:
    a. Residential MSW 48%[16]
    b. Commercial MSW 52%[16]
4. Allocation by Housing Type:
    a. Single family households (1-4 units)
    b. Multi-family households (5+ units)
    c. Proportion based on U.S. Census 2020 housing unit estimates by county.
        i. Ex: Alcona County Michigan 91% single family, 9% multi-family
5. Material Composition by Generator Type and Geography
    a. The county-level totals for single family, multi-family, and commercial MSW were multiplied by the corresponding aggregated composition data to estimate tons disposed by material category.

This five-step methodology enabled the project team to model MSW composition across all Michigan counties, disaggregated into 58 standardized material categories.

[14] Report of Solid Waste Landfilled in Michigan Fiscal Year 2024
[15] U.S. Census 2020
[16] Estimate of residential and commercial MSW proportions based on average proportions from Pennsylvania Waste Characterization Study 2022, Illinois Commodity/Waste Generation Characterization Study Update 2015, MSW Residential/Commercial Percentage Allocation U.S> Environmental Protection Agency 2013.

FULL CHARACTERIZATION

Table 25 below is the results of the full waste characterization, broken out by residential and commercial sectors. The residential and multi-family sector total 4,054,202 tons (48%) and the commercial sector totals 4,454,091 (52%).

Table 25: Residential and Commercial Waste Characterization (tons, 2024)[17]

SPECIFIC CATEGORY	SINGLE-FAMILY	MULTI-FAMILY	COMMERCIAL	TOTAL
Glass Non-Container	20,159	4,531	24,511	49,201
Glass Bottles and Jars	73,281	16,473	54,663	144,416
Glass Broken	0	0	0	0
Aluminum Cans	23,214	5,218	20,182	48,614
Aluminum Foil	9,774	2,197	8,693	20,664
Steel Cans	23,020	5,175	28,561	56,756
Steel Aerosol Cans	821	185	671	1,677
Ferrous Scrap	61,761	13,883	93,897	169,541
Non-ferrous Scrap	14,845	3,337	17,206	35,388
Mixed Ferrous and Non-Ferrous	8,664	1,948	17,228	27,840
Wasted Food	581,277	130,664	821,128	1,533,070
Leaves and Grass	142,717	32,081	71,356	246,155
Pruning and Trimmings	0	0	0	0
Branches and Stumps	0	0	0	0
Dimensional Lumber	7,753	1,743	31,138	40,633
Engineer Wood	2,472	556	13,089	16,117
Pallets and Crates	202	45	3,154	3,402
Painted and Treated Wood	96,419	21,674	130,370	248,463
Other Recyclable Wood	45,551	10,239	190,303	246,094
Manure	98,231	22,081	17,727	138,039
Other Organics	71,992	16,183	88,872	177,047
Corrugated Cardboard	147,427	33,140	431,577	612,144
Paperboard Boxboard	5,775	1,298	8,949	16,022
Kraft Paper	0	0	0	0
Newspaper (ONP)	24,820	5,579	19,537	49,936
Magazines	26,221	5,894	27,964	60,079
White office	22,541	5,067	41,790	69,398
Mixed Paper	171,852	38,630	235,119	445,602
Gable-top and Aseptic Cartons Combined	12,351	2,776	23,306	38,433
Paper Shredded	0	0	0	0

[17] Due to rounding, the totals presented in this table may differ from the sum of individual values by several tons.

SPECIFIC CATEGORY	SINGLE-FAMILY	MULTI-FAMILY	COMMERCIAL	TOTAL
Compostable paper	228,706	51,410	268,825	548,941
Paper Non-Recoverable	85,904	19,310	156,802	262,016
Plastic PET #1 Bottles	48,698	10,947	51,756	111,401
Plastic HDPE #2 Bottles colored	16,977	3,816	17,287	38,080
Plastic HDPE #2 Bottles Natural	13,031	2,929	19,327	35,287
Plastic PET #1 Non-Bottle	10,573	2,377	12,653	25,602
Plastic HDPE #2 Non-Bottle colored and natural combined	409	92	1,498	1,999
Plastic PS #6 Expanded Polystyrene	23,785	5,347	42,901	72,033
Plastic PP#5	15,797	3,551	22,233	41,581
Plastic Mixed Plastics Rigid #3-#7	48,230	10,841	68,345	127,416
Plastic Film	222,086	49,922	446,718	718,726
Plastic Durable and Bulky Plastic Items	48,973	11,008	76,691	136,672
Plastic Other	70,858	15,928	148,176	234,962
Plastic Compostable	202	45	257	504
Textiles	182,168	40,949	137,447	360,564
Electronics	28,718	6,456	22,402	57,576
Construction and Demolition Bricks and Concrete	14,924	3,355	22,068	40,346
Construction and Demolition Asphalt Roofing	6,347	1,427	14,518	22,292
Construction and Demolition Gypsum Board	17,196	3,865	30,073	51,134
Construction and Demolition Other	56,467	12,693	55,821	124,982
HHW	12,864	2,892	24,149	39,905
Bulky Waste Mattresses	2,176	489	2,397	5,062
Bulky Waste Carpet	48,226	10,841	65,901	124,968
Bulky Waste Other Bulky Waste	119,065	26,764	64,457	210,286
Tires	202	45	3,154	3,402
Inert Materials: Rock, Soil, and Fines	96,495	21,691	99,007	217,192
Residue	189,468	42,590	141,970	374,027
Rubber	8,440	1,897	16,267	26,604
Total	**3,310,125**	**744,077**	**4,454,091**	**8,508,293**

SMALL BUSINESS COMMERCIAL CARDBOARD RECYCLING SURVEY

Background Information

Q1. Are you knowledgeable about how waste and recycling is managed at your business?

- ☐ Yes
- ☐ No

Q2. How would you describe your business's NAICS code? *Drop-down of applicable NAICS codes provided.*

Q3. How many locations does your business have in the state of Michigan? *Drop-down options ranging from 1-10+*

Q4. If more than 10, please indicate the number of locations? *Free numerical entry*

Q5. What is the zip code of your flagship location? *Free text for zip code entry*

Q6. Is the flagship business location a standalone property or a part of a shared property complex under property management?

- ☐ Standalone property
- ☐ Shared Property

Q7. Approximately how many people do you employ at your flagship location? *Drop-down options ranging from 1-100.*

Q8. If more than 100, please indicate number of employees. *Free numerical entry*

Q9. How many people do you employ across all business locations? *Drop-down options ranging from 1-100.*

Q10. If more than 100, please indicate number of employees? *Free numerical entry*

Q11. Which best describes how your flagship location manages cardboard waste?

- ☐ My business does not have cardboard.
- ☐ My business throws cardboard out with the trash
- ☐ My business separates cardboard from trash for recycling
- ☐ My business does not separate cardboard from trash, but we are told by our hauler it is recycled.

Q12. What type of container is used for cardboard at the flagship location?

- ☐ Cart
- ☐ Dumpster
- ☐ Baled Separately
- ☐ Roll-off Container
- ☐ Cardboard is self-hauled to a drop-off location

Q13. How is cardboard collection handled at the flagship location?

- ☐ My business does not have cardboard
- ☐ The building owner contracts with a hauler for businesses renting units.
- ☐ The municipality provides collection to businesses.

Cart Collection (if selected in Q12)

Q14. What size of cart(s) do you use for cardboard at the flagship location?

- ☐ 35 Gallon
- ☐ 64 Gallon
- ☐ 95 Gallon
- ☐ Not Sure

Q15. How many carts do you currently have for cardboard at the flagship location? *Drop-down options ranging from 1-10.*

Q16. How frequently are carts picked up at the flagship location?

- ☐ Once a week
- ☐ Twice a week
- ☐ Three times a week
- ☐ Four times a week
- ☐ Five times a week
- ☐ Not sure

Q17. Typically, how much of the cart is full of cardboard at the flagship location right before it is picked up?

- ☐ Quarter full
- ☐ Half full
- ☐ Three quarters full
- ☐ Completely full
- ☐ Not sure

Q18. Are cardboard boxes broken down before being placed in carts at the flagship location?
- ☐ Yes
- ☐ Sometimes
- ☐ Not sure

Q19. If recycling, is cardboard mixed in with other recyclables at the flagship location?
- ☐ Yes, cardboard is mixed with other recyclables
- ☐ No, carts are cardboard recycling only
- ☐ Not sure

Dumpster Cardboard Collection (if selected in Q12)

Q20. What size dumpster does your business use for cardboard at the flagship location?
- ☐ 2 cubic yards
- ☐ 4 cubic yards
- ☐ 6 cubic yards
- ☐ 8 cubic yards
- ☐ Not sure

Q21. How many dumpsters does your business use for cardboard at the flagship location? *Drop-down options ranging from 1-10.*

Q22. How frequently are dumpsters picked up at the flagship location?
- ☐ Once a week
- ☐ Twice a week
- ☐ Three times a week
- ☐ Four times a week
- ☐ Five times a week
- ☐ Not sure

Q23. Typically, how much of the dumpster is full of cardboard at the flagship location right before it is picked up?
- ☐ Quarter full
- ☐ Half full
- ☐ Three quarters full
- ☐ Completely full
- ☐ Not sure

Q24. Are cardboard boxes broken down before placing in dumpsters at the flagship location?
- ☐ Yes
- ☐ Sometimes
- ☐ Not sure

Q25. If cardboard is recycled, is cardboard mixed in with other recyclables at the flagship location?
- ☐ Yes, cardboard is mixed with other recyclables
- ☐ No, carts are cardboard recycling only
- ☐ Not sure

Roll-off Cardboard Collection (if selected in Q12)

Q26. What size roll-off does your business use for cardboard at the flagship location?
- ☐ 15 Yard
- ☐ 20 Yard
- ☐ 30 Yard
- ☐ 40 Yard
- ☐ Not sure

Q27. How many roll-offs does your business use for cardboard at the flagship location? *Drop-down options ranging from 1-10.*

Q28. How frequently are the roll-offs picked up at the flagship location?
- ☐ Once a week
- ☐ Twice a week
- ☐ Three times a week
- ☐ Four times a week
- ☐ Five times a week
- ☐ Not sure

Q29. Typically, how much of the roll-off is full of cardboard at the flagship location right before it is picked up?
- ☐ Quarter full
- ☐ Half full
- ☐ Three quarters full
- ☐ Completely full
- ☐ Not sure

Q30. Are cardboard boxes broken down before placing in roll-offs at the flagship location?
- ☐ Yes
- ☐ Sometimes
- ☐ Not sure

Q31. If cardboard is recycled, is cardboard mixed in with other recyclables at the flagship location?
- ☐ Yes, cardboard is mixed with other recyclables
- ☐ No, carts are cardboard recycling only
- ☐ Not sure

Compactor Cardboard Collection (if selected in Q12)

Q32. What size compactors does your business use for cardboard at the flagship location?
- ☐ Small compactor
- ☐ Large compactor
- ☐ Not sure

Q33. How many compactors does your business use for cardboard at the flagship location? *Drop-down options range from 1-10.*

Q34. How frequently are the compactors picked up at the flagship location?

- ☐ Once a week
- ☐ Twice a week
- ☐ Three times a week
- ☐ Four times a week
- ☐ Five times a week
- ☐ Not sure

Q35. Typically, how much of the compactor is full of cardboard at the flagship location right before it is picked up?

- ☐ Quarter full
- ☐ Half full
- ☐ Three quarters full
- ☐ Completely full
- ☐ Not sure

Q36. If cardboard is recycled, is cardboard mixed in with other recyclables at the flagship location?

- ☐ Yes, cardboard is mixed with other recyclables
- ☐ No, carts are cardboard recycling only
- ☐ Not sure

Baling Cardboard Collection (if selected in Q12)

Q37. How many cardboard bales does your business make in an average week at the flagship location? *Drop-down options ranging from 1-10+*

Q38. If more than 10 bales per week, please write in approximate number of bales at the flagship location. *Free numerical entry*

Q39. Where are your cardboard bales going at the flagship location? *Free text entry*

Self-Haul Cardboard Collection (if selected in Q12)

Q40. Approximately how much cardboard does your flagship location self-haul to a drop-off site?

- ☐ Quarter pickup truck bed
- ☐ Half pickup truck bed
- ☐ Full pickup truck bed
- ☐ Not sure

Q41. Are cardboard boxes broken down before being placed in carts at the flagship location?

- ☐ Yes, cardboard is mixed with other recyclables
- ☐ No, carts are cardboard recycling only
- ☐ Not sure

Behavioral Questions

Q42. If your business does not recycle cardboard, please indicate why. *Select all that apply.*

- ☐ I do not know how to establish a recycling service for my business
- ☐ My business does not generate enough cardboard to warrant recycling it
- ☐ My business rents a location where the landlord is unwilling to provide recycling services
- ☐ Recycling services are not available in my area
- ☐ There is no good location onsite for cardboard to be stored for recycling
- ☐ Other

Q43. If your business does not recycle cardboard, please indicate why. *Free text entry, if answered in Q42.*

Q44. If you do not currently recycle cardboard but would like to, please describe what kind of help you would need to start a recycling program. *Free text entry*

Q45. In general, how could EGLE make recycling easier for small businesses? *Free text entry*

SCRAP METAL SURVEY

Survey Questions

Demographics

Q1. Are you 18 years or older?

- ☐ Yes
- ☐ No

Q2. What is your zip code? *Free text for zip code entry*

Q3. What is your age range?

- ☐ 18 to 24
- ☐ 25 to 44
- ☐ 45 to 59
- ☐ 60 to 74
- ☐ 75 and up

Q4. How would you describe your gender?

- ☐ Female
- ☐ Male
- ☐ Non-binary

Q5. How would you describe your race or ethnicity?

- ☐ American Indian and Alaska Native alone
- ☐ Asian alone
- ☐ Black or African American alone
- ☐ Hispanic or Latino
- ☐ Native Hawaiian and Other Pacific Islander alone
- ☐ White alone
- ☐ Another race alone
- ☐ Two or more races

Large Appliances

Q6. Which (if any) of these large appliances have been replaced or removed in your home in the past year? *Select all that apply.*

- ☐ Air Conditioner
- ☐ Dishwasher
- ☐ Dryer
- ☐ Full Sized Refrigerator
- ☐ Heating, ventilation, and air conditioning (HVAC) unit/ system
- ☐ Large Grill
- ☐ Mini-Refrigerator
- ☐ Stand Alone Freezer
- ☐ Stove
- ☐ Washing Machine
- ☐ None of these Items

Q7. Approximately how many of those items did you replace in the past year? *Large appliance options were carried forward based on the items selected in Q6.*

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5 or more

Q8. For large appliances that your household has replaced or removed in the past year, please indicate what you did with them. *Large appliance options were carried forward based on the items selected in Q6.*

- ☐ Arranged with appliance service provider to haul away after delivery of new item
- ☐ Arranged for curbside trash pickup with hauler
- ☐ Brought to a landfill or transfer station myself
- ☐ Brought to a metal scrap yard myself
- ☐ Gave item to friends or family
- ☐ Kept item at home
- ☐ Sold on Facebook marketplace, Craigslist, curbside give away, or other

Q9. If you kept any of the items, please indicate why. *Large appliance options were carried forward based on the items selected in Q6 if 'kept at home' was selected for item in Q8.*

- ☐ Continued to use the item in my current home
- ☐ Moved the item to a secondary residence for use
- ☐ Did not have a convenient way to get rid of item
- ☐ Did not know how to get rid of item

Metal Furnishings and Small Appliances

Q10 Which (if any) of these metal furnishings and/or small appliances have you replaced in your home the past year? *Select all that apply.*

- ☐ 2-Drawer metal filing cabinet
- ☐ 4-Drawer metal filing cabinet
- ☐ Bicycles
- ☐ Garden equipment such as shovels, clippers, spades
- ☐ Large metal shelves (4 to 6 shelves)
- ☐ Metal bed frames
- ☐ Metal cutlery, scissors, utensils
- ☐ Metal lamps
- ☐ Metal lawn chairs
- ☐ Metal patio furniture and/or accessories
- ☐ Metal power tools
- ☐ Metal sink
- ☐ Metal swings and/or play equipment
- ☐ Metal wheelbarrow
- ☐ Microwave oven
- ☐ Other metal micromobility electric vehicles (e.g., electric scooters, one wheel, segways)
- ☐ Other metal micromobility non-electric vehicles (e.g., unicycles, tricycles, kick scooters, handcycles, walker, crutches, wheelchairs)
- ☐ Other miscellaneous small metal items (keys, chains, etc.)
- ☐ Outdoor equipment such as lawnmowers or snow blowers
- ☐ Pots and pans such as iron skillet, iron pot, stainless steel or copper pan
- ☐ Small appliances such as toaster, metal coffee maker, toaster oven
- ☐ Small metal shelves (1 to 2 shelves)
- ☐ Steel aerosol cans
- ☐ Steel paint cans
- ☐ Televisions and/or radio
- ☐ Wrought iron furniture
- ☐ None of these items

Q11. Approximately how many of those items did you replace in the past year? *Metal Furnishing and Small Appliance options were carried forward based on the items selected in Q10.*

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7
- ☐ 8
- ☐ 9
- ☐ 10 or more

Q12. For metal furnishings and/or small appliances that your household has replaced in the past year, please indicate what you did with them. *Metal Furnishing and Small Appliance options were carried forward based on the items selected in Q10.*

- ☐ Arranged with appliance service provider to haul away after delivery of new item
- ☐ Arranged for curbside trash pickup with hauler
- ☐ Brought to a landfill or transfer station myself
- ☐ Brought to a metal scrap yard myself
- ☐ Gave item to friends or family
- ☐ Kept item at home
- ☐ Sold on Facebook marketplace, Craigslist, curbside give away, or other

Q13. If you kept any of the items, please indicate why. *Metal Furnishing and Small Appliance options were carried forward based on the items selected in Q10 if 'kept at home' was selected for item in Q12.*

- ☐ Continued to use the item in my current home
- ☐ Moved the item to a secondary residence for use
- ☐ Did not have a convenient way to get rid of item
- ☐ Did not know how to get rid of item

Household Remodeling and Demolition

Q14. Which (if any) of these items have been replaced in home in the past year? *Select all that apply.*

- ☐ Aluminum doors or window frames
- ☐ Aluminum siding
- ☐ Copper pipes
- ☐ Copper wiring
- ☐ Metal bathroom pipes
- ☐ Metal doorknobs and hinges
- ☐ Metal light fixtures
- ☐ Metal railings, gates, or fences
- ☐ Other miscellaneous small metal items (brackets, nails, screws, etc.)
- ☐ None of these

Q15. Can you estimate the rough size of the area where these items were replaced? For example, a typical small bathroom may be 20 square feet, a living room 100 to 250 square feet, and a single-family house 2,000 to 3,000 square feet. *Question asked if copper wiring, copper pipes, and/or metal bathroom pipes were selected in Q14*

- ☐ 20 square feet or less (small bathroom)
- ☐ 21 to 50 square feet (small room)
- ☐ 51 to 100 square feet (medium room)
- ☐ 101 to 250 square feet (large room)
- ☐ 251 to 500 square feet (extra-large room)
- ☐ 501 to 1,000 square feet (small home or condo)
- ☐ 1,001 to 2,000 square feet (medium sized single-family home)
- ☐ 2,001 to 3,000 square feet (Large single-family home)
- ☐ 3,001 to 4,000 square feet
- ☐ 4,000+ square feet

Q16. Approximately how many of the items did you replace in the past year? *Question asked if aluminum doors or window frames, metal light fixtures, metal doorknobs and hinges, and/or other miscellaneous small metal items (brackets, nails, screws, etc.) were selected in Q14*

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7
- ☐ 8
- ☐ 9
- ☐ 10
- ☐ 11
- ☐ 12
- ☐ 13
- ☐ 14
- ☐ 15
- ☐ 16
- ☐ 17
- ☐ 18
- ☐ 19
- ☐ 20 or more

Q17. Approximately how many linear feet of metal fencing, railing, or gates were replaced? *Question asked if metal railings, gates, or fences were selected in Q14*

- ☐ 0 to 50 feet
- ☐ 51 to 100 feet
- ☐ 101 to 200 feet
- ☐ 201 to 300 feet
- ☐ 301 to 400 feet
- ☐ 401 to 500 feet
- ☐ 501 to 600 feet
- ☐ 601 to 700 feet
- ☐ 701 to 800 feet
- ☐ 801 to 1,000 feet
- ☐ More than 1,000 feet

Q18. Approximately how large of a home was aluminum siding replaced on? *Question asked if aluminum siding was selected in Q14*

- ☐ 500 square feet
- ☐ 501 to 1,000 square feet
- ☐ 1,001 to 1,500 square feet
- ☐ 1,501 to 2,000 square feet
- ☐ 2,001 to 2,500 square feet
- ☐ 2,501 to 3,000 square feet
- ☐ 3,001 to 3,500 square feet
- ☐ 3,5001 to 4,000 square feet
- ☐ More than 4,000 square feet

Q19. Please indicate what you did with the replaced items? *Household Remodeling and Demolition options were carried forward based on the items selected in Q14*

- ☐ Arranged with appliance service provider to haul away after delivery of new item
- ☐ Arranged for curbside trash pickup with hauler
- ☐ Brought to a landfill or transfer station myself
- ☐ Brought to a metal scrap yard myself
- ☐ Gave item to friends or family
- ☐ Kept item at home
- ☐ Sold on Facebook marketplace, Craigslist, curbside give away, or other

Q20. If you kept any of the items, please indicate why. *Household Remodeling and Demolition options were carried forward based on the items selected in Q14 if 'kept at home' was selected for item in Q19*

- ☐ Continued to use the item in my current home
- ☐ Moved the item to a secondary residence for use
- ☐ Did not have a convenient way to get rid of item
- ☐ Did not know how to get rid of item

Other Scrap Metal Items

Q21. Were there other metal items your household disposed of in the last year that were not mentioned in this survey? If no, please enter "N/A". If yes, could you specify what, approximately how much, and what you did with item(s). For example: bag of iron nails, 10Lbs, Kept at home (don't know how to get rid of them). *Free text entry*

Behavioral Questions

Q22. What factors currently prevent - or would prevent - you from taking scrap metal to a scrap yard? *Select all that apply*

- ☐ I don't have enough metal items to make it worth the trip
- ☐ My items are not easy to transport
- ☐ The financial return is too low to motivate me
- ☐ There isn't a scrap yard close to me, or I don't know where to find one
- ☐ Other
- ☐ None of the above, I find it easy to take my items to the scrap yard

Q23. If you answered other to the above question, please let us know what prevents you from taking metal to a scrap yard. *Free text entry*

Q24. Which of the following would most motivate you to bring your items to a scrap yard? Please rank them from most (1) to least (5) important.

- ☐ Keeping items out of landfill
- ☐ Receiving payment for my scrap metal items
- ☐ Closest or easiest way for me to get rid of it
- ☐ Avoiding pickup fees or hassle
- ☐ Freeing up space at home

Q25. In the future, would you be more willing to take scrap metal to a super drop-off that accepts more materials in addition to a scrap yard?

- ☐ Yes
- ☐ No
- ☐ N/A, this would not have an effect

Q26. Thank you for taking the time to complete this survey! Please let us know if you have any additional comments or questions related to residential scrap metal recycling. *Optional free text entry*

Response Demographics

A total of 802 Michigan residents took part in the Residential Scrap Metal Survey. The age breakdown showed a wide range of participants: 43% of the respondents were 44 or younger, 58% were between the ages of 45 and 74, and 9% reported being 75 or older.

Gender representation was nearly even, with 49% identifying as male and 51% as female. An additional five individuals (1%) identified as non-binary. In terms of race and ethnicity, 77% of respondents identified as white, with all other racial groups represented at lower response rates. The largest single demographic group was white women aged 60 to 74, who made up 15% of total responses.

Geographically, the survey achieved broad statewide coverage, with responses from all Michigan Councils of Government (COGs) and 368 unique ZIP codes. The highest response rates came from COG 1 (41%) and COG 8 (15%), while more rural northern regions, COGs 11 and 13, each accounted for 1% of the total responses.

Material Weight Conversions

The scrap metal survey asked residents to report either the number of items replaced (e.g., large and small appliances and other household scrap metal), the linear feet of materials such as fencing or railing, or the square footage of spaces where materials such as aluminum siding, copper wiring, or copper piping were replaced. These estimates were used to translate the quantity of items replaced into total tons of material replaced. Table 26 provides the average weights used for each item represented in the survey. The sources for all weights are provided below.

Table 26: Quantity to weight conversion table for common household scrap metal items

	MATERIAL	AVG WEIGHT (TONS)
Small Appliances	2-Drawer metal filing cabinet	0.0265
	4-Drawer metal filing cabinet	0.0441
	Bicycles	0.0132
	Garden equipment such as shovels, clippers, spades	0.0011
	Large metal shelves (4 to 6 shelves)	0.0617
	Metal bed frames	0.0141
	Metal cutlery, scissors, utensils	0.0018
	Metal lamps	0.0044
	Metal lawn chairs	0.0062
	Metal patio furniture and/or accessories	0.0062
	Metal power tools	0.0011
	Metal sink	0.0176
	Metal swings and/or play equipment	0.0567
	Metal wheelbarrow	0.0132
	Microwave oven	0.015
	Other metal micromobility electric vehicles (e.g., electric scooters, onewheel, segways)	0.0214
	Other metal micromobility non-electric vehicles (e.g., unicycles, tricycles, kick scooters, handcycles, walker, crutches, wheelchairs)	0.0132
	Other miscellaneous small metal items (keys, chains, etc.)	0.0018
	Outdoor equipment such as lawnmower or snow blower	0.0132
	Pots and pans such as iron skillet, iron pot, stainless steel or copper pan	0.0018
	Small appliances such as toaster, metal coffee maker, toaster oven	0.0009
	Small metal shelves (1 to 2 shelves)	0.0265
	Steel aerosol cans	0.0001
	Steel paint cans	0.0062
	Televisions and/or radios	0.0176
	Wrought iron furniture	0.0132
Large Appliances	Air Conditioner	0.02
	Dishwasher	0.06
	Dryer	0.04
	Full Sized Refrigerator	0.05
	HVAC	0.01
	Large Grill	0.02
	Mini-Refrigerator	0.02
	Stand Alone Freezer	0.03
	Stove	0.07
	Washing Machine	0.08
Construction & Building	Aluminum doors or window frames	0.02205
	Aluminum siding*	0.00035
	Copper Pipes*	0.00032
	Copper Wiring*	0.00032
	Metal Bathroom Pipes*	0.0003
	Metal doorknobs and hinges	0.00057
	Metal light fixtures	0.01102
	Metal fencing, railing, or gates*	0.0004
	Other miscellaneous small metal items (brackets, nails, screws, etc.)	0.0000024

* Indicates estimated weight per sq ft

Weight Conversion Source Links

- https://democracy.york.gov.uk/documents/s2116/Annex%20C%20REcycling%20Report%20frnweights2005.pdf
- https://bax.tools/en/total-%CF%86%CF%84%CF%85%CE%B1%CF%81%CE%B9-%CE%BC%CE%B5%CF%84%CE%B1%CE%BB%CE%BB%CE%B9%CE%BA%CE%BF-%CE%BC%CE%B5-%CE%BE%CF%85%CE%BB%CE%B9%CE%BD%CE%B7-%CF%87%CE%B5%CE%B9%CF%81%CE%BF%CE%BB%CE%B1%CE%B2%CE%B7-(ththw0103)
- https://www.lehighvalleyabrasives.com/tools-and-equipment-sb-18-lt-bl-602316520-cordless-hammer-drill-kit-metabo-met602316520?gad_source=1&gad_campaignid=300333618&gbraid=0AAAAAD-u2noervY1Monaoc4CZZjVRMq82&gclid=Cj0KCQjwmK_CBhCEARIsAMKwcD5eGkjSQDyoACj-t8ypVMRhdxYT-U_i6iQQ1Consu4xtTuKZDqHNswaAnCvEALw_wcB
- https://www.lifetime.com/lifetime-91137-metal-swing-set-earthtone
- https://shop.segway.com/nl_en/ninebot-kickscooter-max-g2-e-powered-by-segway1.html
- https://www.pipelinepackaging.com/22-oz-aerosol-can-unlined-211x713-2p-10#:~:text=aerosol%20cans%2C%20unlined%20are%20made,a%20pallet%20weighs%20406%20lbs.
- https://sfsupplies.com/pdf/AluminumSheet_Weight_Chart.pdf
- https://taylorwalraven.ca/copper-tubing-data-Type-K.php#:~:text=Table_title:%20Copper%20Tubing%20Data%20%2D%20Type%20K,%7C%20Weight%20of%20Tube%20Lb/ft:%200.641%20%7C
- https://www.nessengr.com/technical-data/bare-copper-wire/
- https://allfasteners.com/pipe-weights?srsltid=AfmBOor6B9YkvSEY9NIR5_kQlmVGWPu9QPXoxurZpIClZ1GJn0b7yZE5
- https://www.wheatland.com/wp-content/uploads/2017/12/Chain-Link-Fabric-Average-Weights.pdf
- https://www.homedepot.com/p/Defiant-Brandywine-Stainless-Steel-Keyed-Entry-Door-Knob-32T8600B/323700654
- https://www.asmc.net/10-x-7-8-ft-sheet-metal-screw-phillips-flat-head-type-a-stainless-steel-18-8/

FOOD SCRAPS AND SURPLUS FOOD PROCESSORS

Access to food scraps and surplus food processing facilities is a vital component for creating and maintaining curbside or drop-off food scraps and surplus food programs. **Error! Reference source not found.** below displays the names and locations of each organics processing site (current as of 2024).

Table 27: Food Scraps and Surplus Food Processing Facilities

FACILITY NAME	ADDRESS	CITY	COUNTY
WeCare Denali - Ann Arbor	4170 Platt Road	Ann Arbor	Washtenaw County
Tuthill Farms and Composting, Inc.	10505 Tuthill Rd.	South Lyon	Livingston County
Morgan Composting Inc	4353 U.S. 10	Sears	Osceola County
Marquette County Landfill	600 County Road NP	Marquette	Marquette County
Mackinac Island Solid Waste	3967 Dousman St.	Mackinac Island	Mackinac County
Carter's Compost	841 Washington Street	Traverse City	Grand Traverse County
Country Oaks Atherton	3218 E Atherton Road	Burton	Genesee County
Delta County Landfill Compost Site	5701 19th Avenue North	Escanaba City	Delta County
Emmet County Pleasantview Rd	7363 S Pleasantview Rd	Harbor Springs	Emmet County
Fremont Community Digester	1634 Locust Street	Fremont	Newaygo County
Hammond Farms	5834 N Michigan Road	Dimondale	Eaton County
Park Place Composting	2116 S Elms Rd	Swartz Creek	Genesee County
Spurt Industries - Wixom	2041 Charms Road	Wixom	Oakland County
Bio Works Energy LLC	4652 Beecher Road	Flint	Genesee County
Michigan State University Anaerobic Digester	4075 College	East Lansing	Ingham County
Krull's Composting	857 W Burdickville Road	Maple City	Leelanau County
Cocoa Corporation	4368 60th Street	Holland	Allegan County
Grow Benzie	5885 Frankfort Highway	Benzonia	Benzie County
Tim Young's	10707 Oviatt Road	Honor	Benzie County

Carbon Farm LLC / 1801 Farm	1801 Frankfort Highway	Frankfort	Benzie County
Grand Haven Environmental Sustainability Center - Grand Haven	16850 Comstock Avenue Suite B	Grand Haven	Ottawa County
Environmental Sustainability Center - Holland	14053 Quincy	Holland	Ottawa County
Cornelius Farms Composting	100 R M86	Coldwater	Branch County
Wormies	5745 Whitneyville Ave	Alto	Kent County
Organicycle	4455 5 Mile Rd NW	Grand Rapids	Kent County
5 Heart Earthworm Farm	10161 S Beyer Rd	Birch Run	Saginaw County
Partridge Creek Farm	550 Cleveland Ave	Ishpeming	Marquette County

Food Scraps and Surplus Food Program Access Methodology

Community-level data, including populations and geographic boundaries, were sourced from the 2020 US Census. Community centroids made from the geographic center of each community in Michigan were used to determine access to drop-off programs. Therefore, if a community centroid was within an access area, the entire community population is counted towards having access.

Community With Food Scraps and Surplus Food Processing Capacity But Lacking Program Access

Table 28 provides a list of communities within a 25-minute drive of an organics facility processing food scraps and surplus food but lacking access to either curbside or drop-off food scraps and surplus food organics recycling programs.

Table 28: Communities Within 25-miles of Food scraps and surplus food Processing Facility with No Program Access

COMMUNITY CENSUS NAME			
Clyde township	Hamlin township	Lowell charter township	Lincoln township
Douglas city	Oneida charter township	Lowell city	Marion township
Fennville city	Potterville city	Rockford city	Marion village
Fillmore township	Walton township	Sparta township	Middle Branch township
Ganges township	Waverly CDP	Sparta village	Orient township
Heath township	Windsor charter township	Tyrone township	Osceola township
Laketown township	Argentine CDP	Vergennes township	Reed City
Manlius township	Argentine township	Elba township	Richmond township
Overisel township	Atlas township	Hadley township	Sylvan township
Salem township	Beecher CDP	Lapeer city	Albee township
Saugatuck city	Beecher CDP	Brighton city	Birch Run township
Saugatuck township	Burton city	Brighton township	Blumfield township
Carlton township	Clayton charter township	Genoa township	Bridgeport CDP
Freeport village	Clio city	Green Oak township	Bridgeport charter township
Freeport village	Davison city	Hamburg township	Buena Vista CDP
Irving township	Davison township	Hartland CDP	Buena Vista charter township
Middleville village	Fenton charter township	Hartland township	Burt CDP
Thornapple township	Fenton city (1 of 3)	Howell city	Carrollton township
Algansee township	Flint charter township	Oceola township	Frankenmuth city
Batavia township	Flushing charter township	Pinckney village	Frankenmuth township
Bethel township	Flushing city	Tyrone township	Maple Grove township
Bronson city	Gaines township	Whitmore Lake CDP	Reese village
Bronson township	Gaines village	Barryton village	Robin Glen-Indiantown CDP
Butler township	Genesee charter township	Chippewa township	Taymouth township
California township	Goodrich village	Fork township	Zilwaukee city
Coldwater city	Grand Blanc charter township	Sheridan township	Zilwaukee township
Coldwater township	Grand Blanc city	London township	Bancroft village

Gilead township	Lake Fenton CDP	Milan city	Byron village
Girard township	Lennon village	Milan township	Caledonia charter township
Kinderhook township	Linden city	Blue Lake township	Durand city
Matteson township	Montrose charter township	Casnovia township	Hazelton township
Noble township	Montrose city	Casnovia village	Lennon village
Ovid township	Mount Morris city	Cedar Creek township	New Lothrop village
Quincy township	Mount Morris township	Dalton township	Venice township
Quincy village	Mundy township	Fruitport charter township	Vernon township
Sherwood township	Otisville village	Fruitport village	Vernon village
Sherwood village	Richfield township	Holton township	Burr Oak township
Union City village	Swartz Creek city	Muskegon charter township	Burr Oak village
Union township	Thetford township	Muskegon Heights city	Colon township
Athens township	Vienna charter township	Norton Shores city	Colon village
Athens village	Allen village	Ravenna township	Fawn River township
Burlington township	Alaiedon township	Ravenna village	Arbela township
Burlington village	Aurelius township	Roosevelt Park city	Denmark township
Tekonsha township	Dansville village	Twin Lake CDP	Millington township
Tekonsha village	Delhi charter township	Ashland township	Millington village
Union City village	Edgemont Park CDP	Bridgeton township	Reese village
Bay Shore CDP	Holt CDP	Dayton township	Tuscola township
Bay township	Ingham township	Denver township	Vassar city
Indian River CDP	Lansing charter township	Fremont city	Vassar township
Tuscarora township	Leroy township	Garfield township	Ann Arbor charter township
Farwell village	Leslie township	Grant city	Augusta charter township
Freeman township	Mason city	Hesperia village	Barton Hills village
Garfield township	Meridian charter township	Lincoln township	Dexter city
Redding township	Okemos CDP (1 of 3)	Newaygo city	Dexter township
Surrey township	Onondaga township	Sheridan charter township	Lima township
Eagle township	Vevay township	Sherman township	Lodi township
Eagle village	Webberville village	White Cloud city	Milan city
Wacousta CDP	Wheatfield township	Groveland township	Northfield township
Bark River township	Berlin township	Holly township	Pittsfield charter township
Brampton township	Boston township	Holly village	Salem township
Escanaba city	Campbell township	Northville city	Saline city
Escanaba township	Clarksville village	Orchard Lake Village city	Saline township
Ford River township	Keene township	South Lyon city	Scio township
Gladstone city	Odessa township	Southfield city	Superior charter township
Rapid River CDP	Saranac village	West Bloomfield charter township	Webster township
Wells township	Coldwater township	Greenwood township	Whitmore Lake CDP
Benton township	Rives township	Hesperia village	York charter township
Brookfield township	Ada township	Newfield township	Ypsilanti charter township
Carmel township	Bowne township	Cedar township	Ypsilanti city
Charlotte city	Caledonia village	Evart city	Belleville city
Delta charter township	Cannon township	Evart township	Livonia city
Dimondale village	Cannonsburg CDP	Hartwick township	Northville city
Eaton Rapids city	Casnovia village	Hersey township	Northville township
Eaton Rapids township	Cedar Springs city	Hersey village	Plymouth charter township
Eaton township	Kent City village	Highland township	
Grand Ledge city	Van Buren charter township	Plymouth charter township	

WOOD WASTE DATA SOURCES

Direct data on wood waste generation and recycling in Michigan is limited; therefore, the preceding section was estimated using multiple data sources, as listed below.

Wood Waste Diversion to Reuse, Repair, Recycling

- Data on pallet generation and recycling was derived from (Hobbs 2020).
- The cited data source provided insights on pallet diversion and disposal at the national level. The data was applied to Michigan using 2021 U.S. Census employment factors between the US and Michigan, broken down by NAICS codes 11 through 99. For example, US employment for those NAICS codes in 2021 was 8.15 million compared to 225,000 for Michigan.
- To estimate the proportion of pallets generated from industrial sectors (NAICS 11 – 33) compared to commercial and institutional sectors (NAICS 42 – 99), total pallet generation rates were varied by sector using the California Generator-Based Characterization of Commercial Sector Disposal and Diversion (CalRecycle 2015). For example, the proportion of generated waste estimated as pallets was varied, as shown in Table 29.

Table 29: Proportion of Pallets in the Waste Stream by Generating Sector

NAICS	NAICS DESCRIPTION	%
11	Agriculture, forestry, fishing and hunting	5.3%
21	Mining, quarrying, and oil and gas extraction	0.4%
22	Utilities	0.4%
23	Construction	6.4%
31-33	Manufacturing	6.4%
42	Wholesale trade	3.2%
44-45	Retail trade	5.5%
48-49	Transportation and warehousing	7.2%
51	Information	0.4%
52	Finance and insurance	0.4%
53	Real estate and rental and leasing	0.4%
54	Professional, scientific, and technical services	0.4%
55	Management of companies and enterprises	0.4%
56	Administrative and support and waste management and remediation services	4.3%
61	Educational services	0.4%
62	Health care and social assistance	1.4%
71	Arts, entertainment, and recreation	0.2%
72	Accommodation and food services	0.6%
81	Other services (except public administration)	0.4%
99	Industries not classified	3.9%

Wood Waste Diversion to Compost, Mulching and Chipping

- Minnesota Pollution Control Agency 2023 reported chipped wood and composted waste.
- Pennsylvania 2023 reported chipped wood waste from residential and commercial sources.

Table 30: Wood Waste Diverted from Similar States, Minnesota and Pennsylvania (Tons)

MINNESOTA	2023
Chipped Wood	552,544
Composted Brush/Wood Waste	13,881
Minnesota Square Miles	86,943
Tons of Wood Waste per Square Mile	6.355
PENNSYLVANIA	**2023**
Wood Waste Residential	78,843
Wood Waste Commercial	232,482
Pennsylvania Square Miles	46,055
Tons Wood Waste Per Square Mile	6.760

- Percent of state forested:
  - MI 56%
  - PA 59%
  - MN 34%
- The Pennsylvania factor of 6.76 tons of wood waste per square mile was applied to Michigan as a proxy to manage wood waste in Michigan.
- Because the chipped wood waste reported in Pennsylvania as commercial is likely a mixture of commercial, institutional, and industrial sources, the Sankey diagram shown in Figure 34 excludes all commercial chipped wood waste to avoid counting any industrially generated wood waste as diversion. As such Figure 34 may represent a lower estimate of total mulched/chipped RCI wood waste in Michigan.

Wood Waste Diversion to Combustion

- There are six biomass facilities operating in Michigan with varying capacities. To determine their approximate wood waste consumption, the project team used a Megawatt-to-tons wood waste consumed conversion factor estimating approximately 9,500 tons of wood waste per MW generated (Table 31).

Table 31: Estimated Wood Waste Consumption of Michigan's Biomass Facilities (Tons)

FACILITY NAME	CITY	MEGA WATT (MW) CAPACITY	ESTIMATED WOOD WASTE CONSUMPTION (TONS)
Grayling Generating Station	Filer City	38	362,500
Genesee Power Station	Flint	36	343,400
L'Anse Warden Power Plant	L'Anse	20	190,800
McBain Power Plant	McBain	18	225,000
Lincoln Power Plant	Lincoln	16	152,600
Cadillac Renewable Energy	Cadillac	38	250,000
Total			**1,524,300**

- Due to the economics of operating these facilities, nearly all wood waste consumed is industrially generated.

Wood Waste Disposal in MSW

- Wood waste disposal in MSW landfills is estimated based on the waste characterization analysis described on the MSW Composition and Appendix Waste Characterization Modeling Methodology. Table 32 provides details on wood waste types and estimated proportions in the waste stream.

Table 32: Estimated Disposed Wood Waste in MSW Stream[4]

	RESIDENTIAL		COMMERCIAL		
WOOD TYPE	**PROPORTION**	**TONS**	**PROPORTION**	**TONS**	**TOTAL**
Dimensional Lumber	0.23%	9,495	0.90%	40,107	49,602
Engineer Wood	0.07%	3,028	0.46%	20,387	23,415
Pallets and Crates	0.01%	248	0.11%	4,913	5,161
Painted and Treated Wood	2.91%	118,093	2.81%	125,007	243,100
Other Recyclable Wood	1.38%	55,791	3.79%	168,806	224,597
Total	**4.60%**	**186,655**	**8.06%**	**359,219**	**545,874**

Wood Waste Disposal in C&D Landfills

- Michigan disposed of 1,810,609 tons of waste into C&D landfills in 2024.[14]
- Minnesota 2020 Construction and Demolition Materials Composition Study
    - The Minnesota Pollution Control Agency conducted a statewide composition assessment of debris disposed of in C&D landfills.

Table 33: Estimated Disposed Wood Waste in C&D Landfills[4]

	PERCENT	TONS
Painted and Treated Wood	8%	150,281
Dimensional Lumber	2%	41,644
Engineer Wood	3%	52,508
Pallets and Crates	2%	36,212
Total	**16%**	**280,644**

Storm Wood Waste Methodology

The total tons of storm wood waste were estimated based on reported cubic yard generation in Emmet County, Michigan and extrapolated across the 12-county affected region based on square land miles of each county (Ross, 2025). Cubic yards to tons were converted assuming 0.25 tons per cubic yard (U.S. Environmental Protection Agency, 2016).